ATLANTIUM

Pure Performance

UV Water Treatment

Atlantium Hydro-Optic™ Solutions

User Manual

Guide to the Atlantium Hydro-Optic UV System

RS104 Series

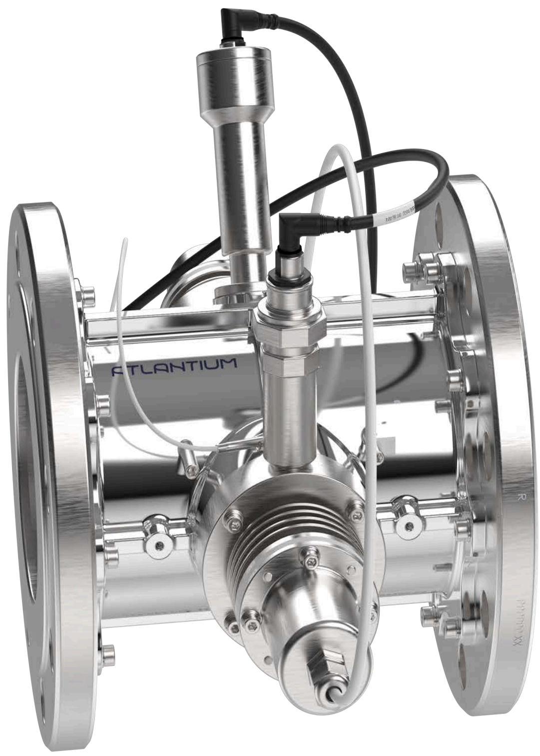
Standard Control Module

For General Applications

Supported Electrical Voltage: 400/440/480VAC

Document No. PS130000E

December 2024



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System and accessory specifications are subject to change without notice.

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IMPORTANT NOTICE

It is strictly forbidden to alter/change the system hardware in any way.

Any change made without written permission from Atlantium shall void the warranty.

For more information, including contact details for your local Atlantium representative, please visit the Atlantium website at: www.atlantium.com

Validations and certifications completed:

TYPE 4X

Certified to NSF/ANSI/CAN 61

UL508 E113027 & CSA C22.2 NO.14-13 Certificate

Revision History

Rev. #	Description	Date
001	Initial version.	Aug 2014
002	Corrections.	Aug 2014
003	CIP procedure updates. Added circuit breaker information.	July 2015
004	Wiring correction	Sept 2015
005	Updates and corrections to electrical and warning icons.	Mar 2017
006	PS130000E contains electrical information for both 400VAC & 440/480VAC.	Apr 2017
007	CIP procedure corrections and minor corrections.	Apr 2018
008	Correction regarding wiring harness length.	May 2018
009	New lamp housing	Feb 2022
010	US market style changes	Jul 2022
011	Updated Modbus	Jan 2024

Table of Contents

ATLANTIUM
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1 Introduction

Congratulations on purchasing an Atlantium RS104 Series Hydro-Optic UV Water Treatment System. This user guide provides you with clear guidelines for preparing the site for installation, correct installation and software configuration procedures, and a good understanding of all system functions for maximum benefit from your Atlantium Unit.

Atlantium systems are designed to provide applications for the following Industries:

- Aquaculture
- Fish Hatcheries
- Dairy
- Municipal
- Power Generation
- Ultra Pure Water
- Marine
- Bio-Pharma
- Bottled Water
- Beverages
- Breweries
- Aquariums
- ...and more

Patented Innovative Technology

- Features Hydro-Optic engineering that optimizes UV efficiency and enables uniform dose distribution.
- Medium Pressure High Intensity lamps provide a UV dose in the 200 to 454 nm spectrum which is much broader than that obtained from low pressure UV lamps (which yield a dose in the 254 nm range only). UV exposure is effective in cold & warm water.
- Sustainable operation provides the security and reliability that the industry demands.
- Uses "total internal reflection" of UV light within the reaction chamber, enabling more effective treatment at a lower dose.
- Some enclosures/components as noted on the Atlantium main website under https://atlantium.com/about.html include:
  - Nema 4x rating for indoor and outdoor use, protection against windblown dust and rain, splashing water and hose directed water.
  - IP56 rating protection from limited dust ingress. Protected from high pressure water jets from any direction

Controlled Dose Delivery

- Monitoring & control software ensures that required UV dose is delivered at all times.
- Dedicated UV Intensity Sensor (UVIS) for each lamp for real-time monitoring.
- Automatically adjusts lamp power based on real-time analyses of critical parameters such as water flow rate, lamp intensity and water Transmission to UV (UVT).
- A user-friendly interface displays the monitored data in real-time including the actual delivered dose as well as the other critical parameters, so you can verify that you are actually getting the dose that you need.

Lower Electrical Consumption

- Optimized light distribution, through Hydraulic and Fiber optic principles applied to the UV system design, enable:
- Optimized UV power and uniform dose distribution.
- Efficient use of power, with lower electrical consumption than low-pressure based UV systems.

Minimum Down Time

- True in-line system.
- Quick lamp replacement (4 minutes) and better safety.

Integrates with your Process Line

- Field proven installation.
- With no additional software, you can integrate the Atlantium system into your current process using Atlantium's customizable settings.
- Automatically generated reports at the push of a button for performance recording and traceability.

Efficiency and Easy Maintenance

- For maximum efficiency and dependable performance, the Atlantium system monitors the lamps for UV output, ensuring that a lamp is replaced only when absolutely needed.
- By following the easy preventive maintenance programs, you keep the system working at peak performance.
- The **Viewport** allows you to safely peek at what's going on inside.

For further information about Atlantium products ad technologies, visit our website at https://atlantium.com/

1.1 Before You Begin

Before you begin using this product, or any installation or service operation, please read the following safety information:

- Attention to these warnings helps prevent personal injuries and damage.
- It is your responsibility to use the product in an appropriate manner.
- This product is designed for use solely indoors.
- You are responsible if the product is used for other than its designated purpose or in disregard of Atlantium instructions. Atlantium shall assume no responsibility in any way for such use of the product.
- The product must be used for its design purpose, based on its product documentation and within its performance limits.
- Using the product requires technical skills. It is therefore essential that only skilled and specialized staff or thoroughly trained personnel be allowed to use the product.
- Keep the basic safety instructions and the product documentation in a safe place and pass them on to the users.
- Applicable local or national safety regulations and rules for the prevention of accidents must be observed at all times.

1.2 Tags and Their Meaning

The following indicators are used in the product documentation to warn the reader about risks and dangers

.

DANGER!

Indicates a hazardous situation which, if not avoided, will result in death or serious injury.

WARNING!

Indicates a hazardous situation which, if not avoided, could result in death or serious injury.

CAUTION!

Indicates a hazardous situation which, if not avoided, could result in minor or moderate injury.

ATTENTION!

Indicates the possibility of incorrect operation which can result in damage to the product.

Indicates a hazardous situation involving electricity which, if not avoided, can result in death or serious injury.

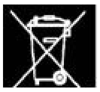
Indicates a hazardous situation involving Electrostatic Discharge (ESD), which, if not avoided, can result in damage to the product.

Indicates a hazardous situation involving UV light which, if not avoided, can result in death or serious injury.

Indicates a hazardous situation involving a hot surface which, if not avoided, can result in death or serious injury.

Indicates a hazardous situation involving working in an enclosure with moving parts, which if not avoided, can result in serious injury to the hands.

Indicates a hazardous situation involving mercury, a poisonous material which, if not avoided, can result in death or serious injury.

	Indicates a hazardous situation involving caustic chemicals which, if not avoided, can result in death or serious damage.
	Indicates that components or equipment are heavy and care is to be taken to avoid lifting incorrectly. Incorrect lifting can be dangerous to the personnel lifting and may result in dropping and damaging the components or equipment.
	Indicates that components or materials to be discarded are classified as hazardous waste and must be disposed of appropriately.
	Indicates information related to installation, safety or use of system.

2 About the RS104 Series Systems

The Atlantium RS104 Series System includes a chamber with one UV lamp and an electrical cabinet incorporating a Ballast Module power supply and a control touch screen. The UV chamber takes advantage of ***Total Internal Reflection*** to reflect UV light back into the water and uniformly distribute the UV dose. Water flows unimpeded, resulting in low head-loss.

The specially designed short (Arc length 95 mm / 3.74 inch) Medium Pressure High Intensity (MPHI) lamp is housed within a thick quartz sleeve that significantly reduces risk of breakage and its consequences. A UV Intensity Sensor (UVIS) provides continuous measurement of each lamp's output.

In addition, a UVT Analyzer reports and tracks the water transparency.

2.1 The RS104 unit

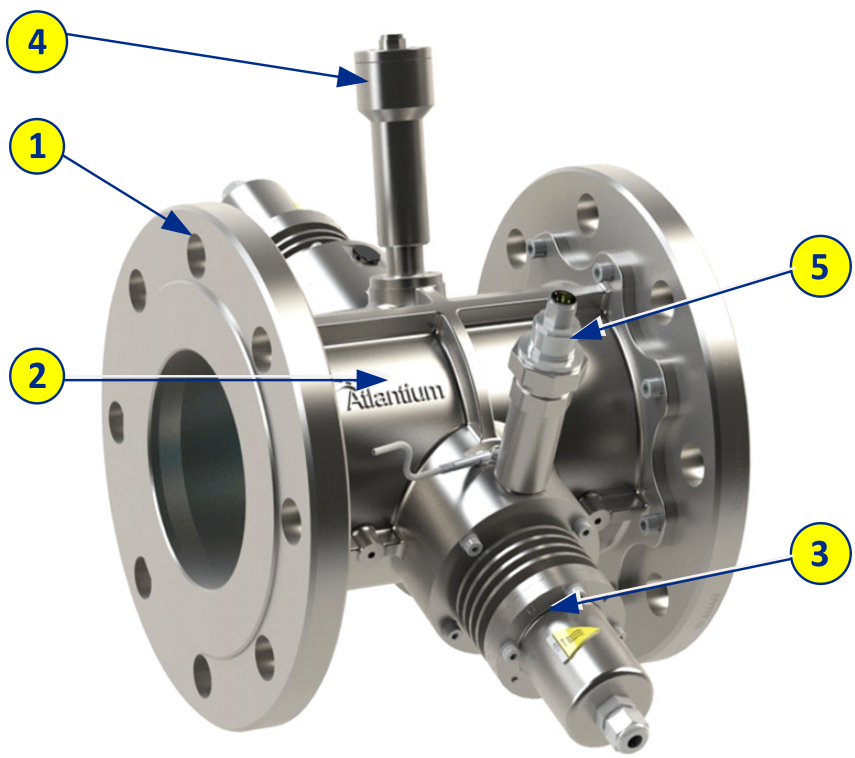


Figure 2-1: RS104 unit

Table 2-1: RS104 Components key and specifications

#	Item	Description
1	Flange interface	The unit connects to the facility piping via a flange interface as per customer order. Options: Flange DIN 2527 DN100 PN16 / Flange ANSI B 16.5 4” 150lb Tri-Clamp Ferrule DIN 32676 DN 100
2	Lamp Chamber	The water flows through a a stainless steel (316L) chamber in which the exposure to UV occurs.
3	Lamp connector	Connector for cable from Ballast power supply in Electrical Cabinet to lamp
4	UVT Analyzer	The sensor that measures the UV transmittance through the water. Together with the UV Intensity sensor and the Control Module, it analyzes, tracks and reports the water quality.
5	UV Intensity Sensor	The sensor that measures the UV intensity in the Lamp Chamber and feeds it to the real-time dose calculation

2.1.1 RS104 Series Interface Control Drawings (ICD)

For full specifications of the supplied system, consult the accompanying ICD files.

2.1.2 Atlantium System Footprint/Specifications

The Atlantium system has a small, compact footprint that facilitates in-line installation in any process train.

Table 2-2: RS104 specifications

Item	Detail		
Number of lamps	1		
Lamp length:	95 mm / 3.74 inch		
Lamp power consumption	1 kW		
Atlantium Unit Construction Materials: Casing: Quartz Chamber:	Electro-polished Stainless steel 316 Optional**: Super-Duplex stainless steel UNS S32750 High grade fused silica (Quartz)		
Standard Pipe Flange Interface Options:	**Flange**		**Ferrule/Tri-Clamp**
	DIN2527 DN100 PIN16	**ANSI B 16.5 4" 150lb**	**DIN 32676 DN100**
Unit Length (mm/inch):	213/8.4	221/8.7	249/9.8
Width-max (mm/inch):	405/16		
Unit weight (kg/lb)	17/38	20/44	13/29
Unit volume (liter/gallon)	0.9/0.2	0.9/0.2	1.1/0.24
Inner ∅ Diameter (mm / inch)	100 / 3.94	101.6 / 4	100 / 3.94
Outer ∅ Diameter (mm / inch)	220 / 8.66	228.6 / 90	119 / 4.7
Bolt Circle Diameter*	180 / 7.08	191 / 7.52	N/A
Bolt Hole Diameter	19/ 0.75		N/A

Item	Detail	
Number of Bolts	8	N/A
Service Clearance on each side (mm / inch)	450 / 17.7	
Minimum Height above Floor Level (mm / inch)	750 / 29.5	
Maximum Working Pressure	7 Bars / 102 psi	
Maximum Flow Rate	Application dependent	
Mounting Brackets	Not required: unit is supported by system pipes	

** When tightening bolts to connect piping, use a torque wrench to apply correct and uniform torque, according to accepted standards and practices.*

*** Super-Duplex is available as an alternative construction material for use with seawater. Consult your Atlantium representative for details.*

2.2 The Electrical Cabinet

The Atlantium system includes a Electrical Cabinet (that may be mounted on the wall or on a suitable anchoring point. The location of the Electrical Cabinet must be designated according to the maximum distance needed for positioning the Cable Harness of the Unit.

The length of the standard Cable Harness that connects the Unit to its electrical cabinet is 5 meters (16.5 ft.). An optional 10 meter (33 ft.) cable is available upon request. This means that you must determine the path that the Cable Harness can be placed and on that basis, calculate where the electrical cabinet can be located. The electrical cabinet can be affixed to a wall or onto a bracket structure.

The electrical cabinet (Figure 2-2) includes:

- Air vents with fan (1) and filters (4)
- Control touch screen (2)
- Cabinet doorhandles (3)
- Cable ducts (5)

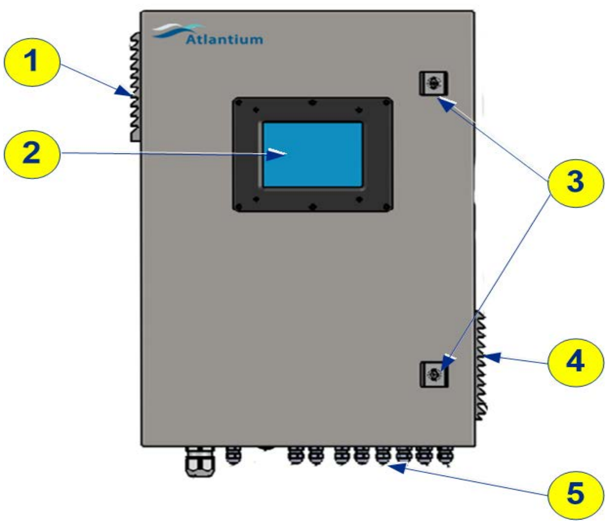


Figure 2-2: RS104 Electrical Cabinet (external)

Table 2-3: Electrical Cabinet Physical Details

Physical	Details
Electrical Cabinet (H x W x D) (mm/ inch):	641.6 / 25.3 x 425 / 16.7 x 255.4 / 10.1
Weight (Kg / lb):	22.2 / 49
Clearance (mm / inch): Filter Removal: Open Door Radius:	500 / 19.7 400 / 15.8
Protection Category:	IP 55

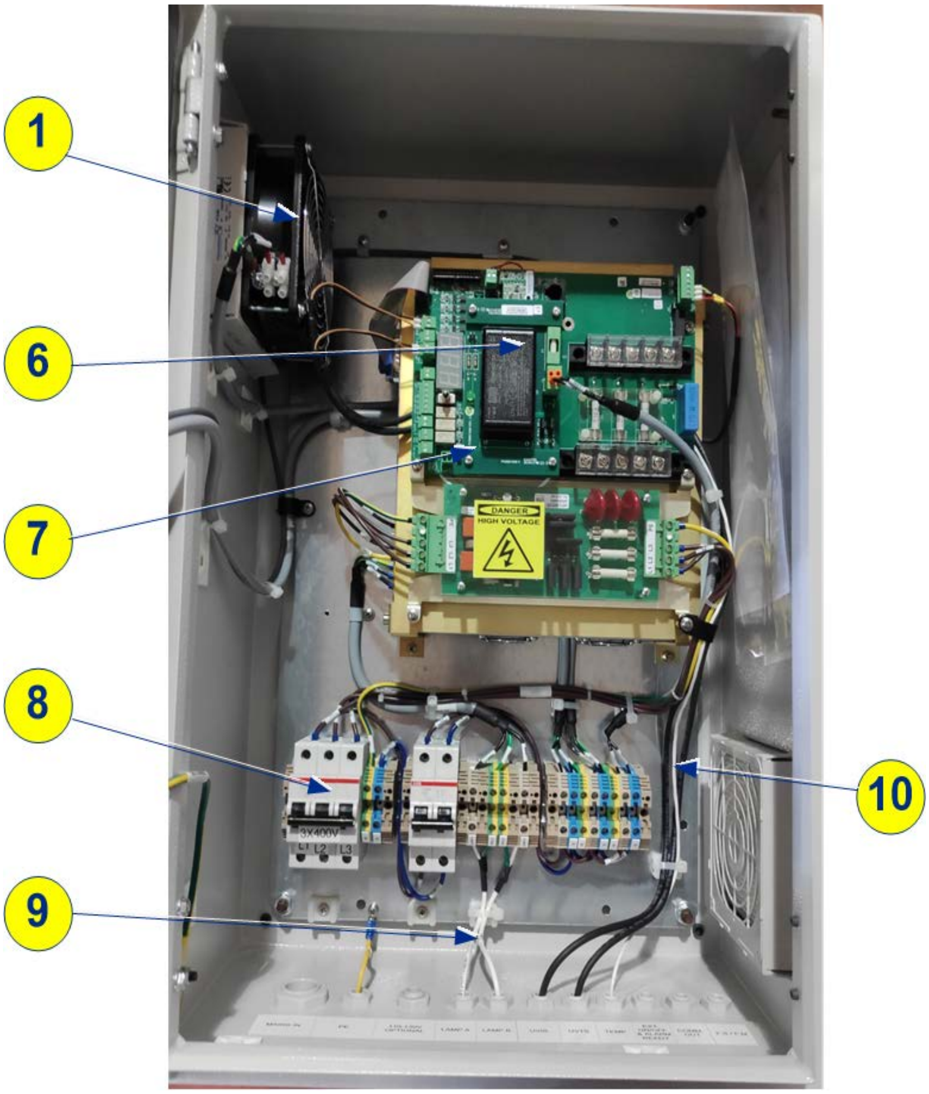


Figure 2-3: RS104 Electrical Cabinet -Internal view

Inside the Electrical Cabinet are the following components:
1 - Fan
6 - Power supply for UV lamp
7 - LM card
8 - Connector for AC and mains circuit breaker
9 - Power cables to lamp
10- Cables to UVIS and UVT sensors

2.3 Electrical Usage

Utilize a Residual Current Device (RCD) that is Type A with 3000A surge resistance.

Atlantium recommends ABB P/N F204 A-xx/0.03 AP-R (xx represents the current rating according to the system load. Select the appropriate RCD accordingly. Calculating the value, takes into account the power consumption of the system (see the table in the section below) and the mains voltage of your locale.

The tables below detail the Atlantium Unit’s electrical details and power consumption.

Table 2-4: RS104 Series Electrical Details

Electrical	Details
Phase: Voltage:	3 phase 400/440/480 VAC
Maximum allowed voltage tolerance:	±10% from the nominal voltage

2.3.1 Power Consumption and Local Voltage Requirements

The table below details the power consumption of the Atlantium system with direct feeding according to local voltage.

# of lamps:		1	
Nominal lamp power (kW):		1	
Feeding type	Feeding voltage (V)	Current consumption each phase (A)	Required Circuit Breaker
Direct Feeding (w/o Power Module)	400	1.6	3x4A

2.3.2 Residual Current Device

In addition to the circuit breakers defined in the electrical ICD provided with the RS104 Series system, you should install a Type A Residual Current Device (RCD) as extra protection for your Atlantium system, while ensuring that local safety regulations are complied with.

Atlantium recommends, for example, using ABB P/N F204 A-xx/0.1 RCD, where xx represents value of current that is at least 25% higher than the current rating of the system, provided in the electrical ICD.

When choosing the proper RCD take into account also the mains voltage and phase configuration of your locale.

The RCD should be tested regularly as per the manufacturers recommendations.

2.3.3 Mains power supply up to 480VAC

Atlantium systems require connection to mains power supplies (PS) of 400 to 480 VAC. If the local mains supply is outside of these limits, transformers or other adapters must be installed in order to provide the necessary PS of ≤ 480VAC.

Throughout this User Manual, it is assumed that the Atlantium system is connected to a PS of 400 - 480VAC.

The tables below detail the Atlantium Unit's electrical usage and power consumption.

Table 2-5: RS104 Series Unit Electrical Usage

Electrical	Details
Main power source	400 VAC 3~ + N + PE 50/60Hz 440Y/254 VAC 3~ + PE 50/60Hz 480Y/277 VAC 3~ + PE 50/60Hz
Maximum allowed voltage tolerance	±10% from the nominal voltage

2.3.4 Power, Voltage and Grounding Requirements

The table below details the power consumption for the Atlantium system, according to local mains voltage.

All circuit breakers are sized according to the tables below and are to be type C.

- The power system must be solidly grounded where the nominal line to ground voltage does not exceed 277V.
- The system is not appropriate for use on corner grounded delta, resistance grounded and ungrounded power systems.
- If the power system at the site is not appropriate, you can create an appropriate power feed by using a 1:1 isolation transformer (by grounding the middle point of the star configuration at the secondary side of the transformer).

2.3.5 Ensuring Power Stability

The voltage of the electrical supply connected to every Atlantium system **must not vary** by more than ± 10% from the correct value (sag, swell and surge) at the system inlet.

To comply with this requirement, you must protect the Atlantium system with a device that prevents electrical surges and fluctuations from causing damage. There are two types of surge protection:

- UPS (Uninterrupted Power Supply) has an internal battery that allows the equipment plugged into it to continue to run in the event of a power outage until power is restored. It also has the benefit of performing the same functionality of surge protectors, protecting against spikes and surges. Make sure to use a UPS that is capable of supplying sufficient current to the system and that the cable length is not more than 10m (33ft) in length. If needed, the UPS allows the system to be shut down cleanly if power is out for a lengthy period.
- Surge protection device protects your equipment from variations in electrical current, such as surges and spikes.

Surge protector devices /UPS must be inspected periodically and replaced if their protection capacity has expired.

2.3.6 Surge Protector Devices

Surge protection devices are designed to protect industrial communication networks. This device recommended by Atlantium uses a combination of 3-electrode gas discharge tubes and fast clamping diodes. Typical applications include industrial processing equipment, transmission systems, I/O cards, probes, actuators, and displays. Surge protection devices are available via Atlantium.

Table 2-6: Surge Protector Details

Atlantium PN	DATA SHEETS	Module PN	Description
EP0002300	DOC002300	8859660000	SURGE PROTECTOR 400V 3P 40KA TNC CONFIG
EP0002600	DOC002600	4983-DS120-402	SURGE PROTECTOR L + N, 115V
EP0002800	DOC002800	1352740000	SURGE PROTECTOR L + N, 230V

2.4 Environmental Requirements

The Atlantium system should be located indoors in an area that is adequately cooled and ventilated with clean airflow. However, there have been many successful Atlantium installations in less friendly environments. For locations with high temperatures - above 45°C (113°F) - air conditioning may be required. Consult with your Atlantium representative.

Table 2-7: RS104 Series System Environmental Details

Environmental	Details
Maximum ambient air temperature:	45ºC (113ºF)
Maximum ambient air temperature in the Ballast Modules environment:	45ºC (113ºF)
Maximum atmospheric relative humidity:	90%
Water Temperature:* Maximum Water Temperature (lamps off) Maximum Water Temperature (lamps on) Minimum – cold temperature:	 90ºC (194ºF)** 60ºC (140ºF) ** No limit - if the unit is fully drained of water before it freezes

* For water temperature above 70ºC (160ºF) consult with Atlantium service engineers
** Any deviation from either of these values require consultation with Atlantium applications department and written approval by Atlantium.

2.5 Regulatory Compliance

Atlantium Systems are designed to comply with the following regulatory standards*:

- EMC directive 2014/30/EU
- European Low Voltage Directive (LVD), 201*4/35/EU for electrical safety
- Council Directive 98 / 83 / EC of 3 November 1998 for the quality of water intended for human consumption
- ISO 9001-2015 2018-2021 Quality Management Standard - Development, Design, Production and Sales of Water Disinfection Systems
- UL 508, CSA C22.2 No, 14-13. By MET Labs (eurofins) NRTL certification
- IP54/IP56 & TYPE4X
- NSF 61. By NSF International.
- Hygienic certificate, National Institute of Public Health, Poland

**Note: Compliance is model dependent, for more details of specific UV Unit model. Consult your Atlantium representative.*

3 Getting Ready for Installation

Using the guidelines and information in this section, you can prepare the piping and electrical setup for installing the Atlantium Hydro Optic Water Treatment System. If you have any questions, consult with an Atlantium application engineer.

3.1 Planning

Consider where to locate the unit from the perspective of your process. Prepare your installation plan according to the Mechanical and Electrical ICD drawings provided with the system and to planning points below, as well as the other guidelines in this manual.

3.1.1 Access

- Installation Access
  - Take into account that sufficient door space is required to transport the Atlantium Unit into the facility where it is to be placed.
  - The Atlantium units are packed and transported in large wooden crates. Consider the need for a place near the installation area for unpacking of the units from the crates, prior to installation.
- Service Access
  - Ensure that sufficient space is required for maintenance personal to access the Atlantium Unit for periodic component replacement and maintenance.
  - The installed units must also be accessible: for example if a unit is installed high above the ground, permanent stairs or walkways may be needed for convenient and safe service access
  - Ensure sufficient space to access the back of the unit, to allow removal and replacement of lamps as well as to do maintenance on cleaning systems (where these are installed), connection boxes, etc.
- Installation: vertical or horizontal
  - Vertical installation of the Atlantium unit and piping, with an upwards direction of flow, is the preferred configuration for installation, where this is possible.
  - If mounting the unit in a horizontal configuration, a slope of at least 5% is recommended to minimize the risk of air pockets developing or remaining trapped in the Atlantium unit. The Customer is to supply the mechanical means for ensuring this slope.

3.1.2 Hydraulics

- Piping
  - Align the upstream and downstream piping with the designated location for the Atlantium unit.
  - Prepare the flanges/ferrules and piping as specified according to ***The Piping Infrastructure*** on page 3-26. To avoid a step in the connection, make sure that the internal diameters where the Atlantium Unit and your piping match.
  - **Do not** use sealing material that contains graphite when sealing the flanges/ferrules and gaskets as it accelerates the corrosive process on stainless-steel components, including those of the Atlantium system.
  - To promote laminar flow of water through the Atlantium unit:
    - The piping connected to either end of the unit should be of the same diameter as the unit itself. Connection of narrower piping may reduce flow and increase water temperature and turbulence.
    - Allow a straight length of 10 pipe diameters (10D) before the Atlantium unit (upstream) and a straight length of 5 pipe diameters (5D) after the unit (downstream). In situations where less space is available, review with your Atlantium representative.

- In order to minimize turbulence and air pocket formation in a vertical installation, install a concentric reducer/expander adapter to streamline the transition between piping dimension and the internal dimension of the unit.
- For a horizontal installation, install an eccentric reducer/expander adapters, with downstream outlet higher than inlet.

- Mechanics.
  - Prepare the supports needed for the Atlantium Unit, ensuring that the minimum height above the floor is 750mm / 29.5 inch. The RS104 unit weighs just 17 kg / 38 lb, so support provide by adjoining pipping is generally enough to hold it up securely.
  - Check that the piping on the inlet and outlet sides has adequate support structures so that there are no mechanical stresses that may endanger the Unit.
  - Check the upstream and downstream piping for vibrations. Make sure that the piping to be attached to the Atlantium Unit is anchored and reinforced to protect the Unit against vibration. A high vibration level can cause damage to any hydraulic system, including the Atlantium unit. High vibration can be caused by an unbalanced pump or by sudden valve close.
- Hydrodynamics
  - Water hammer can cause damage to any hydraulic system, including the Atlantium Unit. Make sure that this phenomenon does not occur in your facility's water line. To prevent water hammer, consult your system engineer and check your facility system procedures, including:
    - When starting the water flow when the lines are empty, keep valves open so that the air in the pipes has a simple release pathway.
    - Sudden valve closures can cause water hammer. Make sure that valves close gradually, enough to avoid this problem.
    - Ensure that all pumps that impact the Atlantium system employ soft start-up procedures.
    - The Atlantium Unit must be totally filled with water. Assure that no air can be trapped in the unit and that air can be released. Verify through the Unit's viewport after installation that there are no bubbles.

3.1.3 Flow Data

- Evaluate your flow trends to determine whether you require a flow meter or flow switch. If the flow varies, a **flow meter** is required to detect and report on the flow coming into the unit. Make sure that there are to be **no** branches on or off the pipes between the flow meter and the Unit inlet. If the flow rate is static, a **flow switch** can be used instead.

3.1.4 Application

- Verify that the dose-related parameters, such as the UVT and the flow, are the same as those defined in the Atlantium application scope of supply.
- Determine what a low flow situation means to your process and whether you need to shut down the Unit (low flow indicates upstream problems) or recirculate (low flow normal) when the flow goes below the normal range.

3.1.5 Feed Water Quality

If the water entering the Atlantium Unit contains debris or particles in suspension, pre-treatment (e.g. filtration) may be required to maximize UV exposure and protect internal parts from blockage or damage. Consult with your Atlantium representative.

3.2 Typical Installation Set-Up

Use the diagram below as a basis for your installation drawings. Vertical installation with upward flow is preferred where it is possible and accessible, as it prevents the accumulation of air pockets. If horizontal installation is required, be sure to include a 5 degree slope (recommended/minimum) in the direction of flow to reduce the potential for air pockets. Be sure to include service access as you fit the Atlantium system into your facility (i.e., nearby pipes, electrical fixtures, support beams, etc.)

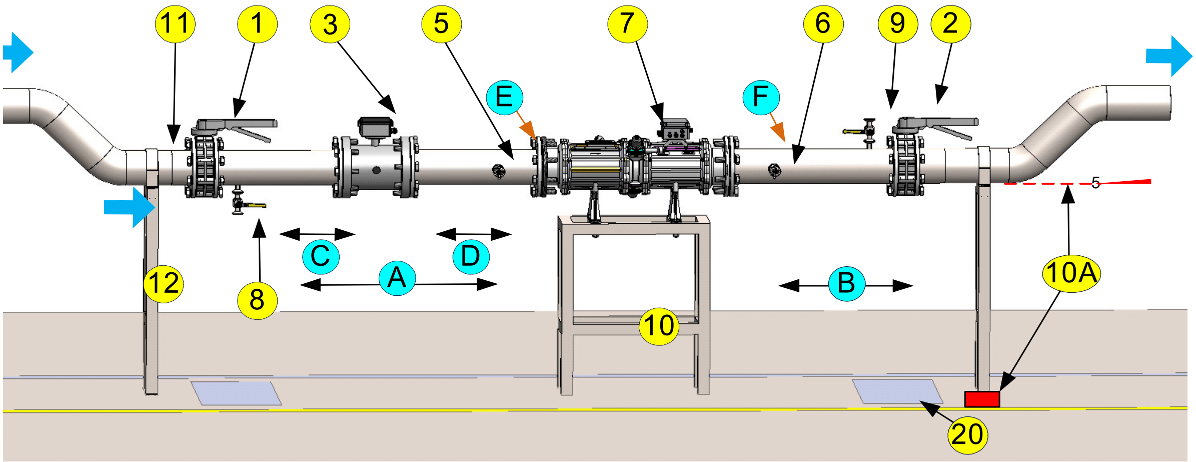


Figure 3-1: Generalized Atlantium system setup, horizontal (no bypass)

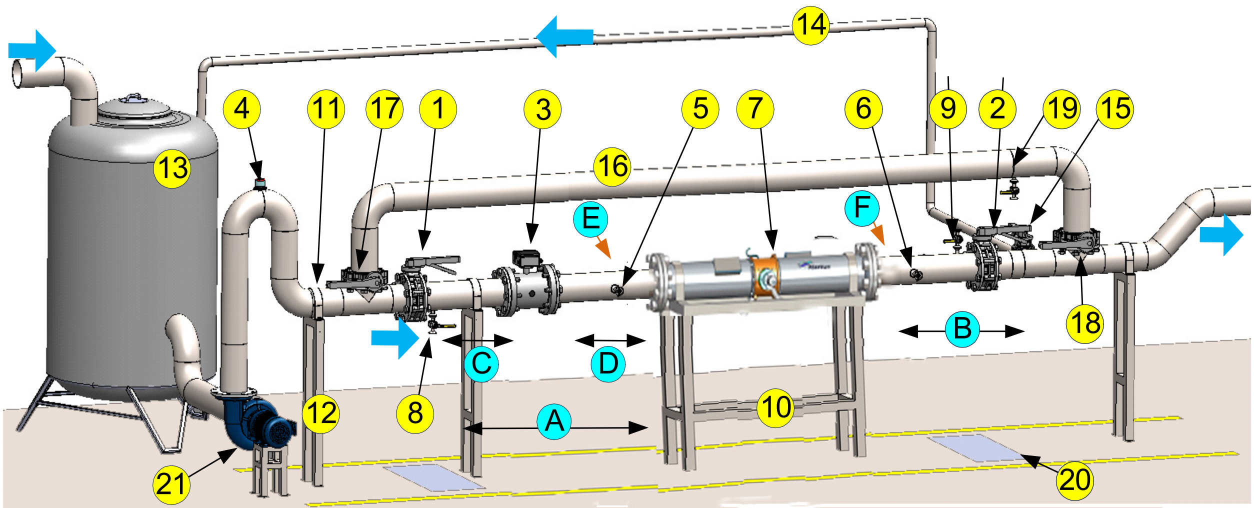


Figure 3-2: Generalized Atlantium system setup, horizontal (with bypass)

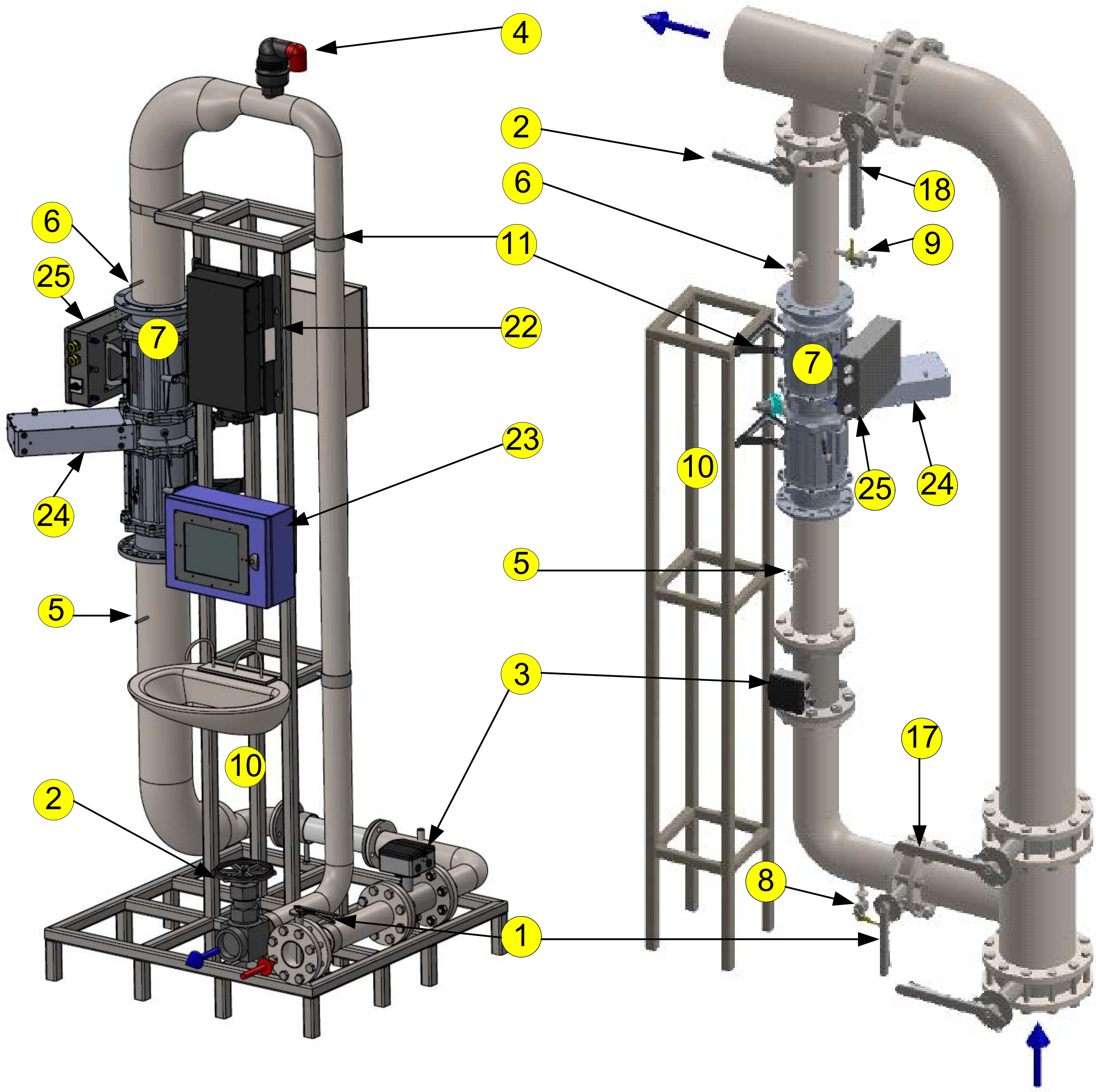


Figure 3-3: Generalized Atlantium system setups, vertical - Left: no bypass, Right: with bypass

Table 3-1: Typical Atlantium System Component Setup Key

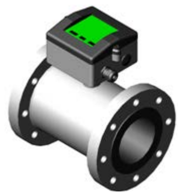
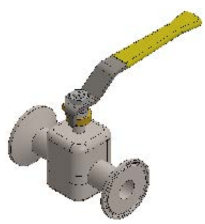
#	Description		Atlantium P/N or recommended and available at Atlantium
	Direction of water flow in the pipes		-
1	Inlet isolation Valve ■ Isolating Valve on the Inlet side		-
2	Outlet isolation Valve ■ Isolating Valve on the Outlet side		-
3	■ Flow Meter flow measurement device with 4-20mA analogue output signal ■ To be installed on the same line as the Atlantium system, with no branching before or after the flow meter ■ Follow the manufacturer's Installation instructions		Available Flow Meters: KROHNE OPTIFLUX 6000 KROHNE OPTIFLUX 2300
4	Air Release valve - To prevent air bubbles where necessary ■ Automatic Air Release valve is used to release air bubbles trapped within the line ■ Install at a high position to allow air bubbles to accumulate and release from this point		-
5/6	Inlet/Outlet Sampling Points ■ Position aseptic sampling valves **horizontally** (90^0) on the pipe center, within 50cm of the Unit ■ Aseptic sampling valve must be a dedicated valve for sampling only and resistant to alcohol and flame		■ Aseptic Sampling Valve Kit for Stainless Steel Pipe - KTB005700 ■ Aseptic Sampling Valve Kit for Plastic Pipe - KTB0011500
7	Generic Atlantium Unit (shown with flange interface. Systems are also supplied with ferrule interface.)		-
8	Draining/CIP Port - For performing CIP on the Atlantium Unit ■ The Draining port should be positioned close to the Inlet valve and between that and the Unit, *on the bottom* of the pipe ■ It is also used as one of the CIP ports and connects to a hose		■ CIP Kit - SAB012900 ■ CIP Valve Kit for Stainless Steel - KT0011600 ■ CIP Valve Kit for Plastic - KT0011700 (the valve itself is not part of the kits) See *CIP Ports and CIP Kits* on page 3-29., for information on CIP equipment.
9	CIP Port - For performing CIP on the Atlantium Unit ■ The CIP port should be positioned between the Unit and the Outlet valve, *on top* of the pipe, at the *highest* point before the isolation valve, in order to allow the pipes to be completely filled with solution		

Table 3-1: Typical Atlantium System Component Setup Key

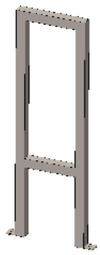
#	Description		Atlantium P/N or recommended and available at Atlantium
10	System Supports ■ Supports are required to assure no mechanical stress on the Atlantium Unit		-
10A	Shim, block or higher downstream support, to achieve 5^0 slope in horizontal installation		supplied by customer
11	Support brackets ■ Brackets are used to hold the Atlantium Unit and pipes attached to the support mounts ■ Is to include internal flexible material to absorb vibrations and assure that the pipes and Atlantium Unit are mounted tightly and securely		-
12	Pipe Supports ■ Pipe Supports are required to assure no vibrations in the pipeline		-
13	Water tank		-
14	Recirculation Line >$5m^3/h$ (>20gpm (Optional) ■ A Recirculation line is highly recommended to keep water flowing constantly.		-
15	Recirculation Line Isolation valve (Optional)		-
16	By-Pass (Optional) ■ If the facility uses quartz corrosive CIP chemicals, such as caustic soda, use a By-Pass line to prevent damage to the Atlantium Unit		-

Table 3-1: Typical Atlantium System Component Setup Key

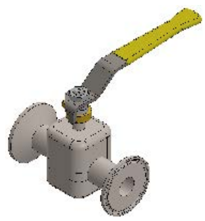
#	Description		Atlantium P/N or recommended and available at Atlantium
17	By-Pass Inlet isolation Valve (Optional)		-
18	By-Pass Outlet isolation Valve (Optional)		-
19	By-pass draining port (Optional) ■ The Draining port should be positioned on the bypass line, at the lowest point of the pipe	 	-
20	Drain ■ It is recommended to install a drain in the floor, close to the draining valve		-
21	Pump ■ To prevent water hammer, the pump must be able to start up gradually.		-
22	Ballast Module (certain Atlantium systems)		
23	Control Module		
24	DPM cleaning system (certain Atlantium systems)		
25	Connection Box		

Table 3-2: Typical Atlantium System Distances Setup Key

#	Description
A	■ The pipe between the Inlet isolation valve and the Atlantium Unit should be straight and horizontal. ■ For most applications, a minimum 10D (mm/ inch) of straight horizontal pipe is needed between the Inlet isolation valve and the Atlantium Unit. ■ For Power Plants, and other applications with space constraints, consult an Atlantium Application Engineer to find appropriate solutions.
B	■ The pipe between the Atlantium Unit and the Outlet isolation valve should be straight and horizontal. ■ For most application, a minimum 5D (mm/inch) of straight horizontal pipe is needed between the Atlantium Unit and the Outlet isolation valve. ■ For Power Plants, and other applications with space constraints, consult an Atlantium Application Engineer to find appropriate solutions.
C	■ Some manufacturers require positioning the Flow Meter at a minimum of 5D (mm/inch) of straight horizontal pipe immediately preceding it on the Inlet side. Refer to the manufacturer's instructions.

Table 3-2: Typical Atlantium System Distances Setup Key

#	Description
D	■ Some manufacturers require positioning the Flow Meter at a minimum of 2D (mm/inch) of straight horizontal pipe immediately following it before the Atlantium Unit. ■ Refer to the manufacturer's instructions.
E	■ Position the Inlet Sampling Point at a maximum distance of 0.5 meters/1.5ft. from the Atlantium System Inlet.
F	■ Position the Outlet Sampling Point at a maximum distance of 0.5 meters/1.5ft. from the Atlantium System Outlet.

3.3 Electrical Requirements

The electrical infrastructure is to be installed and in place prior to the system's delivery.

For information on the footprint of the components of the Atlantium system, see the section, ***About the RZ300 System*** on page 2-10. Use the Electrical ICD provided by Atlantium to help you plan the electrical infrastructure. See also ***Power, Voltage and Grounding Requirements*** on page 2-16 and ***Connecting the Ballast Module Cables*** on page 5-53.

Take into account:

For locations with 440Y/254 VAC 3~ +PE 50/60HZ
or 480Y/277 VAC 3~ +PE 50/60HZ
The Atlantium System requires a balanced voltage supply (WYE configuration). If you do not have such a connection, you must create an appropriate power feeding using a 1:1 isolation transformer (where the ground is connected to the middle point of the star configuration at the secondary side of the transformer).

- Evaluate the quality of your power and the likelihood of blackout/brownouts, electrical surges or interruptions. Determine if you need power surge protection and or uninterrupted power supply if any.
- To verify that you order the correct lengths for cables (i.e., Cable Harness, power cables, data cable, etc.), measure the distance between the proposed location of the relevant Atlantium components and between them and the Power source.
  - Planning the placement of the Ballast Modules, Control Module and the path of the various cables must take into consideration the length of the **Cable Harness** and the other cables. Cables must be placed in a path that does not interfere with passage ways.
  - The power mains connections must be in the vicinity of the Ballast Modules, Control Module.
  - The power cable and connecting plug are to be obtained locally.
  - An extra-long, 10M (32' 9.7") cable is available for the RS104 system only.
- If the location experiences periodic power outages, consider including uninterrupted surge protection devices.
- Prepare cables of sufficient length to connect from the auxiliary equipment (i.e., the flow meter/flow switch, etc.) to the Atlantium system.
- **For Systems Utilizing Seawater** Minimize the corrosion potential of the treated water by ensuring that there is sufficient grounding. Do not leave the Atlantium Unit standing with seawater. Either rinse with fresh water and leave the Atlantium Unit standing with fresh water or empty.

3.3.1 Preparing for the Electrical Wiring to the Mains

You must prepare the elements listed below for connecting the Atlantium system to the mains.

- **For preparing all electrical connections, be sure to use the *Electrical ICD* drawing provided to you by Atlantium with this manual. If no electrical drawings were provided please contact your Atlantium representative.**
- **Atlantium systems require connection to mains power of 400 to 480 VAC. See *Mains power supply up to 480VAC* on page 2-16.**

- Mains system area circuit breaker, according to the total systems' power consumption to be connected to this line. See ***Electrical Usage*** on page 2-15.
- Install the Required Circuit Breakers, as defined in the Electrical ICD.
- If you intend to lengthen or replace the mains power cable connected to the Atlantium Unit with a longer cable, prepare the cables according to your location electrical distribution system requirements.

  - **For locations with 400 VAC: 5-wire system that contains 3 phases, Neutral and Ground (The Neutral and Ground are mandatory)**For sites with 400VAC 3 phase + Neutral, the Control Module's Mains feeding voltage is 230 VAC (L1 to N or L2 to N or L3 to N).
  - For sites with 440/480VAC 3 phase, the Control Module **cannot** be fed from one phase of the 3 phase supply. The Control Module **must** be connected to a separate mains feeding, which is in the voltage range of 100-240 VAC.

3.4 The Piping Infrastructure

To review how to prepare the infrastructure, read through the typical installation section before you begin and refer to the section, *Installing the Atlantium System* on page 48 .

Install the piping infrastructure before the Atlantium system arrives, but to avoid built-in mechanical stresses perform the final pipe adaptations and welds when the unit is actually situated in its permanent position.

ATTENTION!

- When tightening bolts to connect piping, use a torque wrench to apply correct and uniform torque, according to accepted standards and practices. Guidance for bolt tightening techniques may be found at:
  http://www.wermac.org/flanges/flanges_torque-tightening_torque-wrenches.html
- When sealing the flanges/ferrules and gaskets, **do not** use sealing material that contains graphite as it accelerates the corrosive process on stainless-steel components, including those of the Atlantium system.
- Be sure to align the piping on the inlet and outlet sides of the designated location for the Atlantium Unit with the unit to avoid mechanical stress on the Unit.

Be sure to secure the piping around the designated location for the Atlantium Unit so that there is no vibration.

- Be sure to position adequate supports on the piping line needed to hold up and stabilize the process piping on the inlet and outlet side of the proposed location of the Atlantium unit.
- Determination of support requirements and design should be made by an on-site mechanical engineer.

3.5 Flow Measurement/Detection

The Atlantium unit requires a flow signal to calculate the dose and operate properly. The flow signal is a measurement of the actual flow or simply a flow/no-flow indicator. The flow signal is collected from either a flow meter or a flow switch.

- A flow meter is required in facilities where the flow rate is variable for any reason.
- A Flow switch can be used in facilities where the flow is at a fixed, steady rate. In this situation the value of the known steady rate is entered manually into the system Control Module (see *Configuring the Flow Settings* on page 7-88).

The Flow meter or flow switch does not necessarily have to be installed in close proximity to the Atlantium Unit. It may be more remotely positioned, provided that it is located between the water inlet and the Atlantium Unit and there is **no** branching or splitting of the water flow.

ATTENTION!

The Atlantium system cannot be operated when there is no water or no water flow. Any damage caused due to operating the system without water or without water flow invalidates the Atlantium warranty.

3.5.1 Flow Meter

In most cases, a flow meter with a 4-20 mA output is required to measure the flow and automatically adjust the output of the UV system. The flow meter must be able to measure the expected maximum water flow of the specified Atlantium unit. The power source for the flow meter must come from an external source according to the manufacturer's instructions.

ATTENTION!

Be sure to provide the power source for the flow meter as part of the preparations for installing the Atlantium Unit.

Install the flow meter in strict accordance with the instructions supplied by its manufacturer.

Flow meters are available from Atlantium, such as:

- KROHNE OPTIFLUX 6000 flow meter for high-purity water applications
- KROHNE OPTIFLUX 2300 flow meter – OPTIFLUX 2000 flow meter, includes KROHNE IFC 300 signal converter

Consult your Atlantium representative.

Examples of Flow Meters

Figure 3-4: KROHNE OPTIFLUX Flow Meters

3.5.2 Flow Switch

The flow switch tells the Atlantium Unit when water is or is not flowing. The recommended flow switch type features a paddle connected to a Dry Contact micro-switch, with minimum flow sensing of 0.2m^3/hr (1gpm), see **Figure** 3-5. Install it in a vertical (12:00 o'clock) orientation on top of the pipe. Follow the manufacturer's instructions.

Figure 3-5: Example of Flow Switch

3.6 Sampling Valve Components and Kits

Sampling valve kits are available from Atlantium.

One Accessory kit is required for each Sampling point.

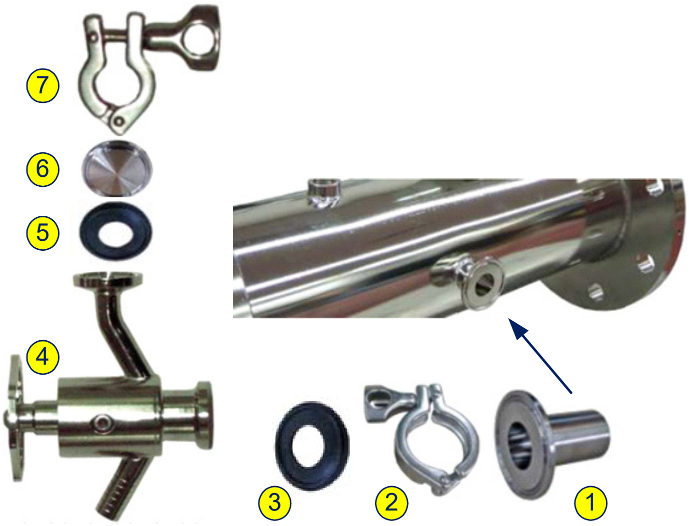


Figure 3-6: Sampling Valve Assembly

The table below details them. For information on installing the Sampling valves, see ***Installing the Inlet/ Outlet Sampling Points*** on page 5-39.

Table 3-3: Sampling Valve Components and Kits

	Catalog number	Description	Purpose	Image	Kit for Stainless Steel Pipe KTB005700	Kit for Plastic Pipe KTB005750
1	FAC003700	Long welding ferrule 1" Adapter	For welding the tri-clamp port to the pipe		**2 Pcs**	**-**
2	FAB008100	Threaded Adapter 1/2" BSP	For connecting the tri -clamp port to the plastic pipe		**-**	**2 Pcs**
3	FA0000045	Clamp 1 1/2", ST.ST 304	For connecting the valve to the welded port		**2 Pcs**	**2 Pcs**
4	FA0000046	Ferrule gasket 1", FJ31.05-E	For sealing the valve connected to the pipe		**2 Pcs**	**2 Pcs**

Table 3-3: Sampling Valve Components and Kits

	Catalog number	Description	Purpose	Image	Kit for Stainless Steel Pipe KTB005700	Kit for Plastic Pipe KTB005750
5	COB000500	Aseptic sampling valve, ST.ST 316L	For taking water samples		**2 Pcs**	**2 Pcs**
6	FA0000048	Gasket 1/2", FJ31/25-E	For sealing the valve's flushing port		**2 Pcs**	**2 Pcs**
7	FA0000049	Solid End Cap, 1/2", ST.ST 304	For covering the valve's flushing port		**2 Pcs**	**2 Pcs**
8	FA00000047	Clamp 1/2", ST.ST 304	For sealing the flushing port of the valve		**2 Pcs**	**2 Pcs**
	FA0008300	Round Brush - 15/60/150	For cleaning the valve during the process		**2 Pcs**	**2 Pcs**

3.7 Draining Valves

It is important to plan for draining the system. Install one draining valve at the lowest point of the pipeline, before or after the Atlantium unit. If you plan to install CIP ports, one of them can double as a draining valve. If there is a Bypass line used occasionally to bypass the Unit in any circumstance, add an additional draining valve to assure that the bypass line does become a potential source of standing water. A bypass line must be empty when not in use.

3.8 CIP Ports and CIP Kits

The CIP (Cleaning-In-Place) recirculation system is used for the periodic cleaning and disinfecting process of the Atlantium unit's inner surfaces and quartz sleeve. The CIP Cart includes a reservoir, a pump and supply and return hoses See the diagram below and refer to **CIP Cart ICD drawing** . **ICB012900** for further details

3.8.1 CIP Cart

For systems with 400VAC: catalog number: SAB012900

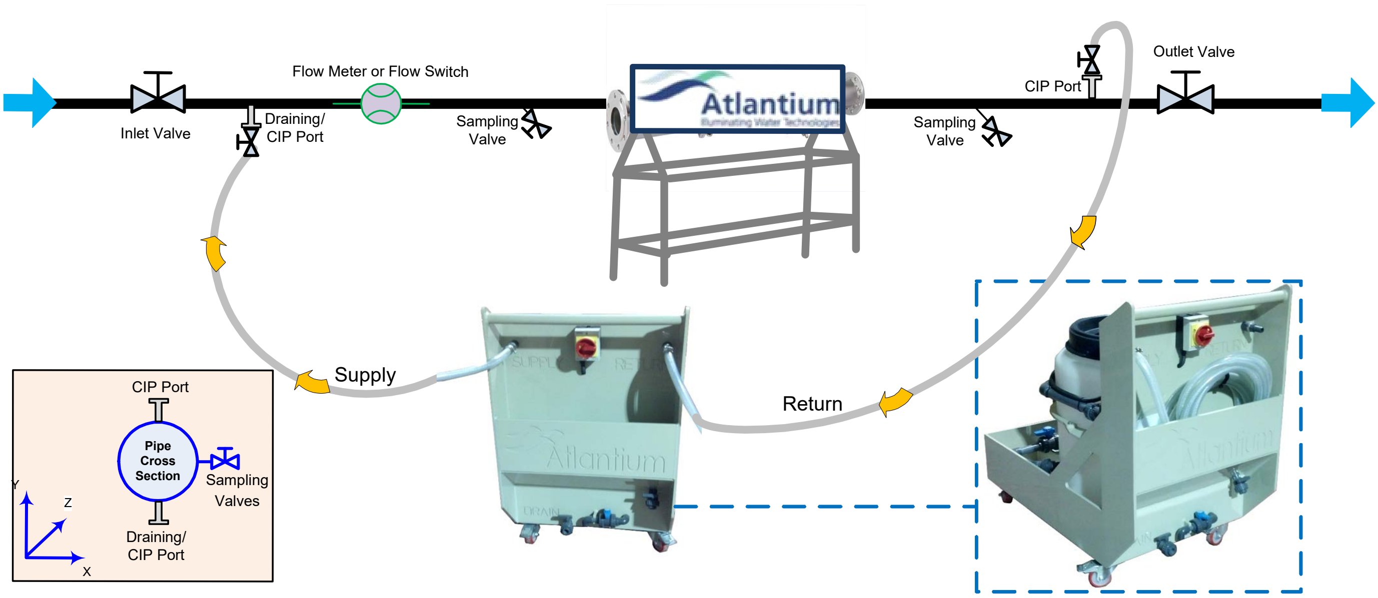


Figure 3-7: CIP Recirculation System Connections

3.9 CIP Accessories and CIP Kits

The CIP (Cleaning-In-Place) recirculation system is used for the periodic cleaning and disinfecting process of the Atlantium unit's inner surfaces and quartz sleeve, see **Figure** 3-7.

Two Accessory kits are required, one for each CIP port.

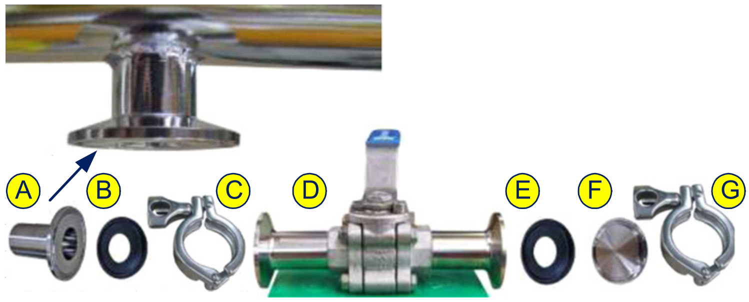


Figure 3-8: CIP Valve Kit

Table 3-4: CIP Accessory Kits

	Catalog number	Description	Purpose	Image	Kit for Stainless Steel Pipe KT0011600	Kit for Plastic Pipe KT0011500
A	FAC003700	Long welding ferrule 1" Adapter	For welding the tri-clamp port to the pipe		**2 Pcs**	-
A	FAB008100	Threaded Male Adapter 1/2" BSP	For connecting the tri clamp port to the plastic pipe		-	**2 Pcs**
B,E	FA0000046	Ferrule Gasket 1", FJ31.05-E	For sealing the valve connected to the pipe		**2 Pcs**	**2 Pcs**
C,G	FA0000045	Clamp 1.5", ST.ST 304	For connecting the valve to the welded port		**2 Pcs**	**2 Pcs**
F	FAC004100	Solid End Cap, 1 1/2", ST.ST 304, EG16A3A 1.54L	For covering the CIP port while not in use		**2 Pcs**	**2 Pcs**

Table 3-4: CIP Accessory Kits

	Catalog number	Description	Purpose	Image	Kit for Stainless Steel Pipe KT0011600	Kit for Plastic Pipe KT0011500
D		Valves	Two high-purity chemical resistant brass ball valve (aseptic ball valves), 1 inch diameter, with tri-clamp fitting connection mechanism (A3 Standard) on both sides.		**2 Pcs** **Not Included** **Purchased by Customer**	**2 Pcs** **Not Included** **Purchased by Customer**
	FAB005700	Hose Adapter 1" to 3/4"	To attach the CIP hose		**2 Pcs**	**2 Pcs**

3.9.1 CIP Safety Equipment

Required but not included in Atlantium's CIP Kits: protective equipment such as goggles, clothing and (chemical-safe) gloves, according to the safety standards specified at your facility.

3.10 Connection to Facility Control

Options are available to set up control of the Atlantium system via your facility control system. Control setup can be made via any of the following:

- **Signal control** - utilizing System Ready, External On/Off, and General Alarm functions, etc. (See *Connection Box* on page 5-55.)

With any of the above control options, you can still utilize the function of the Control Module of the Atlantium system. Whichever control facility is utilized, the system communicates to the other control systems such that the last command issued determines the current status.

4 Safety Overview

The Atlantium System has been designed according to the highest safety standards, assuring the safety of operating personnel, the environment and the treated water.

DANGER!

Improper use of controls or adjustments or performance of procedures other than those specified herein could result in significant hazards.

Therefore, personnel operating or servicing this system must be thoroughly familiar with all safety requirements and operating procedures and are to adhere to them during use of this system.

To guard against injury, observe basic safety precautions, including the following:

All persons operating the Atlantium system must be aware of proper use and the potential hazards of violating safeguards. Be certain that all personnel carefully review the safety information and the procedures specified in this manual. Only authorized individuals with appropriate training and knowledge, including local regulations, should operate, assist in the operation of or provide maintenance to the system.

All service and repair of the Atlantium system must be performed by Atlantium field service technicians or other factory-authorized personnel ONLY.

4.1 UV Lamp Safety

UV Exposure: The system generates ultraviolet (UV) light within the Unit, which can cause serious eye damage or blindness if you stare at it directly when it is operating. Use the Viewport only. Do not look directly into the lamp enclosure during system operation, examining, or servicing the system's internal components during operation, or when lamp's breather caps are open, or if energized testing is required.

The Viewport is specifically designed with a UV filter to prevent UV light from escaping so it is safe to look inside. Nevertheless, personnel should not stare at it or any other brilliant light source for any but short intervals to check on Unit operations. The Viewport cover should also be preserved intact.

If the viewport is broken, DO NOT look directly into it. Be sure to request a replacement part immediately. Do not open the viewport's metal cover until the replacement is installed.

Persons potentially subject to UV light exposure – technicians who are servicing the Unit – must wear appropriate eye protection whenever the system is operating.

During maintenance procedures that are conducted with the UV lamps turned on:

- Use Caution floor signs at a distance of 1m/40inch from the Atlantium Unit to warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.
- Wear white cotton gloves.
- Wear long sleeves rolled down to your wrist.
- Wear appropriate eye protection, such as containing polycarbonate lens that meet EN1 ultrasonic, CAN/CSA-Z94.3-02 and/or Z87.1 standards.
- **DO NOT look directly into openings that emit UV light.**

Protect Hands: Do not touch the UV lamp with bare hands.

- Wait at least 10 minutes until the lamp is cooled down before touching the lamp or starting the replacement procedure.
- Use appropriate protective gloves both to protect your hands and to avoid skin oils that leave fingerprints and/or harm the UV lamp.
- Lamps can reach a temperature of 1,000C° under operating conditions.
- Wait ten minutes to allow the lamp to cool down fully before replacing it.
- Keep materials that are sensitive to heat, or which contain solvents, a safe distance away from the lamp, Unit, and electrical connections.

Electric Shock: Before replacing a UV lamp or other components, and during any maintenance requiring lamps to be turned off, make sure the circuit breaker for the Connection Boxes of the relevant lamps are turned OFF and place a sign on the electrical cabinet alerting others NOT to touch the screen and circuit breakers during maintenance so that no one can inadvertently turn it on while maintenance is in progress.

ATTENTION!

Prevent Damage: **Handle the UV lamp assembly by holding only the contact housings (white ceramic casing).**

When handling the UV lamp, always place it on a flat surface or table, so that it is not accidentally damaged.

If the body of the UV lamp is touched accidentally, clean the fingerprints off with alcohol (at least 70%) and wipe dry with a soft, clean, lint-free cloth (usually provided with the lamp). Do not use cleaning rags or materials that can leave a residue.

Do not use a UV lamp that shows any scratches, cracks, or other damage.

Mercury Poisoning: **Lamps contain mercury, which is a hazardous substance. Always wear gloves and safety glasses when handling or replacing a UV lamp.**

Inhalation of vaporized mercury compounds can be harmful to the lungs, kidneys, and nervous system. Mercury that penetrates the skin or is ingested can also be harmful.

If mercury is inhaled, penetrates the skin, or is ingested accidentally, seek emergency medical treatment immediately.

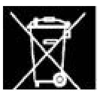
If a UV lamp breaks during handling and releases mercury, the following precautions should be observed to minimize the risk of exposure to mercury:

- Always wear gloves and safety glasses when handling or replacing a UV lamp.
- Leave the area immediately to avoid inhalation of the mercury vapor.
- Thoroughly ventilate the area for at least 30 minutes or until the mercury vapor concentration is in compliance with applicable federal and local health and safety regulations.
- After handling a broken UV lamp, carefully remove gloves, and then wash hands thoroughly with soap and water. Follow all applicable federal and local health, safety, and environmental regulations.
- Do not turn on or operate the system until all mercury contamination has been cleaned up and removed.
- Establish a Hazard Response Plan to deal with all related hazards and provide it to all personnel associated with the Atlantium system.

Refer the Chemical Safety information for mercury at: http://www.inchem.org/documents/icsc/icsc/eics0056.htm

ATTENTION!

Lamp Compatibility: Use original Atlantium UV lamps ONLY. **Other lamps are incompatible.** They **void the warranty** and may damage the system.



Proper Disposal of Lamps

Expired lamps are classified as hazardous waste and must be disposed of appropriately.

- A non-ruptured UV lamp that is no longer used must be disposed of according to specific local or federal environmental and hazardous waste regulations.
- Broken lamps and parts contaminated by the mercury have to be vacuum-packed and disposed of. Broken lamps, as well as contaminated packing material and other parts have to be considered as special waste, which may only be removed by authorized waste disposal companies.

4.2 Electrical Hazards and Safety Considerations

DANGER!

UNDER NO CIRCUMSTANCES

SHALL ELECTRICAL WORK BE CARRIED OUT WITHOUT VERIFYING THAT ALL MAIN CIRCUIT BREAKERS Of THE ELECTRICAL LINE FEEDING THE ATLANTIUM SYSTEM COMPONENTS ARE SET TO THE **OFF** POSITION AND THAT **NO** ELECTICAL FEED IS LIVE.

DO NOT SET ANY MAIN CIRCUIT BREAKER OF THE ELECTRICAL LINES FEEDING THE ATLANTIUM SYSTEM COMPONENTS TO THE ON POSITION UNTIL **ALL** ELECTRICAL WIRING IS PROPERLY CONNECTED. UNCONNECTED WIRES/CABLES IN A LIVE ELECTRICAL FEED IS A HAZARDOUS SITUATION THAT CAN RESULT IN DEATH OR SERIOUS INJURY.

- **DO NOT DRILL INTO THE CONNECTION BOXES OR ALTER THEM IN ANY WAY!!!!**
- **The Connection Box already contains entry glands for ALL of the cabling needs.**
- **Making holes in a Connection Box will destroy its waterproof seal, endangering the components inside.**
- **Do Not draw electricity from the Atlantium Unit's Connection Box as it will damage the Unit's electrical configuration.**

The Atlantium Hydro-Optic system incorporates high-voltage internal components, which can cause serious injury or fatal electrical shock if not used or serviced properly. High voltage components can retain a charge for some period of time even after the system has been turned off.

- Never attempt to remove any system covers or to dismantle any parts, except when performing the maintenance procedures detailed in ***System Maintenance*** on page 10-108.
- No portion of the Atlantium system is be opened or removed by anyone other than a trained and authorized technician.
- Do not spray or pour any type of fluid directly into the Unit or electrical components. Moisture causes damage to the equipment and electrical shock may result.
- Do not operate the system if the power cables or harnesses are frayed or otherwise damaged.

CAUTION!

DO NOT enter **Technician Mode** in the Control Module unless instructed to do so by Atlantium certified personnel or by an Atlantium System User Manual, for a specific task.

If you are in Technician Mode, **DO NOT WALK AWAY!**

NEVER LEAVE THE ATLANTIIUM SYSTEM WHEN IT IS OPERATING IN TECHNICIAN MODE!!

A SYSTEM LEFT IN TECHNICIAN MODE WITHOUT SUPERVISION CAN RESULT IN PERMANENT DAMAGE!!

Utilize best practices regarding electrostatic discharge (ESD).

Avoid direct contact with the electronic circuit boards of the Control Module, which is sensitive to electrostatic discharge (ESD).

Wear an electrostatic discharge (ESD) Wrist Strap when performing procedures where contact with a circuit board is possible.

4.3 Keep the Unit Full of Water

ATTENTION!

The Atlantium system cannot be operated when there is no water. Any damage caused due to operating the system without water invalidates the Atlantium warranty.

4.4 Chemical Use for Cleaning in Place (CIP)

ATTENTION!

- The chemicals selected and the concentrations used must be approved by Atlantium's Application staff, and the ship's quality assurance and safety officials. The CIP process must comply with the written protocols and procedures set in place by those departments. Refer to the section, ***Selecting the Correct Chemicals for CIP*** on page 10-109
- Use and handle all chemicals in strict accordance with their manufacturer's instructions, product information sheets, and Safety Data Sheets (SDS).
- NaOH (Caustic Soda) as it causes irreversible damage to quartz. It is strictly forbidden to use it without a specific analysis of temperatures, concentration, frequency and duration and written approval from Atlantium's application engineer.

Refer to the Chemical Safety information provided with the CIP chemicals which you choose to use, for example:

https://portal.ecolab.com/servlet/PdfServlet?sid=915373&cntry=AU&langid=en-GB&langtype=RFC1766LangCode&locale=ja&pdfname=OXONIA+ACTIVE+150

4.5 Anti-Corrosion Spray Safety

WARNING!

When using OKS 2101 Spray:
Refer to the OKS 2101 Safety Data Sheet according to Regulation (EC) No. 1907/2006 - GB.
Use the following protective gear:

- Safety glasses with side-shields conforming to EN166
- Protective gloves that satisfy the specifications of EU Directive 89/686/EEC and the standard EN 374 derived from it
- Respiratory protection mask with filter type A-P

Follow these precautions for safe handling:

- Do not use in areas without adequate ventilation.
- Do not breathe vapors or spray mist.
- In case of insufficient ventilation, wear suitable respiratory equipment.
- Avoid contact with skin and eyes.
- Keep away from fire, sparks and heated surfaces.
- Persons with a history of skin sensitization problems or asthma, allergies, chronic or recurrent respiratory disease should not be employed in any process in which this mixture is being used.
- Smoking, eating and drinking should be prohibited in the application area.
- Wash hands and face before breaks and immediately after handling the product.
- Do not get in eyes or mouth or on skin.
- Do not get on skin or clothing.
- Do not ingest.
- Do not use sparking tools.
- These safety instructions also apply to empty packaging which may still contain product residues.
- Pressurized container: protect from sunlight and do not expose to temperatures exceeding 50 °C. Do not pierce or burn, even after use.

4.6 System Safety Features

The Atlantium Hydro-Optic system includes the following safety features:

4.6.1 UV Protective Seal

Each lamp is enclosed within a sealed lamp chamber that prevents the UV light from escaping. This means that with regular operation there is no ambient UV escaping from the Unit.

While servicing the UV Unit or checking the Intensity sensor, UV blocking safety glasses should be worn by service staff or others working in the area to prevent exposing the eyes to the UV light.

4.6.2 Software Safety Controls on Control Module

CAUTION!

When the system is started, the Atlantium software performs a series of internal self-testing routines prior to starting system operation.

If the system **does not** pass these tests satisfactorily, UV lamp operation is disabled until the problem is resolved.

If problems arise during normal operation, appropriate error messages with varying levels of severity are generated on the touch-screen of the Control Module.

Control Module screens that provide setup options and compliance reports are password protected.

ATTENTION!

DO NOT leave the Control Module unattended, even for a short time, while setting change screens are accessible. An untrained person could damage the operation of the Atlantium system by making unauthorized setting changes.

4.7 Atlantium Training

As part of the Commissioning and Hand-Off process, Atlantium provides hands-on training on all aspects of the Atlantium system for your designated personnel. Should any component be replaced with upgraded features, Atlantium offers training as needed and is ready to provide additional training for new personnel or any other needs related to its systems.

5 Installing the RS104 Series

5.1 Setting the Main Circuit Breakers to OFF Position

Before beginning work to connect the wires/cables of the Atlantium system components, turn the main circuit breakers of the electrical lines feeding the atlantium system components to the OFF position.
Verify that no electrical feed to the Atlantium system components is live.

WARNING!!!

UNDER NO CIRCUMSTANCES SHALL ELECTRICAL WORK BE CARRIED OUT WITHOUT VERIFYING THAT ALL MAIN CIRCUIT BREAKERS OF THE ELECTRICAL LINES FEEDING THE ATLANTIUM SYSTEM COMPONENTS ARE SET TO THE OFF POSITION AND THAT NO ELECTICAL FEED IS LIVE.

DO NOT SET ANY MAIN CIRCUIT BREAKER OF THE ELECTRICAL LINES FEEDING THE ATLANTIUM SYSTEM COMPONENTS TO THE ON POSITION UNTIL ALL ELECTRICAL WIRING IS PROPERLY CONNECTED. UNCONNECTED WIRES/CABLES IN A LIVE ELECTRICAL FEED IS A HAZARDOUS SITUATION THAT CAN RESULT IN DEATH OR SERIOUS INJURY.

5.2 Unpacking And Checking Package Contents

➲ To unload and check the system:

Step 1. Walk around and inspect the crate for damage. Note if there is any separation in the joints or any other indication of any damage in transit. If there is any indication of damage, take a photograph of it and use extra caution when proceeding to the next step.

Step 2. Unclamp and remove the panels of the crate.

Step 3. As you remove the wrapping and shipping protections, verify that the contents of the crate are correct according to the delivery documents.

5.3 Installing the Electrical Cabinet

Locate the Electrical Cabinet according to the design made for the path that the cables is to be placed.

The Electrical Cabinet can be mounted on a wall or appropriate anchoring point.

The Electrical Cabinet includes fans and filters on the top and bottom of each side. It must have at least 50cm (19.7 inches) of clearance on its right and left sides for proper ventilation. To review the complete clearance requirements of the Electrical Cabinet, see Table 2-3, *Electrical Cabinet Physical Details*, on page 11.

The power cable and connecting plug are to be obtained locally.

➲ To position the Electric Cabinet:

Step 1. Uncrate the Electrical Cabinet package.

The Electrical Cabinet of the Atlantium system is heavy! - Be sure to make the proper provisions for its transport.

Step 2. Move the Electrical Cabinet to its location designated according to installation plan.

Step 3. Lower the Electrical Cabinet to its designated location.

Step 4. As appropriate, bolt the Electrical Cabinet to the wall or anchoring point. Refer to the diagram to the right.

Figure 5-1: RS104 Electrical Cabinet

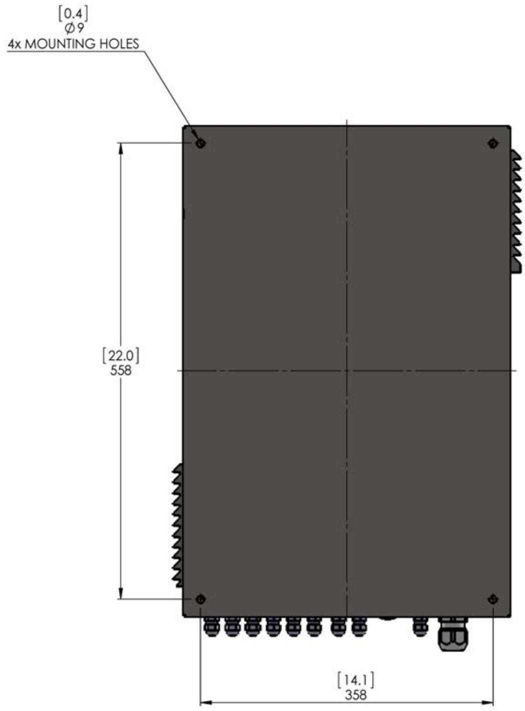
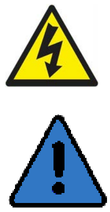


Figure 5-2: RS104 Electrical Cabinet Mounting Holes

5.3.1 Cabling Inside the Electrical Cabinet.

Refer to the appropriate wiring diagram in accompanying ICD



DO NOT DRILL INTO THE ELECTRICAL CABINET OR ALTER THEM IN ANY WAY!!!!

The Electrical Cabinet already contains entry glands for ALL of the cabling needs.

Making holes in a Electrical Cabinet will destroy its waterproof seal, endangering the components inside.

Do not connect any wires or cables to any of the connectors on the terminal blocks or to any other connectors in the Electrical Cabinet that are not specified in your system configuration or without the express instructions from Atlantium personnel. Unauthorized connections may damage the system and void the Atlantium warranty.

Any person involved in handling the Electrical Cabinet's Connection card or its components is required to wear an electrostatic discharge (ESD) Wrist Strap.

Connect the ESD Wrist Strap to any bolt on the body of the Electrical Cabinet using a banana plug or an alligator clip.

Cable Harness

The RS104 comes with the Cable Harness assembly already connected between the UV Reactor and the Electrical Cabinet. You must position the Cable Harness assembly in a path that does not obstruct access to the RS104 component.

In the diagram to the right, **A** identifies the Connection Card. **B** identifies the incoming cables of the Cable Harness.

The Cable Harness contains cables according to the table below.

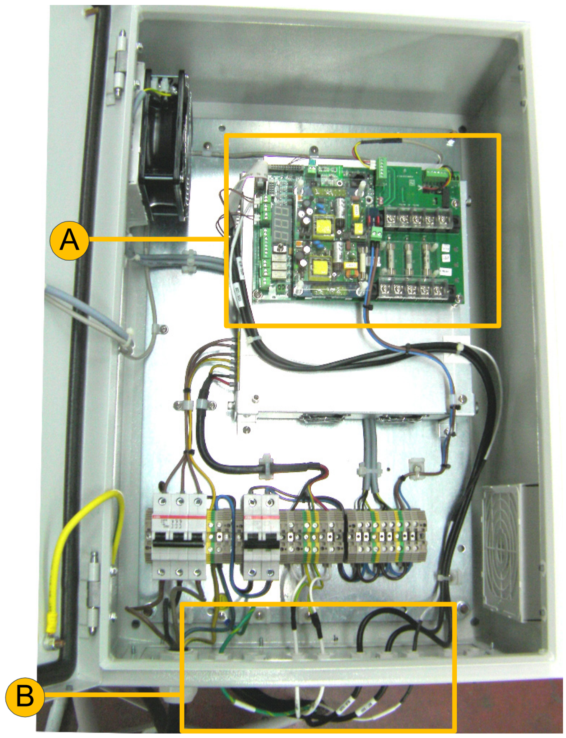


Figure 5-3: Inside the Electrical Cabinet

#	Item
1	Grounding/Earthing wire
2	110V (for locations with 480VAC only)
3	Lamp cable A
4	Lamp cable B
5	UVIS Sensor cable
6	UVT Analyzer cable
7	Temperature Sensor cable

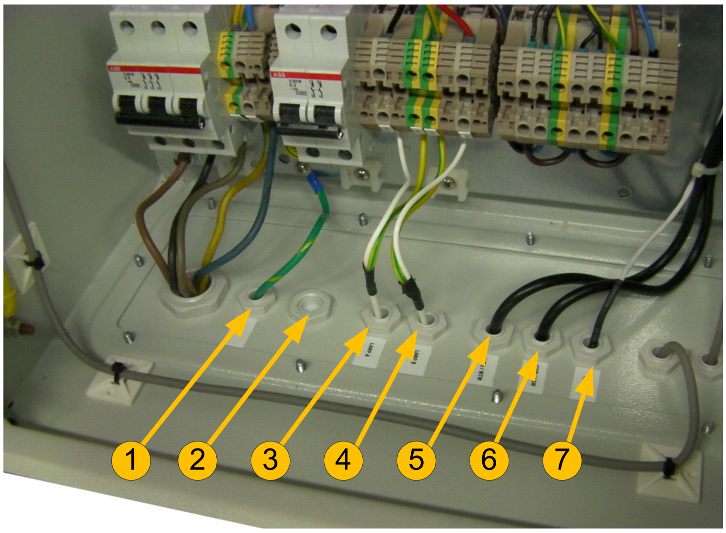


Figure 5-4: Cable Harness

Auxiliary Connections

Connect the cables of the mandatory flow signal, and External On/Off/ Alarm Ready (according to your system configuration) to the Connection card inside the Electrical Cabinet. See the diagram above for how to thread these cables and the diagram below for where to connect the cables.

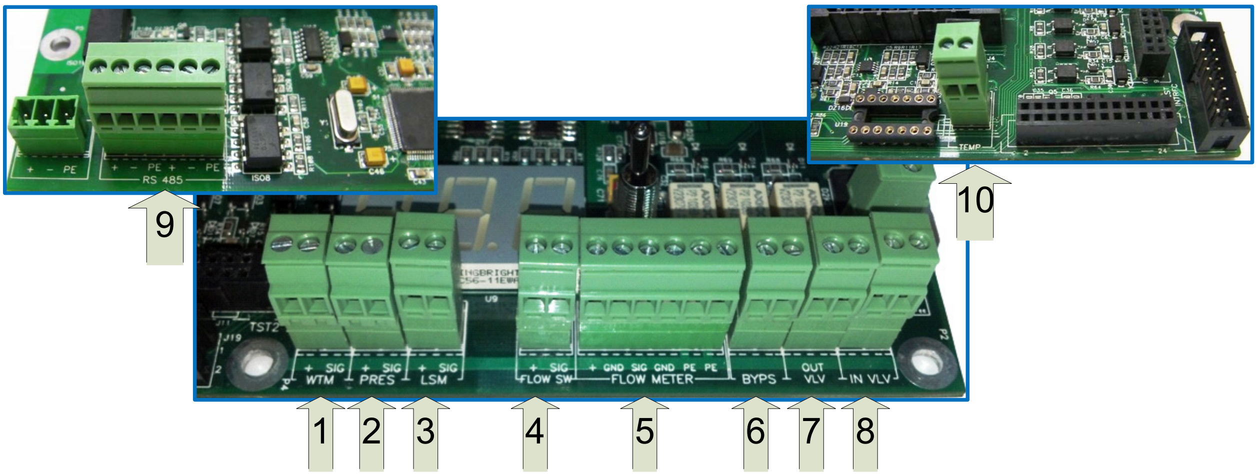


Figure 5-5: Signal Connections on 1300 Connection Card

Table 5-1: Signal Connections Key (on 1300 Connection Card)

#	Text on the Circuit Board	Description	Signal type	Value	Notes
1	WTM	UVT Analyzer input	Analog	4-20mA	Already assembled (Part of Cable Harness)
2	PRES	Pressure Transmitter input (Optional)	Analog	4-20mA	Optional, not implemented
3	LSM	UVIS Sensor input	Analog	4-20mA	Already assembled (Part of Cable Harness)
4	FLOW SW	Flow switch input	Digital	Dry contact / Open collector	Either Flow Switch or Flow Meter
5	FLOW METER	Flow meter input	Analog	4-20mA	
6	BYPS STR	Bypass valve output	Dry contact		Optional
7	OUT VLV	Output valve output	Dry contact		Optional
8	IN VLV	Inlet valve output	Dry contact		Optional
9	RS485	RS485 Port			Already assembled
10	TEMP	Temperature sensor input	Analog		Already assembled

➲ To connect the flow signal:

Step 1. Open the plastic gland marked **A** and thread the Flow signal cable through it until the cable reaches the Connection card terminal.

Step 2. Connect the cables to the appropriate terminal **(D)**. See **#** 4 or **5** in the diagram and table above.

Step 3. Fasten the cable to the anchor point **(B)** using a plastic fastener.

Step 4. Fasten the cable to the other cables **(C)** using plastic fasteners.

Step 5. Cut the excess from the plastic fasteners.

Step 6. Tighten the plastic gland **(A)**.

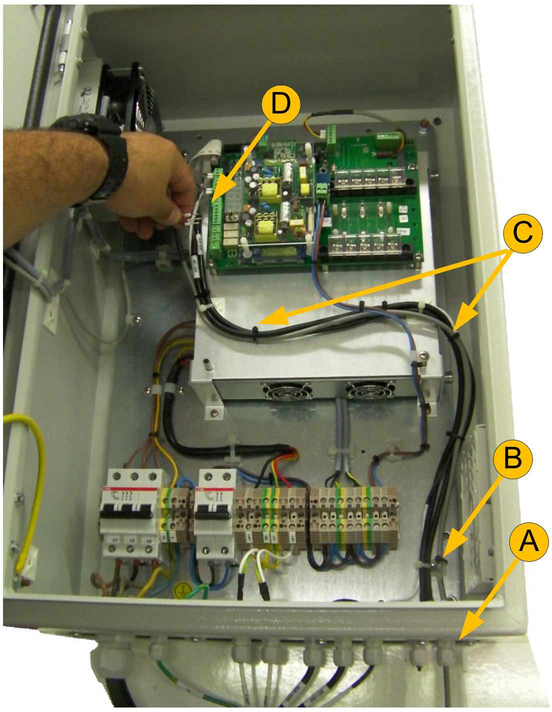


Figure 5-6: Signal Connection Cable to the Connection Card

5.3.2 Connection card

Control of the system is implemented by the 1300 Connection Card and the blue & white touchscreen, which are incorporated into the Electric Cabinet include the following connections:

- Power Cable - already assembled)
- External On/Off Signal
- General Alarm Signal
- Data Cable
- Modbus

Avoid direct contact with the electronic circuit board which is sensitive to electrostatic discharge (ESD). Utilize best practices regarding ESD when replacing the power cable.

➲ To connect peripheral equipment:

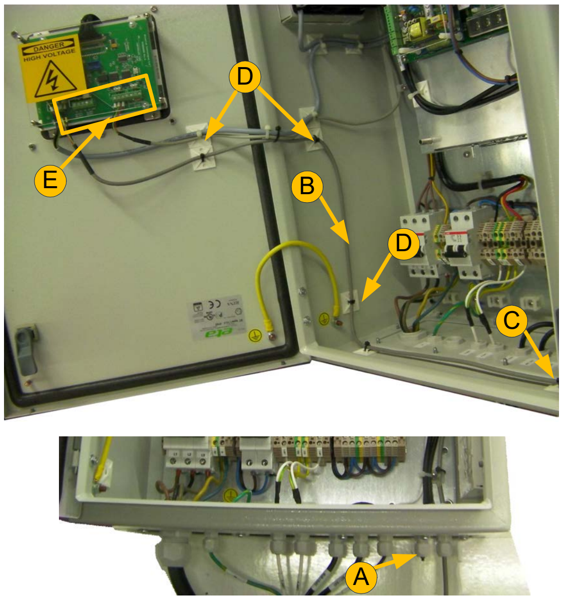


Figure 5-7: Path of Cables to the Connection Card

Step 1. To connect an **External On/Off** signal, thread the signal cable through one of the available holes with plastic glands on the bottom (A) and extend the cable (B) along the bottom and left side of the Electrical Cabinet, along the door until it reaches the appropriate terminal on the Connection Card.

Step 2. Using plastic fasteners, fasten the cable to the anchors (D) along the path.

Step 3. Connect the cable's leads to the terminals according to the diagrams and table below.

Step 4. To connect a.n **General Alarm** signal, follow the directions as for the **External On/Off**.

Step 5. To connect to an external facility control via Modbus, follow the directions as for the **External On/Off**. For more information, see Appendix A, ***Modbus Communication Protocol*** on page 112.

Step 6. Add jumpers to any unused pins.

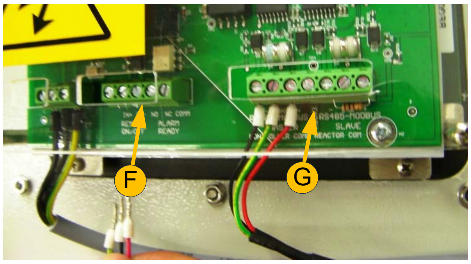


Figure 5-8: Connection Card Terminal Connectors-1

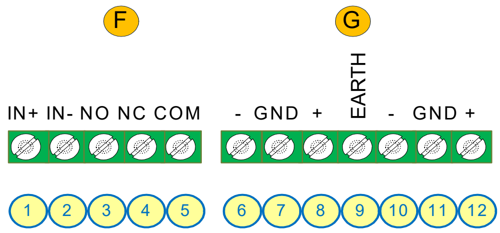


Figure 5-9: Connection Card Terminal Connectors-2

Table 5-2: **Controller Connector Pins**

#	Item	
1	External On/Off IN+	On: 20?28VDC Off: <1VDC See *External On/Off - Additional Notes* below.
2	External On/Off IN-	
3	General Alarm/System Ready Dry Contact Normally Open **NO**	
4	General Alarm/System Ready Dry Contact Normally Closed **NC**	
5	General Alarm/System Ready Dry Contact Common **COM**	
	Data Cable Connectors	
6	RS485 / Modbus Master TX-	Cable is provided already attached.
7	RS485 / Modbus Master Ground	
8	RS485 / Modbus Master TX+	
9	Earthing connector	

Table 5-2: **Controller Connector Pins**

#	Item	
10	RS485 / Modbus Slave TX- to connect to an external control system	The Controller is provided including a **100R, <5% , 0.25W, Through hole resistor** between #13 and #15. This resistor is to be removed in Controllers connected by daisy-chain, but is to be left attached in the last Controller of the daisy-chain. See *Cabling Inside the Electrical Cabinet.* on page 42.
11	RS485 / Modbus Slave Ground	
12	RS485 / Modbus Slave TX+	

5.3.2.1 External On/Off - Additional Notes

For regular wiring:

- Positive voltage must be connected to pin 4 – IN+
- Negative (GND) must be connected to pin 5 – IN-

Table 5-3: External On/Off

Controller Screen More Settings	Physical Connection	Physical Connection	
	IN+	IN-	Lamp Control
NC Unchecked	0 VDC	0 VDC	Off
NC Unchecked	+24 VDC	0 VDC	On
NC Checked	0 VDC	0 VDC	On
NC Checked	+24 VDC	0 VDC	Off

5.3.3 Setting Up the Facility System Communication Properties

To properly configure communications between your Atlantium system and control network, set up the communication properties and **COM1** port settings at the facility control system as follows:

Node Address:	2 to 254 ***
Available Commands	Function codes
Function Codes: *	0x01, 0x02, 0x03, 0x04, 0x05, 0x06, 0xF, 0x10
Support IEEE Floating Point: **	NO (register/100)
Baud Rate:	4800, 9600, 19200, 38400, 57600, 115200 ***
2 Wire	
Parity:	None
# of Stop Bits:	1
# of Data Bits	8

(*) You can read all the registers in the system at once with **0x03** (includes coils, discreet coils, holding registers and input registers).

(**) Divide the desired parameter by 100 to calculate the real flow point.
(***) Must be the same in the controller setting
To view the Modbus Registers, which govern the signaling communication from the Atlantium system to the facility control system, see Appendix A, ***Modbus Communication Protocol*** on page 14.

5.3.4 Connecting the Main Power Cable

The main power cable that you prepared for this installation must be attached to the Electrical Cabinet.

WARNING!

- The electrical connections must be performed by a qualified electrician **ONLY**.
- **Do not** plug in the RS104 Series's Power cable to an outlet until after the hardware installation is complete and the system is ready to be initialized. (See ***Initializing the*** RS104 Electrical Cabinet on page 17.)

➲ To connect the power cable to the Electrical Cabinet:

Step 1. On the bottom-left side of the cabinet, open the cable gland **(A)** and insert the end of the power cable into the cabinet, ensuring that the cable can reach the main power connections **(B)** without tension.

Step 2. Fasten the cable gland's plastic nut.

Step 3. Connect the cable's power leads according to the local voltage:

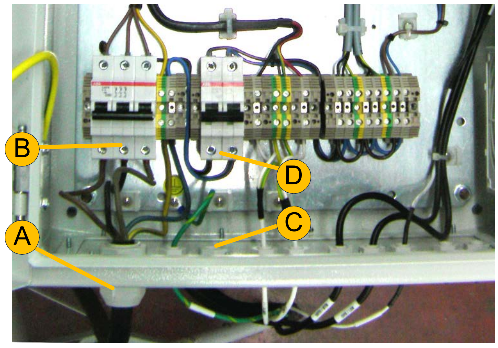


Figure 5-10: Main Power Cable Connections

- 400 VAC 5-wire system:
  - Connect three leads to the terminals marked **L1**, **L2** and **L3**.
    Connect the fourth and fifth lead Neutral and Ground to terminals marked **GN1** and **GN2**.
- 440/480 VAC 4-wire system:
  - Connect three leads to the terminals marked **L1**, **L2** and **L3**.
    Connect the fourth lead Ground to a terminal marked **GN1** OR **GN2**.

Step 4. **For location with 440/480VAC ONLY**, to connect the **110V** cable, on the bottom of the cabinet, open the cable gland **(C)** and insert the end of the **110V** power cable into the cabinet, ensuring that the cable can reach the 110V power connections **(D)** without tension.

Step 5. Connect the leads to the terminals marked **L1** and **N**.

6 Initializing the Atlantium System

The Atlantium system starts up with the initialization of the Control Module on the electrical cabinet.

➲ To start up the Atlantium system:

Step 1. Check that the power cables of the electrical cabinet are all connected to the mains power source.

Step 2. To ignite the lamps, press the power button .

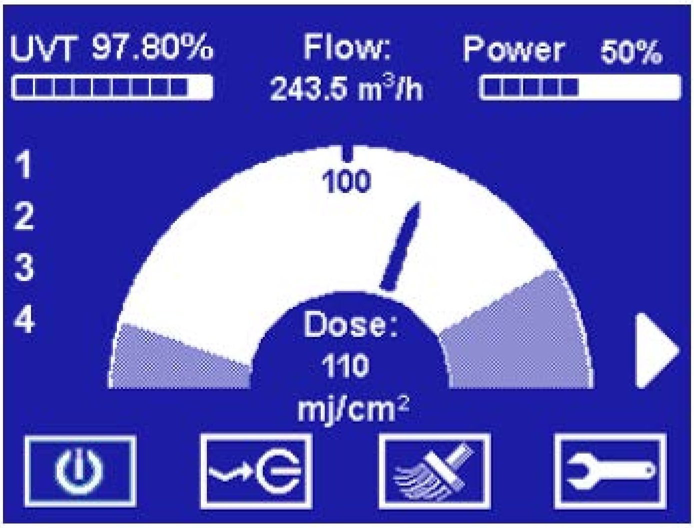


Figure 6-1: Initial Main Operation Screen

7 First Time System Activation

Following Control Module configuration, completing the system commissioning includes the following major steps:

- Cleaning the Atlantium Unit (See below)
- Filling unit with water (See below)
- Igniting the UV Lamps (See below)
- Initiating the Water Flow (See page 7-51)
- Adjusting the Auxiliary Equipment (See page 7-52)
- System Tuning (See page 7-52)
- Installation QA (See page 7-52)

7.1 Cleaning the Atlantium Unit

The inner surfaces of the Atlantium Unit are to be cleaned via the CIP process at installation, before starting up and proceeding with the commissioning phase. Follow the directions in ***Cleaning In Place (CIP)*** on page 10-109.

7.2 Filling unit with water

The unit must be filled with water before full activation.

Water hammer can cause damage to any hydraulic system, including the Atlantium Unit. **Make sure that water hammer is NOT present in your facility's water line.**

To prevent water hammer, consult your system engineer and check your facility system procedures, including:

- When starting the water flow when the lines are empty, keep valves open so that the air in the pipes has a simple release pathway.
- Sudden valve closures can cause water hammer. Make sure that valves close gradually enough to avoid this problem.
- Ensure that all pumps used with the Atlantium system employ soft start-up procedures.

➲ **To introduce water:**

- Open the Inlet valve and, using the Viewport on the Unit, check that the quartz chamber fills with water completely.

7.3 Igniting the Lamps

The lamps of the Atlantium Unit are ignited to commence live system operation.

- **Temporarily, the breather caps on the lamps must be left open to allow any condensation water to evaporate. During this time, set up Caution floor signs at a distance of 1m/40inch from the Atlantium Unit to warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.**
- Wear white cotton gloves.
- Wear long sleeves rolled down to your wrist.
- Where appropriate eye protection, such as containing polycarbonate lens that meet EN166, CAN/CSA-Z94.3-02 and/or Z87.1 standards.
- **DO NOT look directly into openings that emit UV light.**

➲ To ignite the lamps:

Step 1. Post temporary Caution signs on the ground 1 m away from the Atlantium unit to warn bystanders of possible UV light exposure and require the use of appropriate skin and eye protection to approach.

Step 2. Open each of the lamp vent caps (2 on each side of the lamp). For the location of the lamp vent caps, see Figure 10-12 , on page 10-85.

Step 3. Check that the Main screen shows that system is working properly according to spec (see **Figure**)

Step 4. Press the power button and verify that power readings are as required.

Step 5. Press ▶ The System Settings screen appears.

Step 6. Configure system settings as required for your particular system. See ***Configuring the Control Module*** on page 8-53.

Step 7. Let the system operate for at least one hour.

Step 8. Continue with Section 7.5 below while the lamps complete the first hour of operation.

Step 9. After that first hour, close all of the lamps' breather caps and remove the Caution floor signs.

7.4 Initiating the Water Flow

When initiating the water flow, the water may be set to drain out or collect in a tank until the system is fully operational. Once the entire installation and QA testing is complete, the water is set to flow normally as part of the facility operation.

➲ To initiate the water flow:

Step 1. Open the relevant valve that allows water to flow through the Atlantium unit. This may be the draining valve or bypass valve.

Step 2. Open the viewport and verify that the Unit is completely filled with water and that no air gaps and no air bubbles exist.

Step 3. From the **Main Operations** screen, press . The **Password** screen appears.

Step 4. Using the numbered keys, enter the password (default password is **1357**) and press **OK**. The Settings screen appears.

Step 5. For **Flow**, press ▶. The Flow Settings screen appears.

Step 6. Verify that the Flow reading on the right is within normal range. If it is not, check the connections between the Atlantium system and the Flow meter. See ***Connection Box*** on page 5-55.

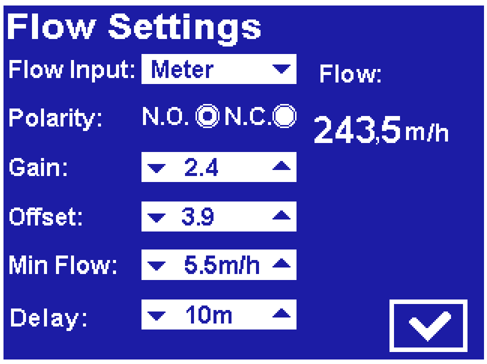


Figure 7-1: Flow Settings Screen

7.5 Adjusting the Auxiliary Equipment

All relevant auxiliary equipment, i.e., flow meter, automatic valves, PLC, etc., must be adjusted at this time. Refer to the manufacturers' user documentation of the relevant equipment.

If your configuration includes a flow meter, check that the value of the flow displayed on the Control Module matches the reading on the flow meter. See page 7-51 For flow meter calibration, see ***Configuring the Flow Settings*** on page 7-88.

7.6 System Tuning

The Atlantium system must be tuned via the Control Module. Follow the instructions in ***Replacing/Checking a UVT Analyzer or UVIS Sensor*** on page 10-125.

7.7 Installation QA

Atlantium personnel performs a quality assurance procedure (see ***Installation Check List*** on page 5-65) following which the Atlantium system is ready for regular facility operation.

8 Configuring the Control Module

8.1 Main Operation Screen

The Standard Control Module controls the operations and dose measurements of the Atlantium system.

The System uses a validated algorithm to measure the dose and display it in real time. Depending on how you choose to run your process, you can configure the Control Module to maintain a specific dose or to simply use a particular power level and inform you if it does not reach the dose you designate.

The Control Module provides a real-time display of operational data.

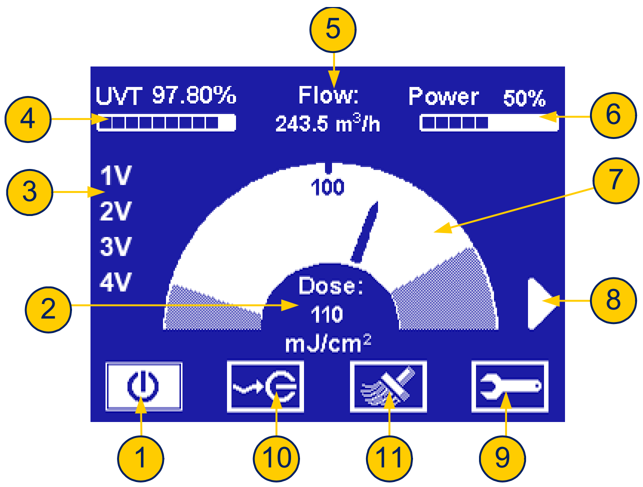


Figure 8-1: Initial Main Operation Screen

Table 8-1: Main screen components

#	Item
1	**Lamp Power Button** - upon Control Module initiation, is on the Off position. After being pressed, turns white, showing that the lamps are working. It can be used to shut lamps down in an orderly manner.
2	**Dose** - this is the dose the user wants the Unit to achieve and maintain. Generally, it should be set about 10-20% above the minimum required dose to assure stability and on-spec operation in the case of small variations of water quality or flow. You should confer with the Atlantium application engineer in setting this number based on the capacity of the Unit, variability at the site and power considerations. When the actual dose is below the **Op**(erate) **At d**ose, the number rounds down. Otherwise, it rounds up.
3	Numbers **1-4** represents up to four lamps included in the System. **V** indicates lamps in operation (see Table 8-2)
4	**UVT** - scale indicates the current quality of the water flowing through the system, and the amount of the UV light absorbed by the water, relative to the intensity of UV light produced by the lamp. If the UVT percentage factor is high, the clarity factor of the water is high, and vice versa.
5	**Water Flow** - Number shows the current rate of water flowing through the Unit, based on a flow signal (for variable flows) or a constant volume (for on/off steady flow applications).
6	**Power** - displays the current power level. The Unit's electronic ballast varies continuously (not in limited steps) to maintain the dose and respond to changing conditions.
7	**Dose Gauge**- represents the total system dose. The gauge has two different linear functions - from the dose setting down and from the dose setting up. The minimum and maximum allowable dosage levels are user-defined.

Table 8-1: Main screen components

#	Item
8	Press to access the System Status screen.
9	Press to access Settings and Technician protected screens via Password entry screen.
10	**External Operation button** - appears on the main operating screen only if the External On/Off feature is activated in the main Settings screen. If the feature is enabled, the system can be turned on or off from the facility's External On/Off switching mechanism. When the feature is disabled, the system may be turned off only from this screen.

8.1.1 Lamp status indication

The symbol next to each lamp number indicates the lamp status:

Table 8-2: Lamp status symbol

Symbol	Lamp Status
	(No symbol) Lamp Off
*****	Lamp in Ignition
V	Lamp On
D	Lamp Disabled
E	Lamp Error

8.2 Planning System Configuration

Plan your Control Module configuration with your Atlantium Application Engineer to assure that the system performs properly and achieve its objectives. Review the Configuration Printout with the Atlantium Technician who is commissioning the site to make sure the settings are understood and correct. The Atlantium Technician configures the initial system settings via the Control Module. The User can afterwards change settings as experience suggests improvements in the operations and processes.

➲ To access the Settings screen:

Step 1. From the **Main Operations** screen, press . The **Password** screen appears.

Step 2. Using the numbered keys, enter the password (default password is **1357**) and press **OK**. The Settings screen appears.

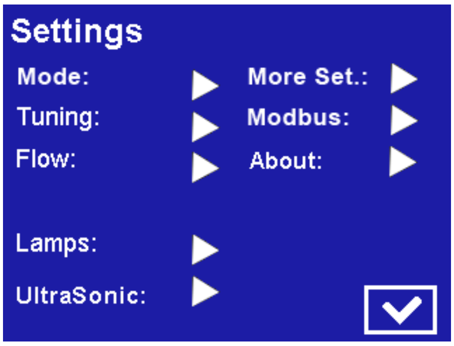


Figure 8-2: Settings Screen

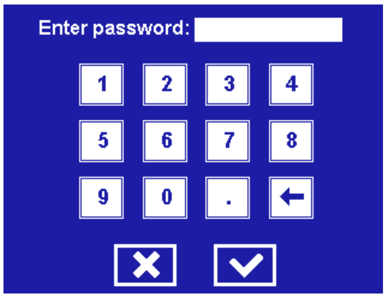


Figure 8-3: Password Screen

8.3 Configuring the Operation Mode

The Control Module provides Atlantium system has two operational modes. Determine which suits your objectives and facility processes: of control:

- Power Mode - The system is set to operate at a particular power level. Under all circumstances, the system operates at the specified designated power level. Power mode is the right choice if maintaining a particular dose is not the objective. It is often chosen where energy consumption concerns are paramount or when the maximum power is needed at all times and concern that low flows might give high doses or that high flows might give lower doses is not an issue (i.e., maintaining a particular dose is not the objective).
- Dose Mode - The system is set to operate at a particular dose. It maintains the designated dose as conditions allow and generally is set to alarm when the dose is not achieved.

➲ To set the Operation Mode:

Step 1. Access the Control Module's Settings screen as detailed above and for **Mode**, press ▶. The Mode Setting screen appears

- Dose mode:
  - a For **Operation Mode**, use the ▲▼ arrows to select **Dose**.
  - b Ignore the **Power%** parameter
  - c To enable an alarm to be triggered should the Dose reach the minimum allowed, press the **Min Dos Alarm** box until a check-mark appears.

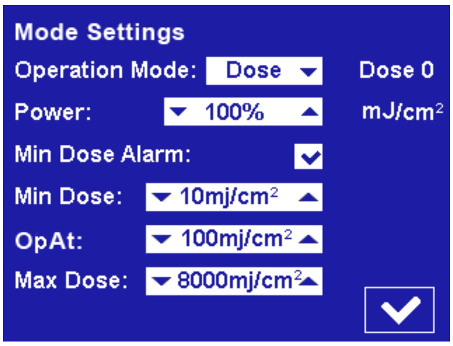


Figure 8-4: Dose Settings Screen

  - d For **Min Dose**, press the ▲▼ arrows to set the lowest allowable dose in mJ/cm^2. This sets the linear function for the left side of the dose gauge and is the dose the Unit maintains. If it is not maintained, it dips into the shaded area, and, if configured to send an alarm, it does so.

e For **Op**(erate) **at**, press the ▲▼ arrows, to set the operating dose. This generally includes a safety factor so that the Unit does not operate at the minimum margin and does not send false alarms when the water quality or flow quivers or changes rapidly.

f For **Max Dose**, press the ▲▼ arrows, to set the maximum UV dose limit. This sets the linear function for the right side of the gauge and defines when the arrow goes into the shady area. In cases where there is low flow, unusual water chemistry or the potential for overheating, high dose can indicate impending issues.

g Press **OK**. The settings are saved and you are returned to the Settings screen.

- Power mode:

a For **Operation Mode**, use the ▲▼ arrows to select **Pwr**.

b Set the **Power %** using the ▲▼ arrows.

c If you want to be alerted to the dose going beneath a value, set it here as the **Min Dose**. Press the ▲▼ arrows, to set the lowest acceptable dose in mJ/cm^2. This configures the left side of the gauge, but the system does not change the power or operation to achieve or maintain this dose. It is simply for information. If you set the **Minimum Dose alarm** (see below) it also triggers an alarm if the Unit falls below this dose.

d Ignore the **Operate at** parameter.

e For **Max Dose**, press the ▲▼ arrows, to set the maximum UV dose limit. This sets the linear function for the right side of the gauge.

f To enable the alarm trigger if the unit goes below the minimum allowed, press the **Min Dos Alarm** box until a check-mark appears.

g To save your changes and return to the Settings screen, press **OK**.

8.4 Configuring the Flow Settings

The Flow signal and related settings are critical for determining the dose and assuring that the system is working within design guidelines. The accuracy and tolerance of the flow meter measurement should match the Control Module interpretation of the flow signal.

➲ To configure the flow settings:

Step 1. Access the Control Module's Settings screen as detailed above.

Step 2. For **Flow**, press ▶. The Flow Settings screen appears.

Step 3. For **Flow Input**, use the ▲▼ arrows to select either **Meter** or **Switch**.

- If you selected **Switch**:
  - Mark the electronic polarity as **Normally Closed** or **Normally Open**.
  - The constant flow number is set by the Atlantium Technician in **Technician 2 Screen**.
- If you chose **Meter**:

a To set Gain, you need the meter's maximum-design flow rate (appears on the flow meter datasheet). You need to enter flow in m^3/h, so if it is written in any other unit of measurement, you must convert it to m^3/h.

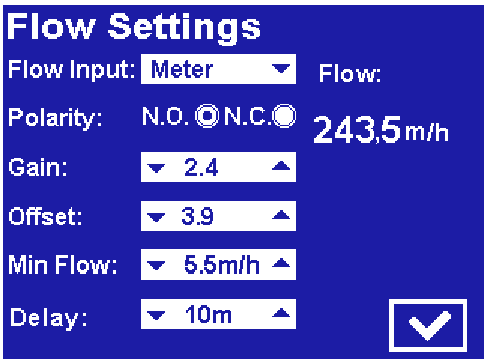


Figure 8-5: Flow Settings Screen

b In the **Gain** field, using the ▲▼ arrows, enter the meter's maximum-allowed flow rate divided by 100. (If the max flow rate is 100 m^3/h, enter the value **1**. Convert any value into m^3/h values to make this calculation.)

c To set **Offset**, turn off the flow meter and note the reading on the screen in the right hand corner. If it is not zero, using the ▲▼ arrows, enter the reading. This assures that the system registers zero in the dose calculation when the flow meter is turned off and compensates for any measurement anomalies.

d For **Min Flow**, using the ▲▼ arrows, enter the minimum flow for standard operations. Set it, making sure to convert your calculations if necessary to use the same units of measurement as are on the screen. Set this number based on process and unit considerations. The Unit cannot run without flow and in some cases, low flow may be indicative of some upstream issue. Once the Low flow number has been reached, the unit reduces power to the lowest possible power setting until the Delay has passed, and then shuts down. If the flow increases before the delay period elapses, the Unit returns to its normal operation. If the Unit shuts down, if the **Restart on Flow** is enabled (see Step 8. on page 58.), it resumes operations when normal flow is restored, otherwise it needs to be turned on manually to return to normal operations.

Step 4. For **Delay**, using the ▲▼ arrows, enter the number of minutes the system should wait before shutting down in the event of flow below the minimum.

Step 5. To save your changes and return to the Settings screen, press ☑ **OK**.

8.5 System Settings

Access the following settings through the More Settings screen:

- Set the maximum temperature for the Atlantium unit.
- Units of Measurement - the default setting is metric; this allows you to change to American units
- Enable/Disable Cleaning- to manually disable/enable the Cleaning system
- General Alarm - to configure a facility general alarm to be triggered when there is a system alarm or if you opt for a minimum dose alarm
- External On/Off - to configure facility control of the Atlantium system

➲ To configure more system settings:

Step 1. Access the Control Module's Settings screen as detailed above.

Step 2. For **More System Settings**, press ▶. The System Settings screen appears.

Step 3. To set the maximum temperature threshold for the Atlantium unit, for **Max Temperature,** using the ▲▼ arrows, set the temperature.

Step 4. To change the units of measurement from Metric units, press the **American Unit** check-box until a check mark appears.

Step 5. To manually disable the Cleaning system, press the **Cleaner En** check-box until the check-mark is removed. To re-enable the Cleaner, press the check-box until a mark appears.

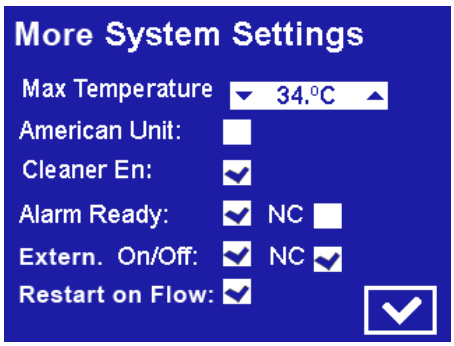


Figure 8-6: More System Settings Screen

Step 6. If your system configuration includes a facility general alarm, or system ready configuration, for **Alarm** mark the left check-box. For **Ready**, leave the left check-box unmarked. For electronic polarity **Normally Closed** mark the **NC** check-box or for **Normally Open** leave the **NC** check-box blank.

Step 7. If your system configuration includes a facility external control, mark the first check-box.

- For electronic polarity **Normally Closed** mark the **NC** check-box or for **Normally Open** leave the **NC** check-box blank.
- For regular wiring, set according to the table in ***External On/Off - Additional Notes*** on page 47

Step 8. To set the system to restart at the configured settings when the flow returns to normal rate, press the **Restart On Flow** check-box until a mark appears. When the check-mark is marked and the system shuts down because of low flow the main power remains on and a flow label on the main screen flashes indicating the system is waiting for a normal flow rate to resume). When normal flow rate returns, the system restarts and resumes operation according to the configured settings automatically.

Step 9. To save your changes and return to the Settings screen, press OK.

8.6 Configuring the Modbus

The Modbus screen displays the software version and other system version information.

Configure the Modbus node address assigned to this system in the Control Module. The Modbus enables communication with the rest of the facility and facilitates integration of the Atlantium Unit into the overall process.

➲ To configure the Modbus:

Step 1. Access the Control Module's Settings screen as detailed above.

Step 2. For **Modbus**, press ▶. The Modbus screen appears.

Step 3. For **Address**, using the ▲▼ arrows, set the Modbus address node to a number as per the serial address setting in your facility.

Step 4. For **BaudRate**, using the ▲▼ arrows, set the appropriate baud rate for this system. The options are:

- 115200
- 57600
- 38400
- 19200
- 9600

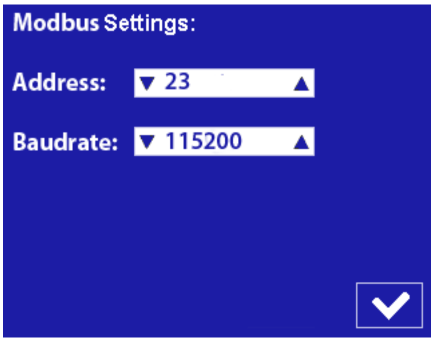


Figure 8-7: Modbus Settings Screen

Step 5. To save your changes and return to the Settings screen, press OK.

The software version and other system version data are displayed on the About screen.

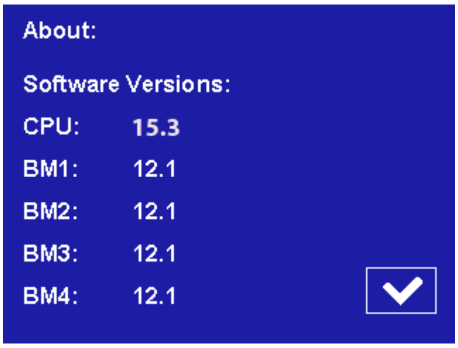


Figure 8-8: About Screen

8.7 Configuring the Technician Settings

As a precautionary measure, the Technician settings in the Control Module should be verified.

➲ To configure the Technician Settings:

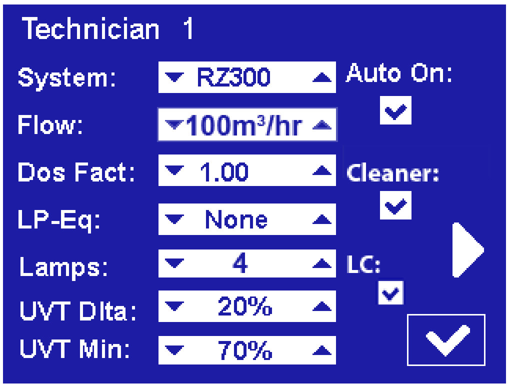


Figure 8-9: Generic Technician 1 Screen

Step 1. From the **Main Operations** screen, press . The **Password** screen appears.

Step 2. Using the numbered keys, enter the password (default password is **8888**) for the Technician Screens and press **OK**. The **Technician 1** screen appears.

Step 3. For **System**, verify that the correct Atlantium system is defined.

Step 4. For **Auto On**, to set the Atlantium system to restart automatically when power is restored after a facility-wide power failure and reactivate according to the most recent settings, press the check-box until a mark appears.

Step 5. If the system includes a cleaning system, press the **Cleaner** check-box until a check mark appears.

Step 6. If the system includes a **flow meter**, for **Flow**, set the maximum flow rate using the ▲▼ arrows. Otherwise, ignore this parameter.

Step 7. If the system uses a **flow switch** instead of a **flow meter**, in the **Flow** field, set the constant maximum flow rate using the ▲▼ arrows.

Be Careful when using this screen as changing the unit time or other errors can substantially impact your system!

Step 8. If your system contains PG quartz, the Dose Factor must be configured. For **Dos Fact**, set the Dose Factor using the ▲▼ arrows.

Step 9. For facilities that require Dose monitoring in Low Pressure Equivalence terms, for **LP-Eq**, use the ▲▼ arrows to set the Low Pressure Equivalence figure. The Dose is expressed as a Low Pressure Equivalent PDM system.

Step 10. Verify that the system is configured correctly for the number of lamps. Use the ▲▼ arrows to set the correct number.

Step 11. The RS104 system has one lamp only.

Step 12. If the system is equipped with a special 10m long cable (applicable to RS104 only) check the **LC** check box

Step 13. To save your changes and return to the Settings screen, press .

Step 14. To access the **Technician 2** screen, press the ▶ arrow on the right of the **Technician 1** screen.

Step 15. Verify that for each lamp, the parameters listed for UVT and UVI appear in the normal range.

Step 16. Ignore the **Power** option. It is used for testing only.

Step 17. Verify that for each lamp the column on the right is set to **ON** (enabling the lamps).

Step 18. To save your changes and return to the Settings screen, press .

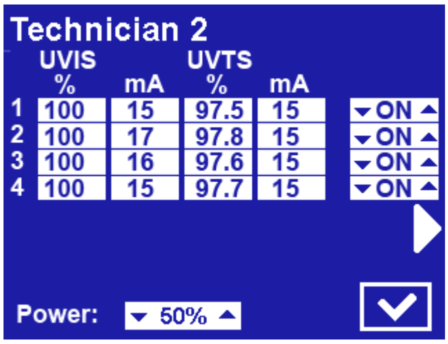


Figure 8-10: Technician 2 Screen*

Step 19. To access the **Technician 3** screen, press the ▶ arrow on the right of the **Technician 2** screen.

Step 20. Verify that the temperature displayed is within the norm. The temperature for each installed and operating lamp is displayed. An **N/A** appears for a lamp that is not operating or not installed.

Step 21. Ignore the **Power** option. It is used for testing only.

Step 22. Verify that for each lamp the column on the right is set to **ON** (enabling the lamps).

Step 23. If a lamp fails, it will be automatically disabled. The Volt column displays the failure code as received from the ballast PS (see **Section 8.7.1.** ***Lamp Failure codes in Technician screen 3***).

Step 24. To save your changes and return to the Settings screen, press .

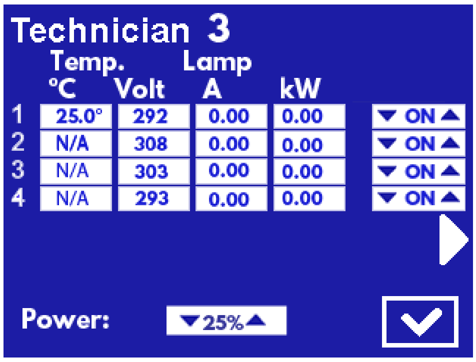


Figure 8-11: Technician 3 Screen**

*For a system with less than four lamps, only the relevant rows are displayed.
**For Temperature, for less than four lamps, only the fields for the installed lamps display the temperature. The other fields display N/A.

8.7.1 Lamp Failure codes in Technician screen 3

The table below provides the various "lamp off" or failure to ignite as identified by the ballast/s. Turn off reason can be identified in two ways:

- By a pop-up message
- By voltage value that acts as a **Failure code** and remains displayed in the voltage column in **Technician screen #3** after the lamp has gone off

Table 8-3: Table of Failure codes displayed in Technician screen #3

#Failure code	Reason for lamp turn off (appears in the pop-up message)
0 (+24)	None
50 (±24)	Lamp nominal voltage too low*
100 (±24)	Lamp nominal voltage too High**
150 (±24)	Device cooling insufficient
200 (±24)	Internal failure / One phase lost
250 (±24)	Lamp earth short circuit
300 (±24)	Lamp not ignited / gone out
350 (±24)	Lamp short circuit
400 (±24)	Phase failure mains voltage
450 (±24)	Mains over-voltage
500 (±24)	Manual turn off
600 (±50)	Loss of ballast's internal parameters***

** Voltage of lamp is low while the current to the lamp is at the maximum allowed value – in this state, the required power cannot be delivered to the lamp.*
*** The voltage required by the lamp cannot be created by the ballast. The current going to the lamp is zero.*
**** In case of loss of internal ballast parameters, consult with Atlantium service personnel for instructions on how to solve this issue.*

9 Operating the Atlantium System

This section details how to operate the system and handle basic and periodic routines to be carried out by Atlantium- trained the local staff.

WARNING!

- Unauthorized servicing or modification of this system in a manner not specified in this manual could expose personnel to potential electrical or other hazards.
- Improper use or adjustment of this system may invalidate the service warranty agreement.

9.1 Operational Guidelines

Follow these guidelines for smooth Atlantium System operations:

- **Incoming Water Supply** - Be alert about incoming water supply. If you do not have pre-treatment, be alert to reports of contamination by upstream slugs of mud, particles, oil, etc.
  If the water source was not previously in operation or the line is new, - flush the line prior to introducing water into the Atlantium system
- **Avoid chemicals that are harmful to UV systems** - Avoid caustic soda and other corrosive chemicals that can degrade gaskets and O-rings. For information on approved chemicals for CIP use, see ***Selecting the Correct Chemicals for CIP*** on page 68
- **Prevent Water Hammer** - Water hammer can cause damage to any hydraulic system, including the Atlantium unit. Take steps to ensure that this phenomenon does not occur in your facility's water line. To prevent water hammer, consult your system engineer and check you facility system procedures for following:
  - Avoid filling empty lines with water without keeping valves open for releasing the air from the pipes and the system.
  - Avoid rapid closure of water lines by sudden closure of valves.
  - Before introducing the water, ensure that all pumps that impact the Atlantium system employ soft start-up procedures.
- **Avoid vibrations on upstream and downstream piping** - Make sure that the piping to be attached to the Atlantium Unit is reinforced to protect the Unit against vibration.
  - A high vibration level can cause damage to any hydraulic system, including the Atlantium unit.
  - High vibration can be caused by an unbalanced pump or by sudden valve close.
- Do not leave the Atlantium Unit filled with stagnant seawater for a long time. Either rinse with fresh water and leave the Atlantium Unit standing with fresh water, or empty. Leaving the Unit empty is a less preferable option, since when refilling the system you must avoid air collection in the Unit and avoid water hammer, as mentioned above.

9.2 Basic Operational Tasks

- Starting up the Atlantium system - See page 63
- Viewing system status - See page 64
- Taking microbial samples - See page 66

9.2.1 Shutting Down the Atlantium System

In a non-emergency situation, you can shut down the Atlantium system from the Control Module. Be sure to enable the **Auto Power On** function in the Control Module if you want the system to save your settings. (See *Igniting the Lamps* on page 50.)

Step 1. To shut down the Atlantium system from the Control Module: On the Main screen of the Control Module, press . The password screen appears.

Step 2. Using the number keys, enter the password **2468**. and press **OK.** The Atlantium system is turned off.

9.2.2 Starting up the Atlantium System

You can start up the Unit via the Control Module and then check the water and system parameters.See page 49.

➲ To start up the Atlantium system:

Step 1. Check that the electrical cabinet's power cable is connected to the main power source.

Step 2. Connect the power cable of the Control Module to the main power source.

Step 3. On the Main screen of the Control Module, tap

Step 4. Wait 10 minutes for lamps to turn on and the system to stabilize.

Step 5. Open the viewport and check that there are no air bubbles in the water flow. If necessary, release all air bubbles from within the system.

Step 6. Check the main screen of the Control Module and verify that the Atlantium system parameters for **Power**, **Flow**, and **UVT** are within wanted acceptable range (use the table below to enter your expected values).

Notify the executive maintenance engineer in case the values are out of range.

Table 9-1: Abbreviation of Table in Appendix C. *Checking the System Parameters*

		Expected Value	
	Parameter	Minimum	Maximum
1	Power		
2	Flow Rate		
3	UVT		
4	UV Dose		
5	Lamp Status		

For periodic checks on Atlantium system parameters, a convenient check list is supplied in **Appendix** C

9.3 Viewing System Status

You can view the real-time status of various System parameters.

➲ To view the system's status:

Step 1. On right side of the Control Module screen, press ▶. The System Status screen appears displaying the real-time status of the following parameters:

- UVT %
- Flow
- Power %
- Per Lamp: **Power**, **Status**, and **Age**
- Temperature

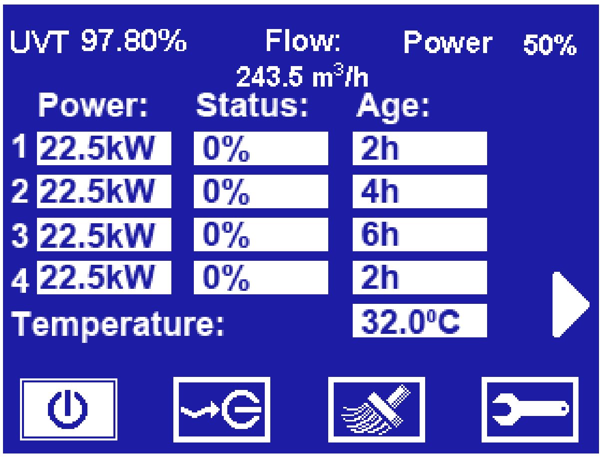


Figure 9-1: Control Module Status Screen

Every defined event is written to the event log of TRACS (if included). You can view the data log of all events in the **Events** tab of TRACS.

9.3.1 .Viewing System Information

View Settings screens and Technician Screens 1, 2 & 3 to see system information (See ***System Settings*** on page 57.)

9.4 Taking Microbial Samples

The designated aseptic sampling valves (i.e., EGMO ESV valves) are the **Inlet** and **Outlet Sampling Points** located on the sides of the Atlantium Unit. Before taking a microbial sample, a valve is sterilized internally with Ethanol 70%, as well as with flame.

Required Equipment

- Sampling containers that are clean, laboratory-grade, sampling bottles with a volume no less than 100ml (3.38oz).
- Ethanol 100-70%
- Bunsen Burner and lighter
- Permanent marker
- Sterile gloves
- Proper Sampler clothing, according to local laboratory protocol

CAUTION!

- Use gloves during sampling and treat them with Ethanol 70%.
- Use **sterile disposable** sampling containers.
- Prevent contact between the sampling containers and the valves' surfaces.
- Be sure that no foreign matter or liquids fall into the sampling containers.
- Perform the sample collection as fast as possible to minimize environmental contamination of the sample.
- **Use open flame to treat the surrounding air while sampling.**

➲ To take a microbial sampling:

Step 1. Open the Sampling valve and allow water to flow for 2-3 minutes. Then close the valve.

Step 2. If the sampling valve contains a cover at the top, open its cover and place it in a clean place.

Step 3. Spray all over the Sampling valve and inside it thoroughly with Ethanol 70%.

Step 4. Using a Bunsen burner, apply the flame comprehensively over the entire Sampling valve surface from its connection to the main pipe to its spout.

Step 5. Spray the cap thoroughly with Ethanol 70% and place it on the Sampling valve.

Step 6. Open the valve and allow the water to flow for 2 minutes. If you have applied the flame properly, vapor emerges from the spout. (If there is no vapor, application of the flame was insufficient. Repeat the previous step.)

Figure 9-2: Cleaning the Sampling Valve with Ethanol

ATTENTION!

Make sure never to touch the opening at the Sampling container's neck.
While the Sampling container is open, do not turn the cap upside down.

Step 7. On the Sampling container, with one hand, remove its cap carefully, avoiding touching the inside of the cap with fingers or any other object. **Do not turn the cap upside down.**

Step 8. With your other hand position the Sampling valve under the water stream and take a water sample. Make sure never to touch the valve's opening or the opening at the container's neck. Fill the sampling container, leaving ample air space in bottle (at least 2.5 cm).

Step 9. Carefully replace the container's cap and tighten it to avoid cross contamination.

Figure 9-3: Taking a Sample

Step 10. Label the Sampling container with the following data:

- Sampling site
- Sampling point
- Sampler name
- Date the sample is taken

Step 11. Close the Sampling valve.

Step 12. Spray the sampling container thoroughly with Ethanol 70%.

Step 13. Store the water samples in a dark, refrigeration unit at a temperature of 2-8°C (35.6-46.4°F) until they are examined.

9.4.1 Microbial Analysis Guidelines

The microbial analysis is to be conducted according to the company guidelines which are used in the local microbiology laboratory.

- Atlantium's recommendation is to use the microbial filtration method for the testing.
- If a number of consecutive anomalous microbiological results are obtained in one of the sampling points, take the samples only after performing CIP with a disinfection chemical (peracetic acid is recommended). It is important to let the chemicals clean and sanitize the aseptic valves, as well by letting the cleaning solution flow through them. See ***Cleaning In Place (CIP)*** on page 109.
- In the event that system outlet counts are higher than the inlet counts, the results are to be considered as a sampling error.
- In the event that the microbial counts are not proportional in different dilutions, the results are to be considered as a laboratory error.
- For further guidance regarding microbial sampling and analysis, contact the Atlantium Application department.

10 System Maintenance

To keep the Atlantium system in peak form, preventive and periodic routine maintenance is essential. During commissioning, Atlantium trains your local staff how to do the tasks on a prescribed schedule or as needed.

WARNING!

- Unauthorized servicing or modification of this system in a manner not specified in this manual could expose personnel to potential electrical or other hazards.
- Improper use or adjustment of this system may invalidate the service warranty agreement.

Before you begin a procedure, carefully read through it so that you can anticipate the steps efficiently.

The periodic routine maintenance schedule includes:

- Weekly Maintenance Tasks:
  - Walk-Around Inspection Tour - see below
- Monthly Maintenance Tasks:
  - Checking the Electrical Cabinet Cooling Fans (if relevant)
- Yearly Maintenance Tasks:
  - Cleaning the Quartz Sleeve (or Replacing as needed) and Replacing the Quartz Sleeve O-rings (See page 10-89)
  - Anti-Corrosion Maintenance (***Anti-Corrosion Spray Safety*** on page 4-38)
- Maintenance Tasks Performed as Needed:
  - Cleaning In Place (CIP) (See page 10-68)
  - Replacing a Lamp (See page 10-78)
  - Checking/Replacing a Sensor (See page 10-93)
  - Tuning the Atlantium System (See page 10-98)
  - Checking/Replacing the Quartz Sleeve (See page 10-89)
  - Cleaning/Replacing a UVIS Wave Guide (See page 10-85)
  - Temperature Sensor Maintenance
    - Replacing the Temperature Sensor (See page 10-101)

10.1 Walk-Around Inspection Tour

Perform Weekly

Walk-Around and inspect the Atlantium system once a month to check for water leaks, visible damage to the cables and harnesses, or any other part of the system. Report any problem you find to an authorized technician.

➲ To perform a Walk-Around inspections:

Step 1. Walk around the Atlantium unit and inspect it. Look for leakage at the incoming and outgoing pipes.

Step 2. Inspect the **Cable Harness** and cables running between the Unit and the Electrical Cabinet for signs of fraying and any other visible damage.

Step 3. Inspect the Electrical Cabinet for any signs of damage.

Step 4. Report any leakage or damage to an authorized technician immediately.

10.2 Cleaning In Place (CIP)

The CIP procedure consists of two parts:

Cleaning to remove the accumulated deposit (particles) that collect on the quartz sleeve over time

(Disinfectant cleaning to disinfect the quartz sleeve (not applicable to some Municipal installations

Clean the Atlantium Unit's inner surfaces and quartz sleeve generally on a periodic basis that is determined by observing that the performance is on target and the UVT Sensor readings are stable. You can clean using this process less frequently, depending on the specific chemistry of your water.

A CIP recirculation system is available from Atlantium as an optional CIP Kit. Connect the recirculation system to the CIP ports to create a closed-loop flow line that pumps a diluted solution through the Unit.

- In facilities where CIP is performed on the plant level, circulating though all piping, the CIP process is the same for the Atlantium System. The inlet and outlet valves referred to in these sections are irrelevant.
- If the Atlantium Unit(s) is/are integrated into the facility's CIP loop, ensure that the water flow rate through the Unit is at least 60 m3 per hour.
- In facilities with plant-wide CIP, perform CIP with the Atlantium System running.
- In facilities that utilized sea water, sea water may be used in the CIP process provided that the chemical manufacturer of the cleaning/disinfectant solution approves.

The estimated time for a CIP process is 1.5 hours.

Prepare the following:

- Protective chemical-safe goggles, clothing and gloves
- An external pump-driven recirculation system with a built-in 100 liter reservoir (26.5 US gallons) or Atlantium's CIP Kit including its accessories (See ***Sampling Valve Components and Kits*** on page 3-28.)
- Scalant cleaning solution
- Disinfectant cleaning solution (if disinfection is planned)

10.2.1 Selecting the Correct Chemicals for CIP

It is **critically important** to select the correct cleaning and disinfecting chemical for the CIP process. It must be powerful enough to clean the inner surfaces of the Atlantium unit, while at the same time:

- Does not harm the components of the Unit that it comes into contact with (stainless steel 316/super-duplex stainless steel UNS S32750, quartz and Viton® O-rings).

- Does not contaminate the product for which the water being treated by the system is used.

WARNING!

- The cleaning/disinfectant solution selected and the concentrations used must be approved for use by the facility's quality assurance and safety officials, and must comply with the written protocols and procedures set in place by those departments.
- It is the responsibility of the owner/operator of the Atlantium system, as applicable, to ensure that the cleaning/disinfectant solution to be used in the CIP process shall be of a type that is permitted for this use in the country/jurisdiction where the CIP process is to be performed.
- Use and handle in strict accordance with their manufacturer's instructions, product information sheets, and material safety data sheets (SDS).
- Use of NaOH (Caustic Soda) may cause irreversible damage to the quartz. Use of this material requires an analysis of temperatures, concentration, frequency and duration and written approval from Atlantium's application engineer.

- All chemicals must be obtained locally.
- Atlantium does not supply chemicals - get them locally from commercial chemical suppliers.

Recommended Scalant Cleaning Solutions

The following chemicals have been tested and found to be very effective for cleaning and are recommended for use when cleaning the Atlantium system:

- Phosphoric acid (H_3PO_4) 2% diluted solution
- Nitric acid (HNO_3) 2% diluted solution
- HCl 2% diluted solution - recommended for metal based (rust-colored) deposits
- Sulfamic acid (H_3NSO_3) 1.5% diluted solution - recommended for calcium based deposits

Recommended Disinfectant Solutions (not applicable to some Municipal installations)

(The Disinfectant CIP process is not applicable to some Municipal installations)

The following have been tested and found to be very effective for disinfection and are recommended for use when disinfecting the Atlantium system:

- Peracetic acid-based chemicals - 30 min. contact time, for example:
  - P-3 Oxonia Active 150 - 0.5% diluted solution according to manufacturer's recommendation
  - Divosan Forte - 0.5% diluted solution according to manufacturer's recommendation
- Hot water at 85°C (185°F) for 30 min.

WARNING!

- **It is the responsibility of the Atlantium system's owner/operator to use and apply the correct chemicals.**
- **The use of any other cleaning and disinfection chemicals requires Atlantium's approval.**

10.2.2 Preparing the Chemical Solution

For both cleaning and disinfecting, the liquid in the reservoir must contain the desired percentage of the chemical additive (see above). Therefore, the purchased chemical solution must be diluted and added to the reservoir. Calculate the correct amount of chemical solution to be used for the CIP process.

Protect yourself from the ill-effects of chemicals with protective chemical-safe goggles, clothing and gloves.

- The concentrations refer to room temperature.
- The diluted solution values denote the required dilution of the chemical circulating through the Atlantium Unit.
- While calculating the diluted volumes, consider all volumes including the Unit, pipes and the recirculation reservoir.

➲ To calculate the correct amount of chemical solution:

Step 1. Calculate the total volume you need for the CIP* by adding the water volumes in:

- The Atlantium Unit volume (measurements in liters/gallons) - Refer to the ***Physical Specifications*** Table in ***Atlantium System Footprint*** on page 2-14
- The pipes between the Atlantium Unit and the Inlet and Outlet Valves
- The CIP recirculation system
- The hoses between the Unit and the recirculation system

Step 2. Calculate the amount of chemical additive to be added to the reservoir to achieve the desired dilution. Use this formula:

$$C = \frac{A * V_t}{B}$$

A - The target concentration for the liquid (chemical substance + water) in %
B - The concentration of the chemical substance to be added in %
C - Volume of the chemical substance to be added in liters/gallons
V_t - Volume of the water in the Unit + piping + CIP Reservoir (as described above*) in liters/gallons

OR:
Assume that you need a 2% diluted solution in the CIP process and assume that your purchased solution is of a 35% concentration in a 4-liter container.
Calculate according to this calculation: [200 liter X 2%] / [35% X 4 liter] = [400] / [140] = 2.9 = 3 containers.

CAUTION!

* If you intend to collect the solution into the CIP reservoir or other container, the total volume of the cleaning
  solution must not exceed the volume of the CIP reservoir or container.

Be sure to obtain from an engineer the correct calculation for the amount of chemicals to be used for the process you intend to use.

10.2.3 Performing CIP

CAUTION!

The anti-scaling procedure is be performed only if necessary. Atlantium recommends that anti-scaling always be followed by disinfection.

The CIP process is performed utilizing fresh water. For systems with sea water, drain the system and rinse before preforming CIP.

- In facilities where CIP is performed on the plant level, circulating though all piping, the CIP process is the same for the Atlantium System. The inlet and outlet valves referred to in these sections are irrelevant.
- If the Atlantium Unit(s) is/are integrated into the facility's CIP loop, ensure that the water flow rate through the Unit is at least 60 m^3 per hour.
- In facilities with plant-wide CIP, perform CIP with the Atlantium System running.

This section details the CIP process with recommended steps for carrying it out. The procedure is to be adapted according to the specific circumstances of your installation.

The CIP process is to follow these major steps:

(Refer to the next section for the detailed description on the CIP process.)

Step 1. Close the inlet and outlet valves and immediately shut down the Atlantium system

Step 2. For systems with sea water, if needed, drain the system

Step 3. Remove the caps covering the CIP ports

Step 4. Connect the supply hose to the bottom CIP port and connect the return hose to the top CIP port

Step 5. Fill the CIP reservoir with fresh water or sea water

Step 6. Starting the circulation

Step 7. Adding the cleaning solution (See ***Selecting the Correct Chemicals for CIP*** on page 10-68.)

Step 8. Running the CIP circulation for at least 30 minutes or as specified according to the chemical manufacturer

Step 9. Rinsing the system and the CIP cart

Step 10. Performing CIP Disinfection with the disinfection solution

Step 11. Ending CIP

The CIP diagram below is followed by the full recommend CIP procedure.

CIP Setup

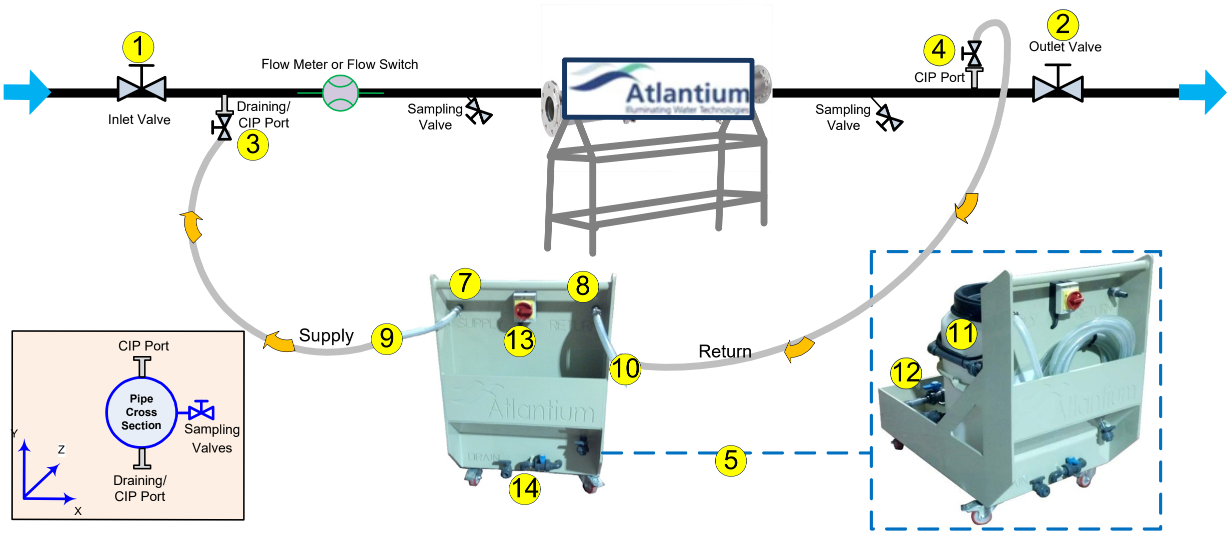


Figure 10-1: Recommended CIP Setup

If the installation is vertical, the water flow is from bottom to top. The Draining/CIP port **(3)** is on the bottom, under the Atlantium Unit and the CIP port **(4)** is above the Atlantium Unit.

➲ To Perform CIP:

ATTENTION!

If you are only performing disinfection: Follow Steps 1 through 20. Then resume with Step 22.
If you are performing both procedures: Follow all of the steps below.

Shutting Down the System

Step 1. Close the Unit's inlet isolating valve **(1)** and outlet isolating valve **(2)**. Immediately proceed to the next step.

Step 2. Shut down the Atlantium system, on the Control Module, by pressing the **Main Unit Operation Button Operation** button in the upper-right corner.

Step 3. On each CIP port, remove the cap's clamp and remove the caps covering the CIP ports **(3 & 4)**. Place the caps in a safe place for reuse later. Verify that the gasket is still inserted into the ferrule.

ATTENTION!

Make sure that the hose fittings are secure and tight.
Water must fill the entire path through the piping on both sides of the Atlantium Unit and the Unit itself, as well as the Supply and Return hoses and the CIP Kit's reservoir.

Connecting the CIP Hoses

Step 4. Remove the hoses from their pocket on the Cart.

Step 5. Using the CIP cap's clamp, connect the CIP Kit's Supply hose to the bottom CIP port **(3)**. Verify that the gasket is present.

Step 6. Using the CIP cap's clamp, connect the CIP Kit's Return hose **(10)** to the top CIP port **(4)**. Verify that the gasket is present.

Filling the CIP Reservoir

Step 7. To fill the CIP reservoir **(A)** with water:

- **Filling option 1:**
  - Slightly open the Unit's inlet isolating valve **(1)** and open the top CIP port on the outlet side **(4)** and allow the facility's water to fill the CIP Kit's reservoir **(11)**.
  - There is a full line indicator on the reservoir **(11)**. Make sure the water in the reservoir is up to the 100 liter (26.5 US gallons) level.
    - Close the Unit's inlet isolating valve **(1)**.
    - Close the CIP port **(4)** valve.
- **Filling option 2:**
  - Open the top cover of the CIP Kit's reservoir **(11)**.
  - Using a hose of an external water source, fill the reservoir with 100 liter (26.5 US gallons) of room-temperature tap water (±25°C/75°F).
  - There is a full line indicator of the reservoir. Make sure the water in the CIP Kit's reservoir **(11)** is filled to the 100 liter (26.5 US gallons) level.
  - Close the cover.

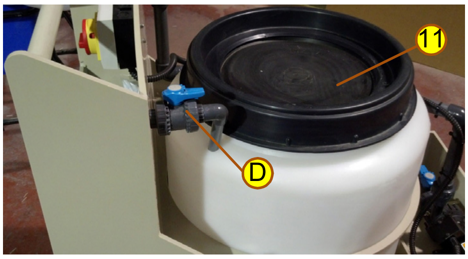


Figure 10-2: CIP Kit's Reservoir

Figure 10-3: CIP Cart's Inlet and Drain Port

WARNING!

Observe chemical safety rules! Follow all safety recommendations in the chemical manufacturer's product information and SDS publications!

Adding the Chemical Solution

Protect yourself from the ill-effects of chemicals with protective chemical-safe goggles, clothing and gloves.

Step 8. Open the top cover of the CIP Kit's reservoir **(11)** and add the prepared chemical solution to the water in the reservoir. (See ***Selecting the Correct Chemicals for CIP*** on page 10-68..

Starting the CIP Circulation Process

ATTENTION!

The Atlantium Unit must be absolutely full for the solution to reach all of the inner surfaces.

Step 9. If the bottom and top CIP ports **(3 & 4)** are valves, open them.

Step 10. To start the CIP recirculation pump, on the CIP cart, turn the switch **(13)** to the **On** position and allow the solution to circulate through the Unit for **the period of time specified in the chemical's Use instructions, but not less than 30 minutes**.

Step 11. Verify that the Atlantium Unit is full of water and that the water is circulating correctly.

Step 12. During the circulation period open and close the sampling valve several times to assure chemical contact with the CIP chemicals.

If the draining valve is separate from the CIP ports, open and close the draining valve.

Rinsing the System and CIP Kit

After the chemical solution has circulated through the Unit for a sufficient duration, the chemicals solution must be rinsed from the system.

Step 13. Open the top cover of the CIP Kit's reservoir **(11)** and add the prepared chemical solution to the water in the reservoir. (See ***Selecting the Correct Chemicals for CIP*** on page 10-68.

Step 14. Connect the CIP recirculation system's drain valve **(B)** to the facility's sewage system. See ***Alternatives for Draining the Cleaning Solution*** on page 10-76.

Step 15. Turn on the Atlantium system, on the Control Module, by pressing the **Main ON/OFF Operation** button in the upper-right corner

.

ATTENTION!

Before performing the next step, verify that the lamp(s) are on and the system has stabilized and indicates it is ready. If not, restart the lamp(s) again and wait until the system has stabilized and indicates it is ready.

Step 16. Close the inlet CIP port **(3)**, stop the recirculation pump **(12)** and open the CIP recirculation system's drain valve **(C)**. Immediately slightly open the inlet valve **(1)** to rinse the Atlantium system thoroughly.

Allow this rinsing process to operate until at least three cycles of water volume have gone through the Atlantium Unit to remove all of the scalant cleaning/disinfectant solution.

Step 17. To assure that all the chemicals are flushed out, use a measurable indicator, such as measuring the pH, or other indicators depending on the chemical being used (consult your chemical supplier).

Step 18. Slightly open the outlet valve **(2)** and close the CIP port on the outlet side (4). To avoid hammer effect if there is air trapped in the system, carefully open the inlet valve **(1)** completely. Resume normal operation.

Rinsing the CIP Reservoir

After the chemical solution has circulated through the Unit for a sufficient duration as specified by the chemical supplier (typically 30 minutes), the chemicals solution must be rinsed from the system by the steps below.

Step 19. Open the CIP cart's drain valve **(C)** and completely drain the CIP reservoir **(11)**.

Step 20. Using an external water source, rinse the CIP cart's reservoir, letting the rinse water drain out of the CIP cart's drain valve **(C)**.

Step 21. Close the CIP cart's drain valve **(C)**.

Performing CIP Disinfection

Step 22. If you are planning to perform an additional CIP process using the disinfectant solution, refer to ***Preparing the Chemical Solution*** on page 10-70 and follow the CIP process again from ***Filling the CIP Reservoir*** in Step 7. above.

Or,

To return to normal operating status, proceed to the next step.

Ending CIP

Step 23. To drain out any water inadvertently collected in the CIP cart's storage pocket or base, use the CIP cart's drain valves **(A** and **B)**.

Step 24. Roll up the CIP Kit's hoses and return them to their storage pocket **(F)**.

Refilling the System

ATTENTION!

Make sure that the Atlantium Unit is filled with water before powering it up and turning on the lamps.

Step 25. Turn on the Atlantium system at 100% power and wait until the system has stabilized and indicates that it is ready.

Returning Power to the Lamps

Step 26. On the Connection Box next to each lamp, turn the switch to the **On** position.

Step 27. Turn on the Atlantium system, on the Control Module, by pressing the **Main ON/OFF Operation** button in the upper-right corner.

Alternatives for Draining the Cleaning Solution

- **If local regulations permit, and a physical drain setup exists,** drain the used cleaning solution to the facility's sewage system:
  - a Connect the CIP recirculation system's drain valve **(B)** to the facility's sewage system.
  - b With the Atlantium System operating, slightly open the Inlet valve.
  - c Close the inlet bottom CIP port **(3)**.
  - d To stop the CIP recirculation, on the CIP cart, turn the switch **(13)** to the **Off** position.
  - e Open the CIP cart's drain valve **(C)**. The water with the chemical solution begins to drain out.
  - f Connect the CIP cart's drain valve **(C)** to the facility's sewage system.

Or,

- **According to local regulations, use the Draining/CIP port to drain the solution in the system into a external container.**

Or,

- **Use the CIP reservoir (11) to collect the chemical solution.**
  - a Connect the CIP recirculation system's drain valve **(B)** to the facility's sewage system.
  - b With the Atlantium System operating, close the inlet bottom CIP port **(3)**.
  - c To stop the CIP recirculation pump, on the CIP cart, turn the switch **(13)** to the **Off** position.
  - d Allow the chemical solution to drain entirely into the CIP reservoir **(11)**.
  - e Dispose of the contents according to regulations. Connect a drainpipe to the CIP cart's drain pipe and open the CIP cart's drain valve **(C)**. The water with the chemical solution begins to drain out.
  - f Disconnect and close the CIP cart's drain valve **(B)**.

10.3 Checking the Cooling Fans

Once a month the cooling fans of the Electrical Cabinet must be checked to verify that the air is flowing out.

➲ To check the cooling fans:

Check that air is being blown out of the fans of the Electrical Cabinet.

10.4 Checking the Temperature Sensors' Readings

Once a month both temperature sensors must be checked to verify that they are working properly.

➲ To check the temperature sensors' readings:

Step 1. On the Control Module, to access the Technician mode, tap . The Password dialog box appears. Enter the password (default password is 8888) and tap **OK**. The Technician Settings 1 screen appears.

Step 2. Tap twice on the ▶at the right of the screen to access Technician 3 screen (Figure 8-11 , on page 8-60). The temperature for each installed and operating lamp is displayed. Verify that the temperatures displayed are within the norm. An **N/A** appears for a lamp that is not operating or not installed.

DO NOT leave the Control Module unattended while in **Technician mode** even for a short time. Unauthorized changes to the **Technician mode** settings could damage the operation of the Atlantium system.

10.4.1 Lamp Failure codes in Technician screen 3

If one of the UV lamps fails, a pop-up message is displayed and the Volt column in Technician screen 3 (see *Configuring the Technician Settings* on page 8-59) displays the failure code as received from the ballast PS.

Step 1. To access the **Technician 3** screen, On the Control Module, to access the Technician mode, tap . The Password dialog box appears.

Step 2. Enter the password (default is 8888) and tap **OK**. The Technician Settings 1 screen appears.

Step 3. Tap twice on the ▶at the right of the screen to access Technician 3 screen.

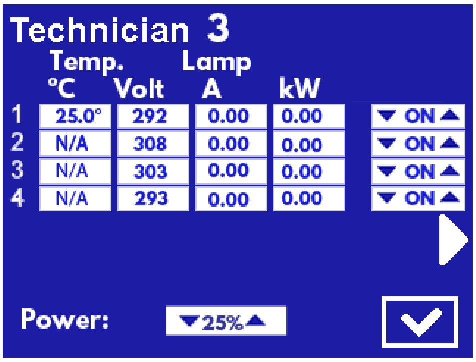


Figure 10-4: Technician 3 Screen

The table below provides the various "lamp off" or failure to ignite as identified by the ballast/s. Turn off reason can be identified in two ways:

- By a pop-up message
- By voltage value that acts as a **Failure code** and remains displayed in the voltage column in **Technician screen #3** after the lamp has gone off and the corresponding pop-up message has been acknowledged. See **Table** 8-3 on page 8-61.

10.5 General Lamps, Sleeves Maintenance Tips

The following are some tips that are meant to help you understand maintenance requirements:

- Lamps and quartz sleeves traverse the width of the Unit and are held in place by holders and connectors at both sides. For maintenance, you can access the lamps and quartz sleeves from either side. Depending on the constraints of the installation (sometimes one side is less accessible) you can usually decide from which side to access the them.
- To access the quartz sleeve, you must remove the lamp housed within it.
- On rare occasions, quartz sleeves and their O-rings may need to be replaced because of a maintenance issue that arises.

10.6 Replacing a UV Lamp

The UV lamp can be used as long as it provides the required dose and efficient service. (Generally, a lamp service status of 80% or more is considered efficient service.) A lamp should be replaced ONLY when necessary.

Each lamp's performance is measured directly by a dedicated UV Intensity Sensor. Replace a lamp when:

- Its status or performance declines and the lamp no longer provides the required performance at the appropriate electrical cost
- Once its rated hours expire, be alert to any changes in efficiency that could indicate the need to change the lamp

In the Control Module, you can set a margin to assure notification as status declines and a warning is sent to the operator at both the user setting level and the internally coded minimum.

For systems with multiple lamps, the system may continue to operate while any one of the lamps is being replaced, however, precautions must be carried out to protect you from the UV.

Before commencing this procedure:

CAUTION!

- Allow the old lamp at least 10 minutes to cool down before starting.
- Read carefully the instructions provided in the new Lamp package.
- Wear white cotton gloves.
- Before installing a new lamp, verify that it is clean and free of grease.
  If not, rub it gently with the provided lens tissue. Never touch the lamp with your bare hands - hold it only by the end-connectors. Oily residues from fingerprints can damage the lamp.

WARNING!

- The UV lamps are designed with high internal positive pressure. Wear protective eye-wear while replacing a lamp. Pay strict attention to the safety warnings and precautions in the ***Safety Overview*** on page 4-33 in the front of this manual.
- In case of lamp breakage, refer to the Safety instructions in ***Safety Overview*** on page 4-33.

When replacing a lamp while the rest of the system is still in operation, you must follow these safety requirements:

- Shutdown the electricity supplied to the lamp to be replaced.
- **During the time when the lamp's breather caps are removed, use Caution floor signs at a distance of 1m/40inch from the Atlantium Unit that warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.**
- Wear white cotton gloves.
- Wear long sleeves rolled down to your wrist.
- Wear appropriate eye protection.
- **Do not look directly into the lamp enclosure or into any openings that emit UV light during the entire lamp replacement procedure.**

CAUTION!

While carrying the steps of this procedure, replace all of the O-rings on the components, such as the lamp housings and holders.

10.6.1 Lamp Replacement Options

- Replacing a Lamp after System Shutdown (10.6.2)
- Replacing a Lamp while System is Operating (10.6.3)

Generally, a lamp is replaced after shutting down the entire Atlantium system. However, in a facility where there is no redundant UV purification, and you require minimizing system downtime, you can replace a lamp while other lamps continue operation provided you follow these safety requirements:

Replacing a Lamp after System Shutdown

Prepare the following:

- Replacement lamp
- Lamp cleaning cloth (comes with the lamp)
- White cotton gloves
- Eye protection
- #3 Allen wrench
- #4 Allen wrench
- Caution signs

10.6.2 Shutting Down the Entire System

Generally, a lamp is replaced after shutting down the entire Atlantium system.

➲ To shut down the entire system:

Step 1. On the Main screen of the Control Module, tap ⏻.

Step 2. Enter the password (the default password is **2468**). The Atlantium system is turned off.

Step 1. Using the viewport on the Atlantium Unit, check that the lamps are indeed turned off.

Step 2. Check the main screen of the Control Module and verify that the system parameter for **Power** is **0%**.

A severe danger of electrocution exists if the lamp is not turned off during replacement. Be absolutely sure you have turned off the correct lamp's circuit breaker before you proceed to replace the lamp.

Step 3. Open the Electrical Cabinet and locate the main circuit breaker and turn it down to the **Off** position.

Step 4. Continue with the procedure, ***Replacing the Lamp*** on page 10-81.

10.6.3 Replacing a Single Lamp/ while System is Operating

If you are replacing a lamp while other lamps are still in operations, follow these safety requirements:

- Shutdown the electricity supplied to the lamp to be replaced.
- **During the time when the lamp's breather caps are removed, use Caution floor signs at a distance of 1m/40inch from the Atlantium Unit that warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.**
- Wear white cotton gloves.
- Wear long sleeves rolled down to your wrist.
- Where appropriate eye protection, such as containing polycarbonate lens that meet EN166, CAN/ CSA-Z94.3-02 and/or Z87.1 standards. The UV lamps are designed with high internal positive pressure.
- **Do not look directly into the lamp enclosure or into any openings that emit UV light during the entire lamp replacement procedure.**

➲ To shut down electricity to a single lamp:

Step 1. Deactivate the lamp to be replaced:

a To access the Control Module's Settings screen, from the **Main Operations** screen, tap . The **Password** screen appears.

b Using the numbered keys, enter the password (default password is 1357) and tap . The Settings screen appears.

c For **Lamps**, tap ▶. The Lamps Settings screen appears.

d To disable a lamp, tap its **Enable** box to remove the checkmark.

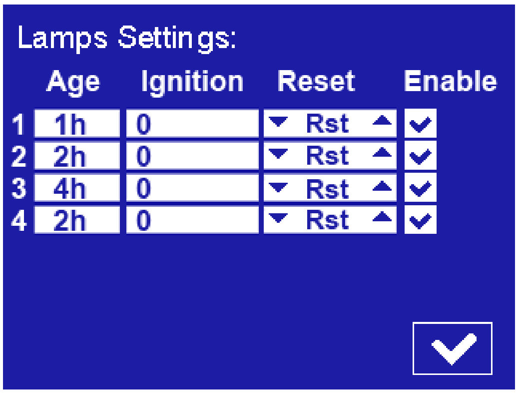


Figure 10-5: Lamp Setting Screen

A severe danger of electrocution exists if the lamp is not turned off during replacement. Be absolutely sure you have turned off the correct lamp before you proceed to replace the lamp.

Step 2. Open the Electrical Cabinet and locate the ballast of the lamp to be replaced. Turn the corresponding circuit breaker down to the **Off** position.

Step 3. Continue with the procedure, ***Replacing the Lamp***, below.

10.6.4 Replacing the Lamp

Once the electricity is turned off for the system or for the specific pair of lamps (at least one of which is to be replaced) and preparations are made (see above) you can replace the lamp.

➲ Before replacing a lamp:

Step 1. If you are replacing a lamp while other lamps are still in operation, follow these safety requirements:

- **During the time when the lamp's breather caps are removed, use Caution floor signs at a distance of 1m/40inch from the Atlantium Unit that warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.**
- Wear white cotton gloves.
- Wear long sleeves rolled down to your wrist.
- Where appropriate eye protection, such as containing polycarbonate lens that meet EN166, CAN/CSA-Z94.3-02 and/or Z87.1 standards.
- **DO NOT look directly into the lamp enclosure during the entire lamp replacement procedure.**

➲ To remove lamp connector assembly

Step 2. Using a #3 Allen wrench, remove the four Allen screws (1) that secure the lamp's connector assembly (2) and remove the connector assembly. Each screw has a washer.

Step 3. **To prevent any damage to the lamp**, make sure you pull the lamp's connector assembly straight out and not at an angle.

Step 4. Using a #4 Allen wrench, remove the 4 screws (3) of the lamp's housing (4). Each screw has a washer.

Step 5. Grasp the lamp's housing and rotate it clockwise and counterclockwise until it loosens. **Make sure the housing is not tilted against the quartz sleeve while rotating it**.

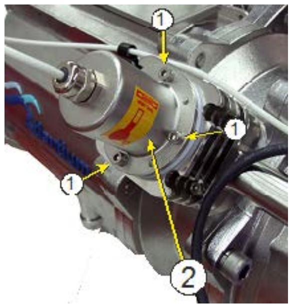


Figure 10-6: Lamp Replacement (Generic View 1)

(The O-Ring on the sleeve stopper between the holder and the stopper may cause the two parts to stick due to the heat. The slight rotation releases it).

Step 6. **To prevent any damage to the lamp and quartz sleeve,** make sure you pull the lamp's housing straight out and not at an angle.

Figure 10-7: Lamp Replacement (Generic View 2)

10.6.5 Removing the Faulty Lamp

The lamp is hot! DO NOT touch the lamp with bare hands, particularly the quartz portion.
Wait at least ten minutes until the lamp has cooled down or use an appropriate pair of long-nose pliers to carefully pull out the lamp.

Step 1. Wearing appropriate cotton gloves and skin covering, and using a suitable long-nose pliers, pull the lamp straight out of the quartz sleeve in the Unit and set it aside on a stable surface.

Step 2. Check the inner surface of the quartz sleeve for residual moisture or high humidity. If any visible moisture exists, using a lens cleaning cloth, dry it.

For Systems with Wave Guide

Every time the lamp is replaced, the wave guide must be cleaned or, if needed, replaced. See ***Cleaning/ Replacing the UV Intensity Wave Guide*** on page 10-85.

Cleaning/Replacing the UV Intensity Wave Guide or Mirror

Step 3. If you need to clean or replace the **UV Intensity Wave Guide** or **Mirror**, do so now before installing the new lamp.

- For wave guide, refer to ***Cleaning/Replacing the UV Intensity Wave Guide*** (on page 10-85)
- For mirror, See page 10-85

Installing the Replacement Lamp

Step 4. Unpack the new UV lamp from its vacuumed sealed package.

Step 5. Make a note of the serial number of the new lamp.

Step 6. Clean the new lamp using the cleaning pad provided with the lamp. Verify that no visible fingerprint or dust is on the lamp.

Step 7. Insert the new lamp into the quartz sleeve in the Unit, all the way until it stops. Ensure that the male contact point (6) on the end of the lamp has meshed with the female contact point in the lamp's housing on the other side of the Unit.

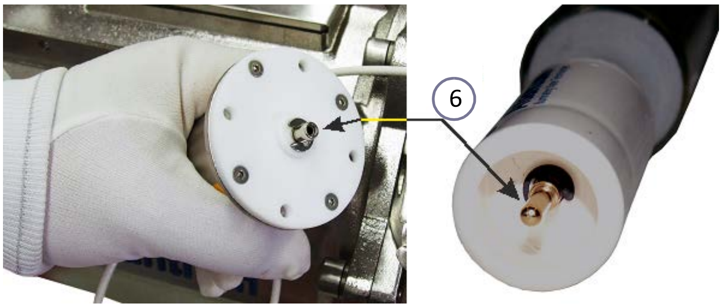


Figure 10-8: Lamp Replacement (Generic View 3)

Assembly

Step 8. Replace the lamp housing's O-ring.

Step 9. Insert the lamp's housing that you removed in above. Ensure that the male contact point on the end of the lamp has meshed with the female contact point in the lamp's housing on the other side of the Unit.

Step 10. Replace the lamp’s housing and, using the # 4 Allen wrench, fasten the four Allen screws with their washers to secure it to the Unit.

Step 11. Replace the lamp's connector assembly pushing it straight in and using the # 4 Allen wrench, fasten the four Allen screws with their washers to secure it to the Unit.

10.6.6 Allowing the Atlantium Unit to Run with Caps Open

Step 12. **Place Caution floor signs at a distance of 1m/40inch from the Atlantium Unit that warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.**

Step 13. Unscrew and remove the two caps (7) one on each side of the lamp. (See the figure to the right.)

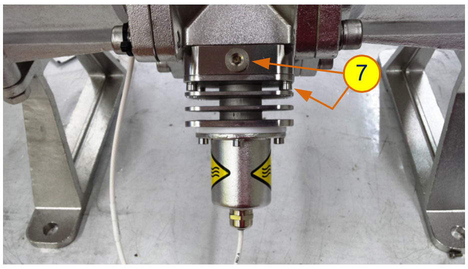


Figure 10-9: Lamp's Caps (Generic View 4)

Reactivating the Lamp

The newly replaced lamp must be tested to verify that it is in working order.

Step 14. Open the Electrical Cabinet and locate the ballast of the lamp to be replaced. Turn the corresponding circuit breaker up to the **On** position.

Step 15. For the lamp that was to be replaced, on the connection box, turn the lamp switch to the **On** position.

If the Entire System is Off

Step 1. Return to the Main screen of the Control Module and tap ⏻.

ATTENTION!

While carrying the steps of this procedure, check that the O-rings on the components, such as the lamp housings and holders, are not damaged. If damaged, replace them before refastening the component in place.

Accessing the Lamp

ATTENTION!

For systems that contain wave guides, **be sure to access the lamp from the side that contains the sensors**. The wave guide is fragile and may be damaged easily if the lamp is inserted from the opposite site.

If only one lamp is off - Reactivation the Lamp

Step 2. Return to the Control Module’s Lamps Settings screen and activate the replaced lamp by using the ▲▼ arrows, set the desired lamp to **ENB**.

Step 3. To reset the lamps’ usage clock, tap .

Step 4. Tap . The settings are saved and you are returned to the Settings screen.

Step 5. Operate the system without the lamp’s caps for one hour.

Step 6. Return the lamp’s caps and remove the Caution floor signs.

Step 7. Perform system tuning according the instructions on page 10-97.

➲ Removing the Faulty Lamp

The lamp is hot! DO NOT touch the lamp with bare hands, particularly the quartz portion.
Wait at least ten minutes until the lamp has cooled down or use an appropriate pair of long-nose pliers to carefully pull out the lamp.

Step 8. Wearing appropriate cotton gloves and skin covering, and using a suitable long-nose pliers, pull the lamp straight out of the quartz sleeve in the Unit and set it aside on a stable surface.

Step 9. Check the inner surface of the quartz sleeve for residual moisture or high humidity. If any visible moisture exists, using a lens cleaning cloth, dry it.

Figure 10-10: Removing the lamp

Installing the Replacement Lamp

Step 10. Unpack the new UV lamp from its vacuumed sealed package.

Step 11. Make a note of the serial number of the new lamp.

Step 12. Clean the new lamp using the cleaning pad provided with the lamp. Verify that no visible fingerprint or dust is on the lamp.

Step 13. Insert the new lamp **(F)** into the quartz sleeve, all the way until it stops. Ensure that the male contact point **(G)** on the end of the lamp has meshed with the female contact point **(G)** in the lamp's housing on the other side of the Unit.

Step 14. Check that the lamp housing's O-ring is in place, clean, dry and undamaged. Replace if necessary.

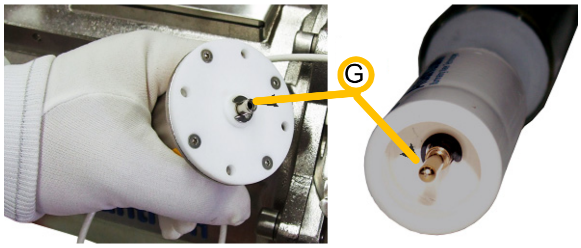


Figure 10-11: Lamp, Connector Assembly and connectors

Step 15. Reinstall the lamp's housing and, using the # 4 Allen wrench, fasten the four Allen screws with their washers to secure it to the Unit.

Step 16. Reinstall the lamp's connector assembly pushing it straight in. Ensure that the male contact point **(G)** on the end of the lamp has meshed with the female contact point **(G)** in the lamp's housing on the other side of the Unit.

Step 17. Using the # 3 Allen wrench, fasten the four Allen screws with their washers to secure it to the Unit.

Step 18. Return to the Control Module's Lamps Settings screen and activate the replaced lamp by using the ▲▼ arrows, set the desired lamp to **ENB**.

Step 19. To reset the lamps' usage clock, tap .

Step 20. Tap . The settings are saved and you are returned to the Settings screen.

Allowing the Atlantium Unit to Run with breather caps Open

Step 21. **Temporarily place Caution floor signs at a distance of 1m/40inch from the Atlantium Unit that warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.**

Step 22. On the side without the DPM (if present), unscrew and remove the two breather caps (H) on each side of the lamp. (See the figure to the right.)

Step 23. Operate the system without the lamp's breather caps for one hour.

Step 24. Return the lamp's breather caps and remove the Caution floor signs.

Perform system tuning according the instructions on page 10-97

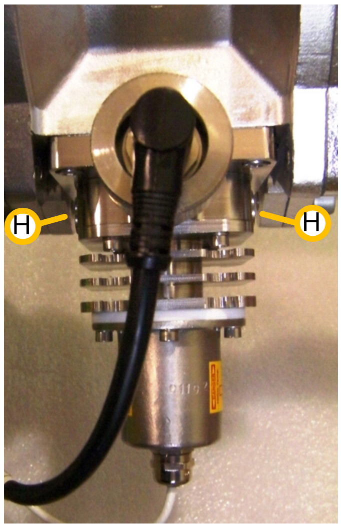


Figure 10-12: Lamp's breather caps (Generic View)

Reactivation the Lamp

Step 25. For the lamp that is replaced, on the connection box, turn the lamp switch to the **On** position. Check that the lamp ignites properly. See **Figure** 10.6

Step 26. Operate the system without the lamp's breather caps for one hour.

Step 27. Return the lamp's breather caps (G).

System Tuning

Step 28. Perform system tuning according the instructions on page 10-98.

Observing Lamp Efficiency

Step 29. Observe the system's efficiency for **100 hours**. If after this period there is a performance degradation of more than 5%, remove the UV intensity sensor and check for humidity. Clean if necessary. For the adjustable UVIS sensor, see the instructions on page 10-94.

Step 30. Perform **System Tuning** according the instructions on page 10-98.

10.7 Cleaning/Replacing the UV Intensity Wave Guide

If your system includes wave guides:

For each UVIS sensor in the Atlantium Unit, there is an internal wave guide for directing UV light into the sensor. The UV Intensity Sensor's wave guide must be cleaned every time the lamps are replaced.

Prepare the following:

- Replacement UV Intensity Sensor's Wave Guide PN SAM000800
- Wave Guide tool PN SA0066000
- White cotton gloves
- Alcohol/isopropanol and lint-free lens cleaning paper
- Appropriate Phillips screwdrivers

➲ To clean/replace the wave guide:

- Allow the old lamp at least 10 minutes to cool down before starting.
- Be very careful when handling the wave guide. The wave guide is made of quartz and is fragile and easily broken.

Accessing the Wave Guide to be Replaced

Step 1. On the side of the Unit where the UVIS sensors are located, identify the lamp whose wave guide is to be cleaned/replaced.

Step 2. Follow the directions for removing a lamp. See ***Replacing a UV Lamp*** on page 10-78.

Removing the Wave Guide

Step 3. Screw the Wave Guide tool (L) onto the wave guide socket (J).

Figure 10-13: Wave Guide Tool

Step 4. While holding the Wave Guide tool (L) now attached to the wave guide (G), using an appropriate Phillips screwdriver, (M) unfasten each of the screws (K).

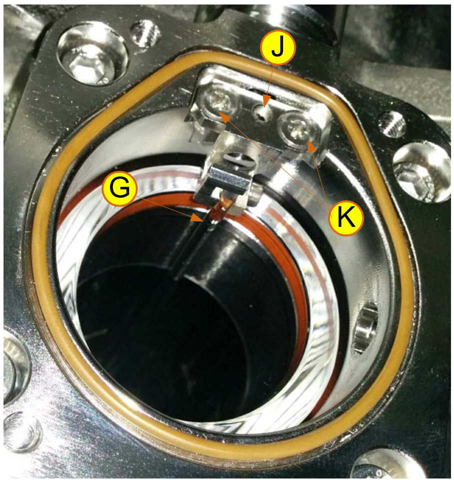


Figure 10-14: Wave Guide Elements (Generic View)

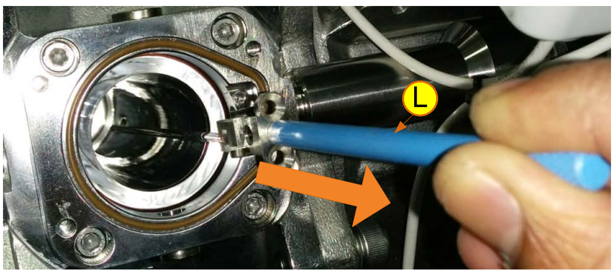


Figure 10-15: Removing the Wave Guide (Generic View)

Step 5. With the Wave Guide tool (L), carefully pull out the wave guide.

Step 6. Unscrew the Wave Guide tool (L) and detach it from the wave guide (G).

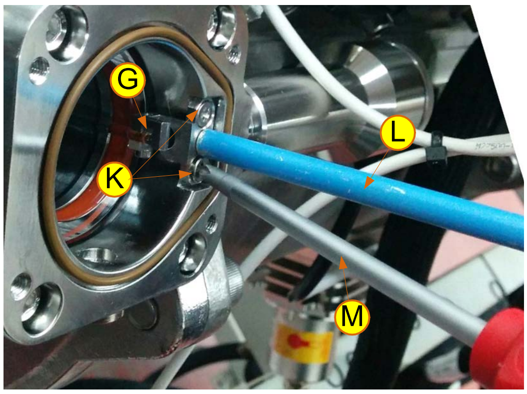


Figure 10-16: Unfastening the Wave Guide (Generic View)

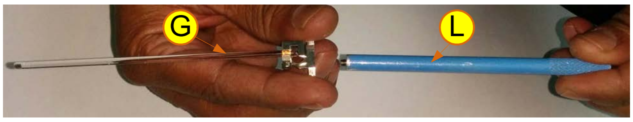


Figure 10-17: Wave Guide Tool and Wave Guide

Cleaning the Wave Guide

Step 7. While wearing white cotton gloves, carefully clean the wave guide with isopropanol and lint-free lens cleaning paper.

Step 8. Ensure that the wave guide is completely dry before returning it to its place.

Step 9. Examine the wave guide for marks and damage. Fingerprints or other stains may cause erroneous UV intensity sensor readings. If the wave guide's surface is damaged, it must be replaced.

Preparing the Replacement Wave Guide

Step 10. Screw the Wave Guide tool (L) onto the wave guide socket (J).

Step 11. With the Wave Guide tool (L), carefully insert the wave guide in place.

Step 12. While holding the Wave Guide tool (L) now attached to the wave guide (G), using an appropriate Phillips screwdriver, (M) fasten each of the screws (K).

Step 13. Check that the wave guide is parallel to the side of the quartz sleeve and not pointing at an angle. If it is at an angle, the quarts sleeve is not installed straight. Remove the wave guide and remove and reinstall the quartz sleeve following the directions in ***Cleaning/Replacing a Lamp's Quartz Sleeve and O-Rings*** on page 10-89. When complete, install the wave guide again.

The correct position of the wave guide is parallel to the side of the quartz sleeve and a distance of 0.75mm from the quartz sleeve.

Step 14. Follow the directions to reinsert the lamp and refasten the lamp's housing and connector See ***Replacing a UV Lamp*** on page 10-78.

10.8 Cleaning/Replacing the UV Intensity Sensor's Mirror

Performed as needed

If your system includes UVIS mirrors:

For each lamp in the Atlantium Unit, there is a small internal mirror **(A)** for directing UV light into the sensor. The UV Intensity Sensor's Mirror must be cleaned every time the lamps are replaced.

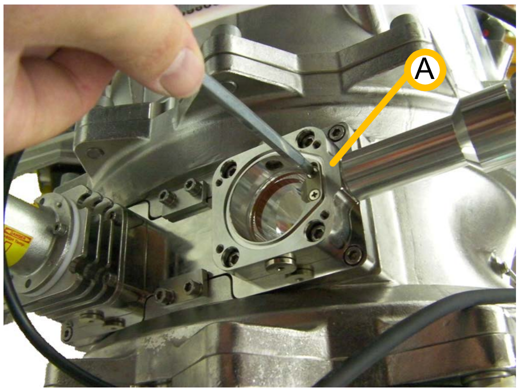


Figure 10-18: (Generic) Cleaning the UV Intensity Sensor's Mirror

Prepare the following:

- Replacement UV Intensity Sensor's Mirror
- Isopropanol and lint-free lens cleaning paper
- Small Phillips screwdriver

➲ To clean the mirror:

Step 1. No particular parts need be removed from the Unit in order to access the mirror for cleaning, once the lamp has been removed. See page 10-78 for instructions on replacing the lamp.

Step 2. Clean the reflective side of the mirror – the side facing into the Unit – with isopropanol and lint-free lens cleaning paper.

Step 3. Ensure that the mirror is completely dry before re-assembling the UV lamp.

Step 4. Examine the mirror's surface after cleaning and before you close the UV lamp' housing. Fingerprints or other stains may cause erroneous UV intensity sensor readings. If the mirror's surface is damaged, it must be replaced.

➲ To replace the mirror:

Step 5. Once the lamp has been removed, you can replace a damaged mirror. See page 10-78 for instructions on replacing the lamp.

Step 6. Using a small Phillips screwdriver, remove the two screws holding the mirror in place and remove the mirror.

Step 7. Clean the reflective side of the replacement mirror with isopropanol and lint-free lens cleaning paper.

Step 8. Examine the mirror's surface after cleaning and before you install it to make sure it is damage-free.

Figure 10-19: Removing the Mirror

Step 9. Carefully place the new mirror in position and using the screwdriver, fasten it securely.

Step 10. Before you close the UV lamp' housing, examine the mirror's surface it make sure it is free of fingerprints or other stains, which may cause erroneous UV intensity sensor readings. Clean it again if necessary.

Step 11. Ensure that the mirror is completely dry before re-assembling the UV lamp.

Figure 10-20: Cleaning the Replacement Mirror

10.9 Cleaning/Replacing a Lamp's Quartz Sleeve and O-Rings

The quartz sleeve is cleaned as needed according to the UVT Sensor readings and when deposit collects on the quartz sleeve. **The quartz sleeve's rubber O-rings must be replaced when dismantling the quartz sleeve.**

Depending on the quality of the water being treated by the Atlantium Unit, the exterior surface of the quartz sleeve that houses a lamp may require periodic cleaning or replacement if after the optional Cleaning system and the CIP process the quartz sleeve is still not sufficiently clean. This is to be determined by the quality control officer at the facility.

Under normal environmental conditions, the quartz sleeve's rubber O-rings (as shown in Section) must be replaced once a year even if they pass visual inspection. For environments where the water temperature is above 70°C (160°F), the quartz sleeve's O-rings must be replaced yearly.

Prepare the following:

- Replacement quartz sleeve (if replacing)
- Two O-rings (See view B in **Figure** 10-21 below)
- #3 Allen wrench
- #4 Allen wrench
- Protective gloves
- Appropriate cleaning cloth. A mild, household cleaning solution
- Alcohol

ATTENTION!

While carrying the steps of this procedure, check that the O-rings on the components, such as the lamp housings and holders, are not damaged. If damaged, replace them before refastening the component in place.

➲ To clean or replace the quartz sleeve:

Powering Down the System

Step 1. On the Main screen of the Control Module, tap ⏻.

Step 2. Enter the password (the default password is **2468**). The Atlantium system is turned off.

A severe danger of electrocution exists if the lamp is not turned off during replacement. Be absolutely sure you have turned off the correct lampcircuit breaker before you proceed to replace the lamp.

Step 3. Open the Electrical Cabinet and locate the main circuit breaker and turn it down to the **Off** position.

Step 4. Using the viewport on the Atlantium Unit, check that the lamps are indeed turned off.

Step 5. Check the main screen of the Control Module and verify that the system parameter for **Power** is **0%.**

Isolating the Unit

Step 6. To isolate the Unit, close the Inlet and Outlet valves.

Step 7. Using the Drain valve, drain the water from the Unit.

Accessing the Quartz Sleeve

Quartz sleeve removal requires that the lamp connector assemblies, lamp housings (on both ends of the lamp) and the lamp be removed, as well as the quartz sleeve holder.

Step 8. Quartz sleeve removal requires that the lamp connector assemblies, lamp housings (on both ends of the lamp) for systems that have them, and the lamp be removed. In the section, 10.6.4. Replacing the Lamp, follow from Step 2. on page 10-86 and remove the lamp connector assembly and lamp housing **from both sides of the Unit,** as well as the lamp, placing them in a secure location until they need to be reinstalled.

While pulling the lamp housing away, DO NOT tilt it and DO NOT pull it away from the lamp/sleeve axis at an angle. For systems with wave guide, the wave guide is fragile. Take care not to damage it when removing the lamp housing.

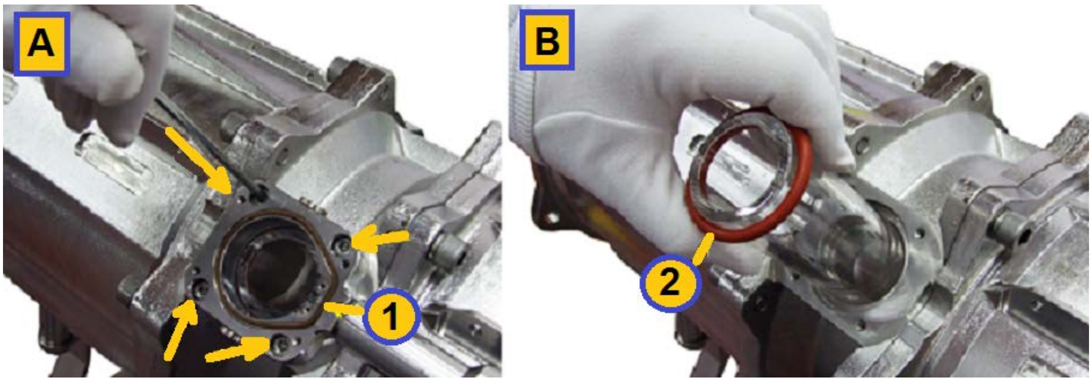


Figure 10-21: Replacing the Quartz Sleeve (Generic View 1)

Step 9. On one side of the quarts sleeve, using a #4 Allen wrench, remove the four Allen screws, opening them in a diagonal pattern (i.e. upper left, lower right, upper right, lower left) (arrow indicate the screws in view A) that secure the sleeve holder to the Unit. Loosen one screw gradually (2 turns) on one side and do the same on the screw at the diagonal opposite side of the sleeve (2 turns).

Step 10. Continue to alternate until both sleeve holders are loose, and then remove the screws completely.

Step 11. On the other side of the quarts sleeve, repeat Step 8. to Step 10.

Removing the Quartz Sleeve

Wear protective gloves. To protect the quartz sleeve, do not touch the internal surface with your bare hands.

Step 12. On one side of the quartz sleeve, place the gloved palm of your hand over the quartz sleeve and gently push in. The quartz sleeve shifts inward. The o-ring on the side where you pushed slips out (if not, take it off).

Step 13. On the other side of the quartz sleeve, the sleeve sticks out slightly. **To prevent any damage to the quartz sleeve, make sure you pull the lamp's connector assembly straight out.** Carefully pull out the quartz sleeve and remove the O-ring (shown in view B above).

Cleaning the Quartz Sleeve

While pulling the connector assembly away, DO NOT tilt it and DO NOT pull it away at an angle.

Step 14. If there is a deposit buildup on the sleeve's exterior surface, clean it with a mild, household cleaning solution.

Step 15. Clean the exterior and interior surfaces with alcohol. If the sleeve cannot be cleaned to a pristine condition, replace it with a new one.

Step 16. Ensure that there is no residual moisture or high humidity on the inner surface of the quartz sleeve. If any visible moisture exists, dry it before inserting it in place.

Installing the Quartz Sleeve

Step 17. Mount the new O-ring onto the quartz sleeve (as shown in view B in **Figure** 10-21 above) and insert the sleeve into the Unit. Make sure it is centered.

Step 18. Add the second new O-ring on the other side of the quartz sleeve.

Step 19. Install the quartz sleeve holder on the side without the UVIS mirror and close it loosely with 4 screws in a diagonal pattern (upper left, lower right, upper right, lower left). **Do not tighten the screws!!**

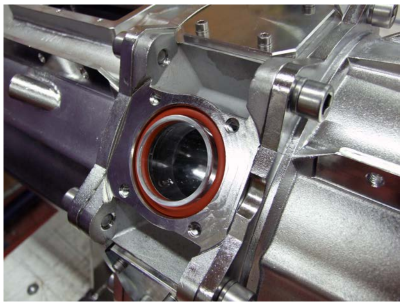


Figure 10-22: Replacing the Lamp's quartz sleeve (Generic View 2)

Assembly

Step 20. Install the quartz sleeve holder on the opposite side – the one with the UVIS mirror or wave guide.

For systems with wave guide, while reinstalling the sleeve holder, DO NOT tilt it and DO NOT insert it at an angle. the wave guide is fragile. Take care not to damage it when inserting the lamp housing.

Step 21. Install the quartz sleeve holder on the lower side and close it loosely with 4 screws in a diagonal pattern (upper left, lower right, upper right, lower left). **Do not tighten the screws!!**

Step 22. Install the quartz sleeve holder on the upper side.

Step 23. Using a #4 Allen wrench, close the screws in diagonal pattern and tighten them.

Step 24. To reinstall the lamp, See ***Replacing a UV Lamp*** on page 10-78.

Step 25. Reinstall the lamps' connector assemblies pushing each straight in and using the # 4 Allen wrench, fasten the four Allen screws with their washers to secure them to the Unit.

Step 26. Connect the UVIS sensors' cables.

Step 27. On the corresponding Connection Box to where the sleeves/lamps are located, turn the switch to the **On** position.

Reintroducing the Water

Step 28. Carefully open the Inlet and Outlet valves (Be sure to avoid water hammer). Follow the directions in the section ***Filling unit with water*** on page 7-50.

Powering Up the System

Step 29. On the Main screen of the Control Module, tap ⏻.

Step 30. Enter the password (the default password is **2468**). The Atlantium system is turned off.

Verifying System Performance

Step 31. Wait 10 minutes for lamps to turn on and the system to stabilize.

Step 32. Open the viewport and check that there are no air bubbles in the water flow. If necessary, release all air bubbles from within the system.

Step 33. Run the system with the breather caps of the lamp off as described in the section, 10.6.4. Replacing the Lamp.

Step 34. Check the main screen of the Control Module and verify that the Atlantium system parameters for **Power**, **Flow**, and **UVT** are within wanted acceptable range and record is using the Table in **Appendix** C. ***Checking the System Parameters*** on page C-119.

Step 35. If the values are out of range, notify the executive maintenance engineer

10.10 Checking/Replacing UVIS Sensors

- From time to time, sensors may need to be checked and wiped clean (most frequently, from condensation).
- During the Calibration process, the UVIS sensor also requires to be removed and temporarily replaced with a Reference Master sensor.
- Sometimes a sensor has to be replaced.

There are two types of UVIS sensors. The sensor itself is the same in both types. The casings of both types are marked with a millimeter scale (**Figure** 10-23) which enables the sensor to be removed for cleaning and replaced in precisely the same position.

The casing markings also include the sensor's sensitivity range (4 - 20 mA / 10Wm^2) part number and serial number.

Figure 10-23: Sensor casing, showing **millimeter scale**

The difference between sensor types is in how they are fastened to the Atlantium Unit. The two types are:

- **Adjustable UVIS sensor** - The Sensor's position is adjustable and its casing is fastened with nuts. See below.
- **Fixed UVIS sensor** - The sensor's casing screws into a fixed position. This type of sensor may include a cover with a pinhole. See ***Replacing/Checking a UVT Sensor*** on page 10-97.

Prepare the following:

- Wrenches # 24 and # 30

➲ Checking position of a UVIS sensor

Step 1. Before removing the UVIS sensor, look at the point where the millimeter scale on the sensor touches the securing nut (**A** in **Figure** 10-24).

Step 2. Record this point on mm scale as a reference for returning the sensor to same position after cleaning.

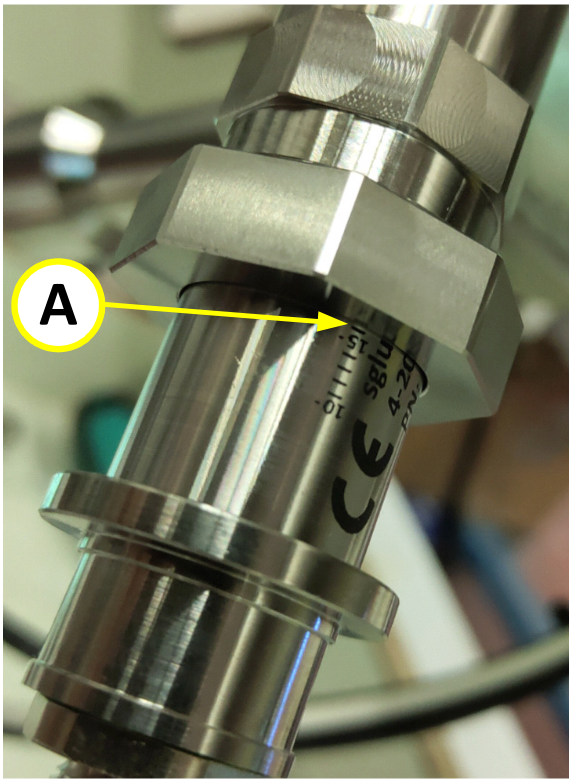


Figure 10-24: Using mm scale as position reference

➲ To remove a fixed sensor:.

Step 3. To disconnect the harness connector (shown by the arrow in view A below), unscrew the sensor's threaded locking nut

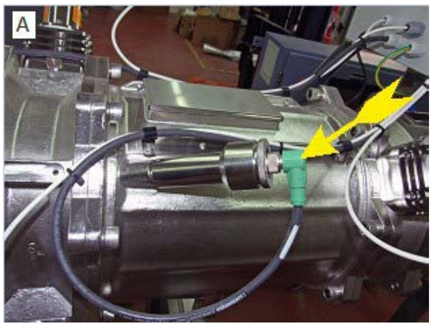


Figure 10-25: Removing a Sensor A(Generic View)

Step 4. Unscrew the sensor's threaded locking nut from the bottom of the sensor housing, as shown in view **B**

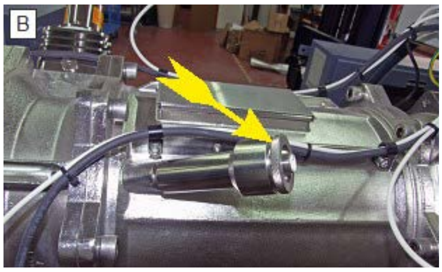


Figure 10-26: Removing a Sensor B (Generic View)

Step 5. Carefully pull the sensor out of the housing (C). If a pinhole (D) is present it slides out. Ensure that the rubber washer stays in place.

Step 6. Proceed to ***Returning cleaned sensor into place*** on page 10-97

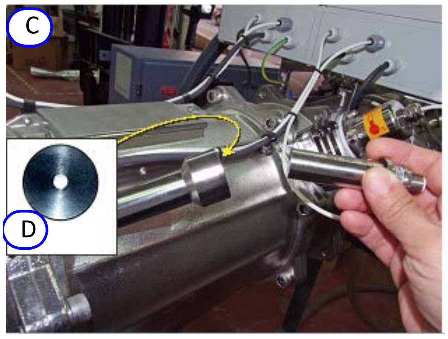


Figure 10-27: Removing a Sensor C 2 (Generic View)

➲ To remove an adjustable UVIS sensor:

Step 1. To disconnect the harness connector (shown in view A, Figure 10-28), unscrew the sensor's threaded locking nut.

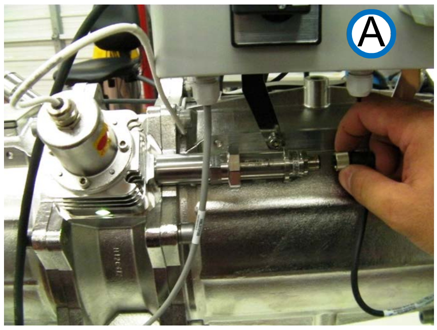


Figure 10-28: Removing connector A (Generic View)

Step 2. Grip the Sensor casing with wrenches **# 24** and **# 30**, as shown in view B, **Figure** 10-29. The inset shows the wrench **#24** in the left hand gripping the casing's left band and wrench **#30** in the right hand gripping the casing's large nut on the right.

Step 3. Carefully pull the sensor out of the housing (Figure 10-27). If a pinhole (D) is present it slides out. The large nut comes off together with the sensor.

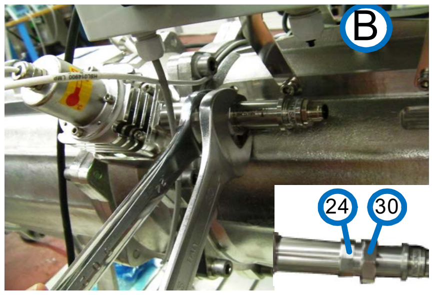


Figure 10-29: Loosening nuts B (Generic View)

➲ Cleaning a sensor

Step 4. Check the sensor's lens (1 in **Figure** 10-30) for condensation. If present, using a dry lens cloth, dry the lens thoroughly.

Be sure that the lens is completely dry before it is returned to its place.

Step 5. Ensure that threaded locking nut (2) and washer (3) are in place

Step 6. If replacing a sensor with a new one, go to Step 13.

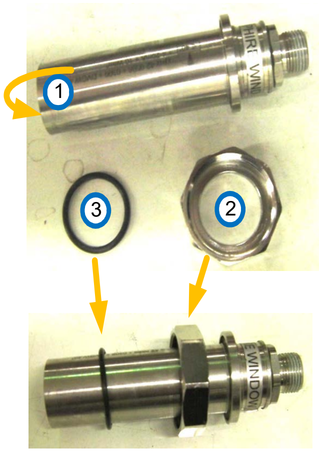


Figure 10-30: Sensor, nut and washer

➲ Returning cleaned sensor into place

Step 7. Slide pinhole (if present) back into place over the end of sensor housing.

Step 8. When you return the sensor to its place, make sure to insert it to the same reference point on the mm scale (**A** in **Figure** 10-24) recorded in Step 1. on page 10-94.

Step 9. Ensure that the washer on the sensor screw thread is in place. Replace as necessary.

Step 10. To re-install an adjustable sensor, tighten both nuts by reversing process in Step 2.on page 10-96.

Step 11. To re-install a fixed sensor, tighten the threaded cap back onto the housing, **Figure** 10-26.

Step 12. In the Control Module, check that the value for UVIS for the lamp on which the sensor was replaced is identical to the value before the sensor was removed.

➲ Replacement with new sensor

Step 13. Unpack the new sensor (**Figure** 10-30) and the sensor pinhole (D in **Figure** 10-27, if supplied).

Step 14. To enable the system to track relevant data about the sensor, make a note of its serial number.

Step 15. Verify that the sensor lens (F) is clean. If needed, using a lens cloth, clean it.

Step 16. If a pinhole (E) is present, insert it.

Step 17. Insert the sensor into the housing and reinstall as in Step 10./Step 11. above.

Step 18. Reconnect the harness's threaded cap to the new sensor.

Step 19. In the Control Module, perform **System Tuning**, on page 10-98.

10.10.1 Replacing/Checking a UVT Sensor

UVT sensors should be cleaned/replaced as required. The procedure is the same as that described for fixed UVI sensors in ***Replacing/Checking a UVT Sensor*** on page 10-97. For some of the sensors, there is a pinhole located on the sensor's window. This pinhole is calibrated by Atlantium for the specific measurement point. Do not switch disks between ports. Before inserting the sensor, be sure to include the pinhole if present.

➲ To replace a sensor:

Step 1. To disconnect the harness connector (shown by the arrow in view A below), unscrew the sensor's threaded locking nut.

Step 2. Unscrew the sensor's threaded locking nut from the bottom of the sensor housing, as shown in view B.

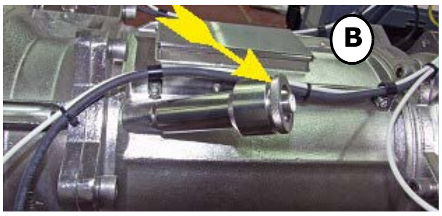


Figure 10-31: Replacing the Sensors 2 (Generic View)

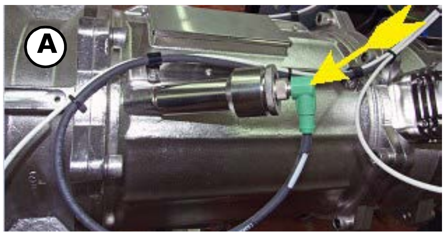


Figure 10-32: Replacing the Sensors 2 (Generic View)

Step 3. Carefully pull the sensor out of the housing (C).

Step 4. If an pinhole (E) is present it slides out.

Step 5. Unpack the new sensor (D) and (if it is supplied) the sensor pinhole.

Figure 10-33: Replacing the Sensors 4 (Generic View)

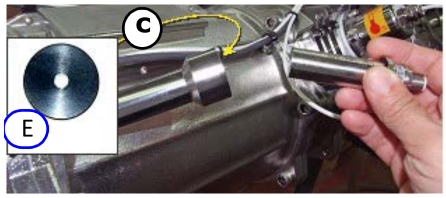


Figure 10-34: Replacing the Sensors 3 (Generic View)

Step 6. To enables the system to track relevant data about the sensor, make a note of its serial number.

Step 7. Verify that the sensor lens (F) is clean. If needed, using a lens cloth, clean it.

Step 8. If an pinhole (E) is present, insert it.

Step 9. Insert the sensor into the housing, and screw the threaded cap back into the housing to secure the sensor.

Step 10. Reconnect the harness's threaded cap to the new sensor.

Step 11. In the Control Module, perform **System Tuning**, see the section, ***System Tuning*** on page 10-98.

10.11 System Tuning

System tuning is performed on these occasions:

- As the last step in system installation
- As the last step when lamps are changed and after 100 operating hours (as needed)
- Upon advice of an Atlantium Service Representative when the system is reset

➲ To tune the Atlantium system:

Step 1. Just prior to tuning, obtain a water sample from the sampling valve on the inlet side of the Atlantium Unit. Test the water sample in an external spectrophotometer. Write down the results. If a spectrophotometer in not available, record the current UVT value shown on the system's display.

Step 2. For each lamp to be tuned, according to **Lamp ID** in column **1**, make a mark in the appropriate checkbox in the last column on the right.

Step 3. Under **Status T0 %** , using the **-/+** buttons, set the **Status T0 percentage** up to **100%**.

Step 4. Wait a few minutes to allow the system to stabilize.

Step 5. To verify that the Sensor readings are stable, observe the reading for a few seconds to see that they remain the same numbers. For the UVIS sensor, see the **UVIS mA** column above. For the UVT Intensity sensor, see the **UVTS mA** column above. If the numbers are not stable, unscrew and remove the two breather caps on each side of the lamp. (See Figure 10-12 , on page 10-85.)

Step 6. Leave the breather caps open for 30 minutes to make sure no humidity is present in the lamp housing. Then replace the breather caps and check the sensor reading again.

When the Lamp's breather caps are removed, do not look directly into the opening without appropriate eye protection. The UV light emitted via the opening is a danger to the eyes.

Step 7. At the bottom of the screen, for **UVT To**, enter the current UVT value obtained from the spectrophotometer that you wrote down in Step 9. above (or the recorded current UVT value).

Step 8. To record the reason for tuning the system, At the bottom of the screen, for **Tuning Reason**, from the dropdown list, select the reason.

Step 9. To shutdown the Atlantium system, on the Control Module main screen, tap and wait approximately three minutes for the **Ready Lamps Off** signal to transmit.

Step 10. On the Atlantium Unit, locate each UVIS and UVT Intensity Sensor.

Step 11. On each sensor (refer to the figures in the section, ***Replacing/Checking a UVT Sensor*** on page 10-97):

a Unscrew the locking nut and gently pull out the sensor.

b With a dry and clean lint free cloth, wipe the UVIS sensor's quartz window.

c Reinsert the sensor and replace the locking nut.

Step 12. To turn on the Atlantium system, on the main screen, tap and wait approximately three minutes for the **Ready Lamps On** signal to transmit.

Step 13. To access the Control Module's Settings screen, from the **Main Operations** screen, tap . The **Password** screen appears.

Step 14. Using the numbered keys, enter the password (default password is **1357**) and tap . The Settings screen appears.

Step 15. To verify that the lamps are already on, for **Lamps**, tap ▶. The Lamps Settings screen appears. (Figure 10-5 , on page 10-80.) Note that the lamps are set to **On**.

Step 16. Tap . The settings are saved and you are returned to the Settings screen.

Step 17. For **Mode**, tap ▶. The Mode Settings screen appears.

Step 18. For **Power**, using the ▲▼ arrows, turn each lamp up to **100%** power.

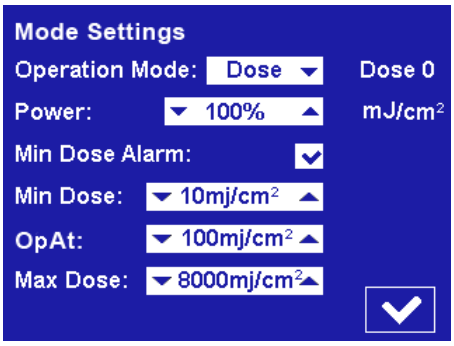


Figure 10-35: Mode Settings Screen

Step 19. Tap . The settings are saved and you are returned to the Settings screen.

Step 20. For **Tuning**, tap ▶. The System Tuning screen appears.

Step 21. For **UVT**, use the ▲▼ arrows to enter the water test results.

Step 22. For **UVIS 1** and **2** (if your system has two lamps):

- For tuning when replacing a lamp, use the ▲▼ arrows, to enter 100%.
- For tuning with existing lamps, leave the settings as is.

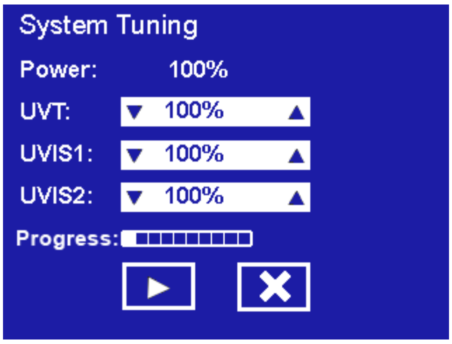


Figure 10-36: System Settings Screen

Step 23. Tap ▶. The systems tunes and the **Tune** bar displays the progress.

Step 24. When tuning is complete, the activate arrow turns to ✓. Tap it. The settings are saved and you are returned to the Settings screen.

Step 25.

10.11.1

10.12 Advanced Maintenance

The advanced maintenance procedures require opening the Connection Box. **Local regulations may require a certified electrician to perform any of these procedures.**

This section contains:

- *Replacing the Temperature Sensor* on page 10-101

- *Replacing the Temperature Converter Card* on page 10-102

10.12.1 Replacing the Temperature Sensor

In the event that the Temperature sensor requires replacement, this section details the procedure.

Follow your local regulations regarding whether a certified electrician may be required to perform this procedure.

Any person involved in handling the Connection Box card or its components is required to wear an electrostatic discharge (ESD) Wrist Strap.
Connect the ESD Wrist Strap to any bolt on the body of the Atlantium Unit or to the metal bracket under the Connection Box using a banana plug or an alligator clip.

Prepare the following:

- Replacement Temperature sensor
- #5 Allen wrench
- #3 Flat screwdriver
- #2 Phillips screwdriver

➲ To replace the Temperature sensor:

Step 1. Turn the Mains circuit breaker to the **Off** position.

Step 2. Using the appropriate Allen wrench, open the Connection Box cover.

Step 3. Verify with volt meter that there is no power in first Connection Box card and that no LED is lit on the card.

Step 4. Locate and disconnect the Temperature Sensor connector **(A)**.

Step 5. Detach the wire from the terminal connector and remove the wire from the Connection Box via the gland.

Step 6. Using a #2 Phillips screwdriver, detach the temperature sensor **(C)** from the Atlantium Unit.

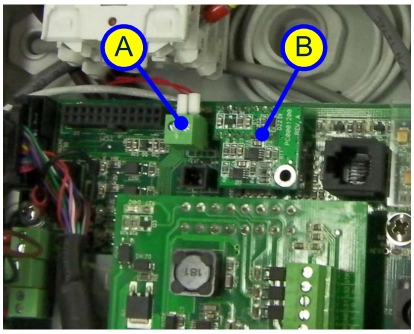


Figure 10-37: Location of the Temperature Sensor Connector (Generic)

Step 7. Attach the replacement Temperature sensor (C) to the Atlantium Unit.

Step 8. Thread the wire through the gland and attach it to the terminal connector.

Step 9. Insert the connector (A) to its socket on the terminal block.

Step 10. Return the Connection Box cover and using the Allen wrench, fasten the screws. Tighten them in a diagonal pattern.

Step 11. Turn the Mains circuit breaker to the **On** position.

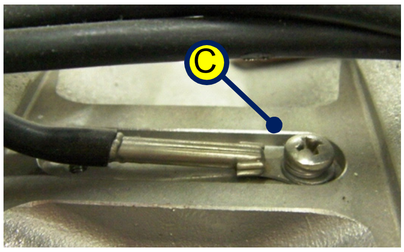


Figure 10-38: Location of the Temperature Sensor on the Atlantium Unit

10.12.2 Replacing the Temperature Converter Card

In the event that the Temperature sensor card requires replacement, this section details the procedure.

Follow your local regulations regarding whether a certified electrician may be required to perform this procedure.

Any person involved in handling the Connection Box card or its components is required to wear an electrostatic discharge (ESD) Wrist Strap.

Connect the ESD Wrist Strap to any bolt on the body of the Atlantium Unit or to the metal bracket under the Connection Box using a banana plug or an alligator clip.

Prepare the following:

- Replacement Temperature Converter card
- #5 Allen wrench
- #2 Phillips screwdriver

➲ To replace the Temperature sensor card:

Step 1. Turn the Mains circuit breaker to the **Off** position.

Step 2. Using the appropriate Allen wrench, on the first Connection Box, open the Connection Box cover.

Step 3. Verify with volt meter that there is no power in first Connection Box card and that no LED is lit on the card.

Step 4. Locate and carefully pull off the Temperature Sensor card (C).

Step 5. Aligning the pins of the replacement Temperature Sensor card (C) with the connector (on the board) on the Connection Box card, carefully insert the small card into the connector of the Connection Box card.

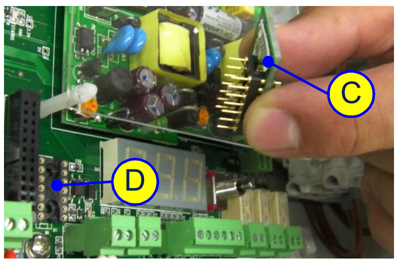


Figure 10-39: Replacing the Temperature Sensor Card (Generic)

Step 6. Return the Connection Box cover and using the Allen wrench, fasten the screws. Tighten them in a diagonal pattern.

10.12.3

11 Troubleshooting with Standard Control Module

The sections here are comprised of messages headers that appear as error messages in the Standard Control Module.

Utilize the following safety notes when reviewing the troubleshooting issues:

The solution to the problem might involve exposure to UV light.

The system generates ultraviolet (UV) light within the Atlantium Unit, which can cause serious eye damage or blindness if you stare at it directly when it is working.

Do not look directly into the lamp enclosure during system operation, or while examining or servicing the system's internal components when the system is operating, or when a lamp's caps are open.

The solution to the problem might involve working in a high-voltage environment.

Before replacing a UV lamp or other components, and during any maintenance requiring lamps to be turned off, **make sure the switch on the ballast of the relevant lamps are turned OFF.**

Place a sign on the relevant Electrical Cabinet alerting others **NOT** to touch the switch and touch screen during maintenance so that no one can inadvertently turn it on while maintenance is in progress.

The solution to the problem might involve working with hot surface.

Do not touch the UV lamp with your bare hands. Lamps can reach a temperature of 1,000°C under operating conditions.

Wait at least 10 minutes until the lamp is cooled down before replacing it.

Use appropriate protective gloves both to protect your hands and to avoid skin oils that leave fingerprints and/or harm the UV lamp.

11.1 Voltage Error Codes

For Ballast Module version C, the Voltage parameter provides a value that functions as an additional Error code and indicates additional information about the fault. See *Lamp Failure codes in Technician screen 3* on page 61.

Troubleshooting issues

11.2 The UV dose is below the allowable threshold

Control Module displays this warning message
UV Dose Below Allowable Threshold

!

	Possible Cause	Solution
1	The Water flow rate is too high.	Check the water flow on the Control Module's main screen. If it is higher than normal, decrease the rate of the water flow through the system.
2	UVT has declined because of chemicals (chlorine, etc.) or solids (sand, silt, etc.) in the water flow.	Carry out cleaning procedures. ***Cleaning In Place (CIP)*** on page 10-68. Try to improve the water condition. Check on the status of upstream filtering equipment and upstream chemical processes.
3	UV lamp efficiency has dropped.	 Check the UVIS sensors and clean them if necessary. Check the lamp's efficiency reading. If it is low, replace the lamp. See ***Replacing a UV Lamp*** on page 78.
4	The UVT sensor has malfunctioned.	 Switch between the UVT and UVIS sensors and see if the reading increased. If the reading is increased, switch the sensors back and replace the UVT sensor. See ***Checking/Replacing UVIS Sensors*** on page 93.
5	The system is operating on **Power mode** instead of **Dose mode**.	If the system settings are in Power mode, in the Control Module, access the Settings screen and change the **Operation Mode** from **Power** to **Dose** setting. Or, change the Power mode settings to allow higher power to respond to operating conditions.

11.3 Unit temperature is too high

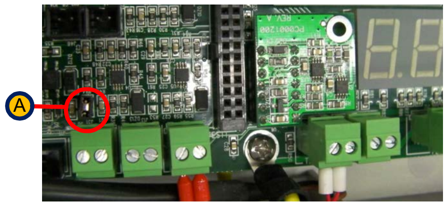
Control Module displays this warning message
Unit Temperature Is Too High

	Possible Cause	Solution
1	The Max temperature value is too low.	Check the system's **Max Temperature** settings.
2	The water flow has significantly decreased.	Check the flow reading in the Control Module's main screen to see if it has deviated from normal flow. Check the flow via the flowmeter. Check the water source for a problem.
3	There are air bubbles or the system is not fully filled with water.	Verify this by looking through the Atlantium Unit's view port.
4	You did not install a designated air release valve at the highest point on the inlet side of the system.	Release the air by installing a designated air release valve at the highest point on the inlet side of the system.

11.4 Unit temperature is critically high!

Control Module displays this warning message
Unit Temperature Critically High

	Possible Cause	Solution
1	If this message appeared right after a message indicating loss of communication with the Ballast Module (and both messages refer to the same lamp) and the temperature display on the monitor is 165°C/365°F, then most likely the communication between the Control Module and the Connection box is lost.	Follow the solutions offered in ***Ballast temperature is too high***
2	The temperature reading dropped to 185°C / 365°F due to a short-circuited in the Temperature sensor, Temperature converter card or Connection Box Card	Replace the temperature sensor. See ***Replacing the Temperature Sensor*** on page 101. If the problem persists replace the Temperature Converter card. See ***Replacing the Temperature Converter Card*** on page 102. If the problem still persists replace the Connection Box card. Consult your Atlantium Representative.
3	The Max temperature value is too low.	Check the system's **Max Temperature** settings. See ***Configuring More Settings*** on page 80.
4	There is no water flow.	Check the flow reading in the Control Module's main screen to see if it has deviated from normal flow. Check the flow via the flow meter. Check the water source for a problem.

	Possible Cause	Solution	
5	There are air bubbles or the system is not filled with water to capacity.	Verify this by looking through the Atlantium Unit's view port.	
6	You did not install a designated air release valve at the highest point on the inlet side of the system.	Release the air by installing designated air release valve at the highest point on the inlet side of the system.	
7	The Temperature jumper is missing.		Verify the existence of the Temperature jumper in its designated location on the Connection Box card #1. On the last Connection Box, open the top cover and locate Jumper **J32** as shown marked A in the figure.  Figure 11-1: Jumper Location
8	The Temperature sensor is faulty.		Replace the temperature sensor. See ***Replacing the Temperature Sensor*** on page 101.
9	The Temperature Converter card is faulty.		If the problem persists replace the Temperature Converter card. See ***Replacing the Temperature Converter Card*** on page 102.

11.5 The rate of the water flow is too low

Control Module displays this warning message
Water Flow Rate: Too Low

	Possible Cause	Solution
1	There is an obstruction in the upstream system or pump malfunction.	If this is a planned decrease or temporary stop, change the **Restart on Flow** setting from **Shutdown mode** to **Restart**. This sets the system to Standby. Check pumps, filters and system feed and increase the flow rate.
2	There is no water running through the system.	If this is a planned stop, change the **Restart on Flow setting** to **Restart**. This sets the system to Standby.
3	The flow meter/flow switch is faulty or a wire has been disconnected.	Check the wires connecting the flow meter/flow switch and reconnect or replace if damaged. Replace the flow meter/flow switch is faulty.

11.6 UV lamp ignition failed

Control Module displays this warning message
UV Lamp ID Ignition Failed

	Possible Cause	Solution	
1	The system is set to the wrong model.	In the Control Module, access **Technician screen #2**. Verify that the system is set to the correct model, number of lamps and branches.	
2	One or more connectors are burnt out.		Turn off the system's main power source On each Connection Box, turn the switch to the Off position. Disconnect and thoroughly inspect the Ballast Module connectors. IF replacement is needed, contact Atlantium Technical Support.
3	The power supply inside the Ballast Module is faulty.	 	In a system with more than one lamp, if possible, switch between the Ballast Module connectors in order to verify the Ballast is faulty, if so replace the Ballast Module. In a system with only one lamp, replace the Ballast Module with a new one.
4	A lamp is faulty/burned out.	 	In a system with more than one lamp, switch between the lamps in order to verify the lamp is faulty, if so replace the lamp. In a system with only one lamp, replace the lamp with a new one. See ***Replacing a UV Lamp*** on page 78.
5	The ID number of the Connection Box card does not match ID number of the Ballast Module.		On the each Connection Box, open the top cover and locate the 3-digit display (as shown on the right). Check that the set number is identical to the number marked on corresponding Ballast Module. If not identical, use the switch to change it to the correct number. 
6	One or more of the fuses in the Connection Box is burned out.		Replace the burned out fuse in the Connection Box. Consult your Atlantium representative. Contact Atlantium support for further instructions.
7	The Connection Box card is faulty.		On the Connection Box, open the top cover and check that the 3-digit display and LEDs are lit. If they are dark, replace the Connection Box card. Contact Atlantium support for further instructions.

11.7 Temperature sensor fault

Control Module displays this warning message
Temperature Sensor Fault

	Possible Cause	Solution
1	If this message appeared right after a message indicating loss of communication with the Ballast Module (and both messages refer to the same lamp) and the temperature display on the monitor is 165°C/365°F, then most likely the communication between the Control Module and the Connection box is lost.	Follow the solutions offered in ***Ballast temperature is too high***.
2	The temperature reading dropped to 0°C / 32°F due to a short-circuited in the Temperature sensor, Temperature converter card or Connection Box Card	Replace the temperature sensor. See ***Replacing the Temperature Sensor*** on page 101. If the problem persists replace the Temperature Converter card. See ***Replacing the Temperature Converter Card*** on page 102. If the problem still persists replace the Connection Box card. Contact Atlantium support for further instructions.

11.8 Ballast temperature is too high

Control Module displays this warning message
Ballast Temp Too high Lamp ID Fault

	Possible Cause	Solution
1	The Connection Box environment's temperature is above 80 degrees Celsius.	Check the environment temperature in the Connection Box area, open the connection box and look for a possible cause for the heating.

11.9 System type settings error

Control Module displays this warning message
Unset Error Fault at Lamp ID

	Possible Cause	Solution
1	**Lamp Unset:** The System is set to the wrong system model. (Example: System is RZ163, but the Control Module settings read RZ300.)	In the Control Module, access the **Technician screen #2**. Verify that the system is set to the correct system model, number of lamps and branches. If the system is already set to the correct model, you might need to refresh the selection: Step 1. Change the model to a different one (e.g. if your system is RZ163, change the selection to RZ300). Step 2. Exit the **Technician** screen and close the Control Module circuit breaker. Step 3. Open the Control Module circuit breaker and enter **Technician** screen. Step 4. Change the model back to the correct model of your system. Step 5. Exit the technician screen and close the Control Module circuit breaker. Step 6. Open the Control Module circuit breaker and turn the lamps on.
2	Your system is supposed to contain High Power lamps, but actually does not contain this type of lamp(s).	Make sure the lamps inside the system are the High Power type. Refer to ***Replacing a UV Lamp*** on page 10-78 for details on checking the lamps.

11.10 Lamp mains failure

Control Module displays this warning message
Lamp Mains Fail Lamp ID Fault

	Possible Cause	Solution	
1	**Mains Fail issue:** The ballast does not receive enough power from the mains.		A certified electrician is required to verify that the Ballast Module receives the proper voltage feed from the Mains via the Connection Box.
2	One or more of the fuses in the Connection Box is burned out.		A certified electrician is required to replace the burned out fuse in the Connection Box. Contact Atlantium support for further instructions.
3	A combined problem inside the Ballast Module regarding power feed and temperature.	Contact Atlantium support for further instructions.	

11.11 Unexpected failure for an unknown reason

Control Module displays this warning message
Unexpected Failure Unknown Reason ID Fault

	Possible Cause		Solution
	Lamp Unknown: The lamp was unexpectedly shutdown due to one of the following reasons:		
1	There is a power interruption (for micro seconds) that caused the Ballast Module to stop working while the Connection Box Card still held power.		Turn the lamps back on using the On/Off button on the Control Module's main screen.
2	The Lamp burned out during operation.	 	Replace the Lamp. See ***Replacing a UV Lamp*** on page 78.
3	The Lamp's connector(s) burned during operation.		Replace the connectors. Contact Atlantium support for further instructions.
4	Ballast Module malfunctioned during operation.		Replace the Ballast Module.

11.12 Lost communication with ballast module

Control Module displays this warning message
Lost Communication with Ballast Module

	Possible Cause		Solution
1	**Communication Error:** The communication between the Control Module's display card and Connection Box card was lost while the lamp was on or during the ignition process.		Verify that all Connection Box cards have power and the communication wire is connected properly on both ends. If the problem persists, contact Atlantium support for further instructions.
2	No power is feeding into the Connection Box.		Open the Connection boxes and verify that all Connection Box s have power.
3	A wire is loose.		With a small flat screwdriver go over all the communication screws in the Control Module and Connections Box s and make sure all wires are located properly in their place and are well tightened. .

	Possible Cause	Solution
4	A jumper is missing.	A communication jumper might need to be installed, contact Atlantium Support for further instruction.
5	Ground is not connected to the Control Module's power inlet.	A certified electrician is required to verify that the Control Module is properly grounded.
6	There is a Hardware problem in the Control Module's display card or in the Connection Box.	Contact Atlantium support for further instructions.

11.13 Clean system deposit

Control Module displays this warning message
Please Clean System Deposit

		Possible Cause
1	Excessive mineral deposit has collected on the quartz sleeve.	Perform the Cleaning in Place procedure. See *Cleaning In Place (CIP)* on page 68.
2	The **UVT Min** value in the Technician screen is set too high.	Set the **UVT Min** value to **0**.

Appendix A Modbus Communication Protocol

Modbus is an application layer messaging protocol for client/server communication between devices connected on different types of buses or networks.

Modbus is often used to connect a supervisory computer with a remote terminal unit (RTU) in supervisory control and data acquisition systems.

For information on how to configure the communication properties and make the cable connections, see *See Control Module Connections on page* 5-47. on page 59.

ATTENTION!

Modbus registers are meant to be used for specific commands in a command algorithm. Do not write to any register if you do not intend to use it.

A.1 Modbus Registry Map

The table on the following pages provides the information you need to help you configure your network to properly communicate with the Atlantium system.

Table A-1: BW Modbus registry address list

Registry address	Registry name	Divide	Type	Comment	Units	Read/ Write
215	UVTS lamp 4	:100	Register		mA	R
205	UVTS lamp 3	:100	Register		mA	R
121	UVTS lamp 2	:100	Register		mA	R
113	UVTS lamp 1	:100	Register		mA	R
210	UVT lamp 4	:100	Register		%	R
200	UVT lamp 3	:100	Register		%	R
115	UVT lamp 2	:100	Register		%	R
107	UVT lamp 1	:100	Register		%	R
100	UVT	:100	Register		%	R
214	UVIS lamp 4	:100	Register		mA	R
204	UVIS lamp 3	:100	Register		mA	R
120	UVIS lamp 2	:100	Register		mA	R

112	UVIS lamp 1	:100	Register		mA	R
110	Temperature °C	:100	Register		°C	R
19	System Ready		Coil Bit	0-> Not Ready		R
103	System Power	:1	Register		%	R/W
0	System On/Off		Coil Bit	0 -> Off, 1 -> On		R/W
217	Status lamp 4	:1	Register		%	R
216	Status lamp 3	:1	Register		%	R
114	Status lamp 2	:1	Register		%	R
106	Status lamp 1	:1	Register		%	R

Registry address	Registry name	Divide	Type	Comment	Units	Read/ Write
125	Set for (dose)	:10	Register		mJ/cm²	R/W
213	Power lamp 4	:100	Register		KW	R
203	Power lamp 3	:100	Register		KW	R
119	Power lamp 2	:100	Register		KW	R
111	Power lamp 1	:100	Register		KW	R
11	Over Temp		Coil Bit	1 -> Fault, 0 -> Ok		R
2	Operation Mode Dose/ Power		Coil Bit	1 -> Dose, 0 -> Power		R
127	Minimum Dose Alarm		Coil Bit	UNCHECK - 0 , CHECHED = 1		R/W
126	Minimum Dose	:10	Register		mJ/cm²	R/W
14	Min flow alarm		Coil Bit	1 -> Fault, 0 -> Ok While Restart Flow isn't checked		R
128	Maximum Dose	:10	Register		mJ/cm²	R/W
26	Lamp 4 Enable/ Disable		Coil Bit	1 -> Enable, 0 -> Disable		R/W
25	Lamp 3 Enable/ Disable		Coil Bit	1 -> Enable, 0 -> Disable		R/W
24	Lamp 2 Enable/ Disable		Coil Bit	1 -> Enable, 0 -> Disable		R/W
23	Lamp 1 Enable/Disable		Coil Bit	1 -> Enable, 0 -> Disable		R/W

22	DPM in CIP (lamps off)		Coil Bit	1-> Wiper moving, 0->Wiper not moving		R
16	General alarm		Coil Bit	Found valid when high temp system off shows 0 . Low dose while system ON Reg. changed to 0 (min dose alarm enable) 1 -> OK Dose below Min Dose 0 -> Fault Dose above Min Dose		R
101	Flow	:10	Register		m³/h	R
147	DPM 4 Counter High	:1	Register			R
146	DPM 4 Counter Low	:1	Register			R
145	DPM 3 Counter High	:1	Register			R
144	DPM 3 Counter Low	:1	Register			R
143	DPM 2 Counter High	:1	Register			R
142	DPM 2 Counter Low	:1	Register			R
141	DPM 1 Counter High	:1	Register			R
140	DPM 1 Counter Low	:1	Register			R
9	Dose < min dose		Coil Bit	While Min.Dose alarm is enabled -> 1 ->register > 1 -> Fault, 0 -> Ok		R
102	Dose	:1	Register		mJ/cm²	R
Registry address	**Registry name**	**Divide**	**Type**	**Comment**	**Units**	**Read/ Write**
18	Critical Over Temp		Coil Bit	1 -> OK temp under Max Temp 0-> Fault temp over Max Temp		R

17	Cleaning System exist		Coil Bit	Fail -At 8888 setting checkbox influence register 15 (1- check,0- uncheck) 1 --> DPM exists, 0-->DPM does not exist		R
15	Cleaner System Off/ On		Coil Bit	Fail - influence register 17 * Enable -- > 1, Disable -->0		R
6	Alarm/Ready output		Coil Bit	At 1357 setting -->More Set. Alarm /Ready: Checked -->1, Unchecked --> 0 0 -> Alarm, 1 -> Ready		R
211	Age hours lamp 4 MSB	:1	Regist er		Hours	R
201	Age hours lamp 3 MSB	:1	Regist er		Hours	R
116	Age hours lamp 2 MSB	:1	Regist er		Hours	R
108	Age hours lamp 1 MSB	:1	Regist er		Hours	R

Appendix B System Messages

The table below details the message that can appear on the Control Module's screen or in the Event logs. A number of parameters are User Defined and include setting a value and upon reaching, exceeding or falling below it, an Action is set to be triggered. The Action is also configurable. The options are:

- **None** - The event is not considered an error, so no corrective action is taken. No warning message is issued.
- **Caution** - The caution alerts you to an anomaly based on the user preferences.
- **Warning** - One or more warning messages are issued and it triggers a pop-up window message and a General Alarm (if configured, see *Configuring the General Alarm* on page 7-73).
- **Shut-Down** - In case of a serious error or dangerous condition, the system shuts down and triggers a pop-up window message and a General Alarm (if configured, see *Configuring the General Alarm* on page 7-73).

The other parameters are system defined. The type column in the table below details the type of parameter.

Table B-1: Control Module's Messages

Key	Type	Event Log	Line Message/Popup Title	Popup Message
1	The value is User Defined	Over Max. Dose	The dose is above the maximum dose	The dose is above the maximum defined in the system settings.
2	The value is User Defined (The default is None)	Dose Too Low	The UV dose is too low	Dose too low! Below minimum!
4	Warning - The default temperature threshold is 5°C (9°F) below what is defined for Over Temp	Temp. Too High	Temperature is Too High	The lamp's temperature getting too high -- check water flow.
5	Shutdown - - The default temperature threshold is User Defined	Over Temp	Temperature is Critically High!	The lamp's temperature is too high and the unit will shut down.
6	Shutdown - Hardware malfunction	Temperature Sensor Fault	Temperature Sensor Fault	One of the temperature sensors has malfunctioned or the communication with the Connection Box card was lost.
9	Shutdown ---The Minimum Flow is User Defined	Low Flow	The rate of the water flow is too low	The water flow rate is lower than the minimum flow rate. Check the water flow.
10	The value is User Defined	High Flow	The rate of the water flow is too high	The water flow rate is higher than the maximum flow rate allowed. Check the water flow.
11	The value is User Defined	Transmission Low	UVT is too low!	UVT is below the required value
14	Warning	Lamp Ignition	UV Lamp Ignition Failed!	The UV lamp has failed to ignite due to an electronic fault. The UV lamp may be burnt out or a wire may be disconnected.

Table B-1: Control Module's Messages

Key	Type	Event Log	Line Message/Popup Title	Popup Message
16	Appears in the Event Log only - pops up from additional info from Event Log	User On		The Control Module screen or an external signal turned on the system.
17	Appears in the Event Log only - pops up from additional info from Event Log	User Off		The Control Module screen or an external signal shut down the system.
18	Appears in the Event Log only - pops up from additional info from Event Log	Auto On		System power was restored or flow raise above minimum. The lamps reignite.
19	Appears in the Event Log only - pops up from additional info from Event Log	Auto Off		Flow rate below user low flow setting; automatic shutdown.
20	Appears in the Event Log only - pops up from additional info from Event Log	Lamps On		System Ready. All the enabled lamps are on.
24	Caution	Dose On Spec	Dose On Spec	The UV dose being delivered by the system is above the minimum validated value
25	Warning	Dose Off Spec	Dose Off Spec	The UV dose being delivered by the system is below the minimum validated value
26	Caution	UVT On Spec	UVT On Spec	The UVT value is above the minimum validated UVT
27	Warning	UVT Off Spec	UVT Off Spec	The UVT value is below the minimum validated UVT
28	Caution	Flow On Spec	Flow On Spec	The flow rate is below the maximum validated flow
29	Warning	Flow Off Spec	Flow Off Spec	The flow rate is above the maximum validated flow
33	Warning	UVT 4th calibration failed	UVT Analyzer 4th calibration check failed	UVT Analyzer calibration check failed 4 times in a row. System is off-spec
34	The value is User Defined (Default as None)	Validated Min UVT	Validated Min UVT	UVT is below Validated Min UVT
35	The value is User Defined (Default as None)	Validated Max Flow	Validated Max Flow	Flow is above Validated Max Flow
39	Appears in the Event Log only - pops up from additional info from Event Log	Power Failure	Power Failure	The system was turned off due to power failure
41	The value is User Defined	UV Dose	The UV Dose is below the set level	The UV Dose is below the Operate At dose

Table B-1: Control Module's Messages

Key	Type	Event Log	Line Message/Popup Title	Popup Message
42	Warning - Sensor needs to be replaced or there may be a loose wire	Temp. Sensor Fault	Temperature Sensor Fault	Fault on lamp temperature sensor
43	Shutdown	Lamp Shutdown	The UV lamp unexpectedly shutdown	Lamp shutdown
45	Shutdown	PS Fault	Power supply fault	Ballast fault [value displayed], error code: [number displayed].
46	Warning	Communication Error	Communication Error	Lost communication with the Ballast Module
54	Caution	UVIS is calibrated successfully	UVIS is calibrated successfully	UVIS is calibrated successfully
55	Caution	UVIS calibration failed	UVIS Calibration Failed	UVIS calibration failed, Sensor is out of calibration. Correction Factor is implemented
56	Caution	UVT successful calibration check	UVT successful calibration check	UVT Analyzer successful calibration check
57	Caution	UVT check failed	UVT Analyzer check failed	UVT Analyzer check failed; Auto recalibration was done

Appendix C Checking the System Parameters

You can use the checklist below to keep a record of your checks on Atlantium system parameters. **Print this page**.

Table C-1: Atlantium System Parameters Checklist

Date	Time	Lamp ID	Power	Flow Rate	UVT	UV Dose	Lamp Status	OK Y/N	Remarks/ Alarms/Events
			Add Expected Value ________	Add Expected Value ________	Add Expected Value ________	Add Expected Value ________	Add Expected Value ________		

Appendix D Glossary of Terms

Term	Definition
Auxiliaries	Valves, flow devices, etc. essential to the workings of the Atlantium system but not directly a part of the Atlantium system.
Ballast	The power supply that translates the electrical power to UV power.
Connection Box	The box that sits on top of each lamp unit and holds the circuit board for the Electrical Cabinet or Ballast Module electrical configurations. Not relevant for Ballast Module/Cable-style.
Control Module	Control Module that controls operations and dose measurement. Comes in two versions: Standard and Premium. The Standard has a monochromatic screen and several key features. The Premium has a color screen can accommodate a full complement of reporting and trending features.
External On/Off	This enables an external source (SCADA, PLC, Process) to turn the Atlantium system on and off.
IP56	Protected from limited dust ingress. Protected from high pressure water jets from any direction
Electrical configuration styles: Atlantium provides a choice of electrical configurations.	
Electrical Cabinet-style (EC)	**EC** - Electrical Cabinet.
Ballast Module	Waterproof case for the ballast which connects via a consolidated plug from the circuit board which sits on top of each lamp unit.
General Alarm	This enables the Atlantium unit to send an alarm to a buzzer, cell phone or other destination.
Inlet/Outlet Valves	Isolating Valves on the Inlet/outlet sides.
Lamp Chamber	The chamber which holds the lamp.
Lifting Support	The lifting element which can also be used to help support the Unit.
Pinhole	This is a small disc that is fitted for each sensor port to ensure that the right amount of light penetrates.
Portable CIP System	The portable unit Atlantium sells that connects to the Unit to clean and disinfect it without disassembling. It includes several kits – a valve kit, an accessories kit and the CIP kit itself.
Quartz Chamber	The **quartz pipe** in which the UV exposure occurs.
System Ready Signal	The signal the Atlantium unit provides to tell the production system that the Atlantium System is ready to supply properly treated water.
System Tuning	This is process is performed when key components are replaced or changed to synchronize them.

Term	Definition
Triangle Support Brackets	RZ-Series only- brackets that can be used to support and balance a unit.
UV Intensity Sensor	The sensor that measures the UV intensity in the Lamp chamber and feeds the real time dose calculation.
UVT Analyzer	The sensor that measures the lamp UV intensity through the water. Together with the UV Intensity sensor and the Control Module, it analyzes, tracks and reports the water quality.

Appendix E Consumables and Spares

The Part Numbers and Descriptions of the Consumables & Spare Parts shown in these tables are subject to change.

Table E-1: Spare parts for RS104

Part Number	Description	Qty
Consumables - Required		
MP0023000	O-RING 2-327 V0680-70 FOR SLEEVE	2
OPE000400	UV LAMP 95mm 1kW	2
Recommended Spare parts		
OPE004200	QUARTZ SLEEVE RZ104/RS104	1
OPB000100	UV SENSOR 10W/m2	1
PS0000240	BALLAST MODULE 1.0-2.3KW, 380-480V	1
AP0001200	TEMPERATURE CONVERTER CARD	1
AP0001350	LM CARD, RZB SYSTEMS	1
AP0001500	AC/DC MODULE, 100-240VAC TO 24VDC	1
SAM000800	WAVE GUIDE ASSY	1